ANKARA YILDIRIM BEYAZIT UNIVERSITY

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Fatigue Life Estimation of Welded Structures: A New Modified Hot-spot Stress Method

PhD Thesis by

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Ph.D. THESIS EXAMINATION RESULT FORM

We have read the thesis entitled "**Fatigue Life Estimation of Welded Structures: A New Modified Hot-spot Stress Method**" completed by **OĞUZHAN MÜLKOĞLU** under the supervisions of **Dr. FATİH GÖNCÜ** and **Prof. Dr. MEHMET ALİ GÜLER** and we certify that in our opinion it is fully adequate, in scope and in quality, as a thesis for the degree of Doctor of Philosophy.

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FATIGUE LIFE ESTIMATION OF WELDED STRUCTURES: A NEW MODIFIED HOT-SPOT STRESS METHOD

ABSTRACT

Welding is a joining technique that still maintains its importance in today's automotive, transportation, civil and industrial structure designs. Fatigue life of welded structures is generally estimated by considering the stress field near the weld. However, even for the same loading conditions, the stress gradient near the weld toe will vary from one specimen to another due to imperfections. This in turn causes large scatters in the S-N curves of fatigue life estimation methods. A successful fatigue life estimation is considered to have a reduced scatter of the corresponding S-N curve under variable geometry and loading conditions. The Hot-spot Stress (HSS) method is a commonly used estimation method owing to its practical approach and ability to include the effect of geometric details. In this paper, an improvement of this method using optimization techniques is presented. The modified HSS method is then used to estimate the fatigue life of welded structures with different thicknesses and loading conditions. The results show that the proposed method will noticeably reduce the scatter in S-N curves by increasing the coefficient of determination value (R^2) 9.1% compared to the classical method.

Keywords: *Fatigue Life Estimation, Hot-spot Stress Method, Extrapolation Points, Surrogate Models, Optimization*

KAYNAKLI YAPILARIN YORULMA ÖMRÜ TAHMİNİ: YENİ BİR DEĞİŞTİRİLMİŞ SICAK NOKTA GERİLME YÖNTEMİ

ÖZ

Kaynaklı birleştirmeler, günümüz otomotiv, ulaşım, sivil ve endüstriyel yapı tasarımlarında hala önemini koruyan bir birleştirme tekniğidir. Kaynaklı yapıların yorulma ömrü, genellikle kaynağa yakın gerilme alanı dikkate alınarak tahmin edilir. Bununla birlikte, aynı yükleme koşulları için bile, kaynak ucu yakınındaki gerilme gradyanı, kusurlar nedeniyle bir numuneden diğerine değişecektir. Bu da yorulma ömrü tahmin yöntemlerinin S-N eğrilerinde büyük dağılımlara neden olur. Başarılı bir yorulma ömrü tahmininin, değişken geometri ve yükleme koşulları altında karşılık gelen S-N eğrisinin azaltılmış bir dağılıma sahip olduğu kabul edilir. Sıcak nokta gerilme (HSS) yöntemi, pratik yaklaşımı ve geometrik detayların etkisini içermesi nedeniyle yaygın olarak kullanılan bir tahmin yöntemidir. Bu tezde, optimizasyon teknikleri kullanılarak bu yöntemin bir iyileştirmesi sunulmaktadır. Daha sonra modifiye edilmiş HSS yöntemi, farklı kalınlık ve yükleme koşullarına sahip kaynaklı yapıların yorulma ömrünü tahmin etmek için kullanılır. Sonuçlar, önerilen yöntemin klasik yönteme göre belirleme katsayısı değerini (R^2) %9,1 artırarak S-N eğrilerindeki saçılımı belirgin şekilde azaltacağını göstermektedir.

Anahtar Kelimeler: *Yorulma Ömrü Tahmini, Sıcak-nokta Gerilme Metodu, Ekstrapolasyon Noktaları, Vekil Modeller, Optimizasyon*

CONTENTS

NOMENCLATURE

R^2	Coefficient of determination
$\Delta\sigma$	Stress range
$\sigma_{\max}$	Maximum stress
$\sigma_{\min}$	Minimum stress
R	Stress ratio
N	Number of cycles
m	slope
t	thickness
w	width
$\Delta\sigma_{HSS}$	Hot-spot stress range
C	Design value of the FAT category
σ_{ss}	Structural stress
σ_m	Membrane stress
σ_b	Bending stress
$f_{x'}$	Line forces acting on x' coordinate
$m_{y'}$	Line moment acting on y' coordinate
$I(r)^{1/m}$	Dimensionless polynomial function
r	Bending ratio
R_{local}	Local stress ratio
E	Young’s modulus
v	Poisson’s ratio
$\hat{Y}$	Response value
X	Predictor value
a	Intercept
b	Slope of the regression line
X_i	i^{th} value of the predictor variable
Y_i	i^{th} value of the response variable
$\bar{X}$	Mean value of the predictor variable

$\bar{Y}$	Mean value of the response variable
e_i	Residual term
$\hat{Y}_i$	i^{th} value of the predicted response variable
s^2	Standard deviation of calculated errors
s	Standard error of the regression
X^*	A random predictor variable
Y_u^*	Response variable on the upper CI for a predictor variable
Y_l^*	Response variable on the lower CI for a predictor variable
$t_{\alpha/2}$	Constant of the t-normal distribution for a desired confidence
$s_{\hat{Y}^*}$	Estimated standard deviation of the predicted values
γ	Confidence level
$\Phi^{-1}_{(P_s)}$	Ps-percentile from the inverse standard normal distribution
σ_{mhss}	Modified hot-spot stress

Acronyms

FE	Finite element
S-N	Stress-Number of cycles
HSS	Hot-spot stress
SS	Structural stress
2D	Two-dimensional
3D	Three-dimensional
FEM	Finite element method
FEA	Finite element analysis
ASTM	American Society for Testing and Materials
FAT	Fatigue strength values
IIW	International Institute of Welding
JSSC	Japanese Society of Steel Construction
DNV	Det Norske Veritas
FKM	Forschungskuratorium Maschinenbau
FPSO	Floating production storage and offloading
CHP	Circular hollow profiles

CAD	Computer aided drawing
ENS	Effective notch stress
SWT	Smith-Watson-Topper
SHELL181	4-Noded shell elements
SHELL281	8-Noded quadrilateral elements
SOLID185	8-Noded brick elements
SOLID186	20-Noded hexahedral elements
Quad.	Quadratic
VDV	Verband Deutscher Verkehrsunternehmen
PRS	Polynomial response surface
DoE	Design of experiments
LHS	Latic hypercube sampling
SSE	Sum of squared errors
MSE	Mean squared errors
RMSE	Root mean squared error
CI	Confidence interval
PI	Prediction interval
Ps	Probability of survival
C3D8R	8-node linear brick elements

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CHAPTER 1

INTRODUCTION

1.1 Background

Welding is the most common joining technique in steel structures such as railway and road bridges, railway vehicles and offshore platforms. These structures are subjected to many number of repetitive cyclic loadings during their lifetime causing fatigue damage. It has been demonstrated by various studies in the literature [1]–[4] that fatigue-induced damage is a type of deformation frequently encountered, especially in steel structures with welded joints.

Fatigue-induced deformations usually occur in regions where there are sharp changes in the geometry of the structures resulting in an increase in stress concentration and thus a decrease in the fatigue life of the component. Therefore, fatigue life estimation of welded structures in the highly localized stress concentration is crucial in the design and maintenance stages of structures.

Fatigue life estimation is not only affected by sharp changes in geometries; it is also affected by loading conditions and other geometrical properties of the components such as thickness and shape of the geometries (T-joint, X-joint etc.). The inclusion of the effects of the above-mentioned variables in welded structures in the fatigue life estimation is decisive for the service life of the structures. Accordingly, the effects of all these variables are included in the fatigue life estimation under various methods.

Considering the widespread use of the welding process in various industries, a great deal of research has been conducted on fatigue life assessment, and as a result, various fatigue assessment methods have been proposed. These methods have some advantages and limitations in fatigue evaluation of complex structures, which will be discussed later. It is crucial to understand these methods correctly and to use a "correct" fatigue assessment method in fatigue design.

A proper fatigue design involves both testing and analysis [4]. The use of finite element (FE) analyzes provides accurate estimation of the stress state of welded structures with complex geometry and loading conditions. Hence, it is quite common to use the FE method to estimate the fatigue life of welded structures. On the other hand, in regions of high stress concentrations, the results of finite element analysis are sensitive to FE modeling techniques. The resulting stresses may differ significantly depending on the size and type of the elements. Therefore, constructing a proper finite element model is crucial for fatigue life estimation.

There can be huge scatter in the S-N curves of fatigue life estimations even under the same loading and geometry conditions. The narrowness of this scatter is the most significant success criterion as it means that the fatigue life estimation is less affected by various parameters. Therefore, this variable, which is called coefficient of determination, R^2, has been considered as the success criterion in the new methods to be developed.

1.2 Research Objectives

The thesis can be divided into two main parts. The first part focuses on the existing fatigue life estimation methods. Among these methods, the modeling techniques of Hot-spot Stress (HSS) and Structural Stress (SS) method were studied in more detail and a comparison between HSS and SS was made on a real life industrial problem on the tram example. In addition, discretization with 2D shell & 3D solid element in the HSS method was compared under different loading conditions. To ascertain the accuracy of the fatigue lives obtained using shell element models, the hot-spot stresses had computed for three commonly encountered welded attachments and compared the results with hot-spot stresses obtained from solid element models. Within the scope of this part, the objectives were to:

- Understand the fatigue life estimation methods correctly.
- Investigate the modelling techniques and examine the suitability of using 2D shell elements in the HSS method.
- Illustrate the application of the Structural Stress approach on a workbench geometry which is meshed with different element sizes.

- Obtain which method gives more conservative results from HSS and SS methods on the tram example.
- Investigate the mesh sensitivity of HSS method.

In the second part of this thesis, a new modified HSS method was proposed. The HSS method is one of the most common estimation methods since it is very practical approach and can include the effect of geometric details. When the studies in the literature and in the first part of the thesis are examined, it is seen that the fatigue life estimation according to the HSS method can have a large scatter in the S-N curves even under same loading conditions. The improvement of this method is discussed with optimization techniques and new findings are presented as a modified HSS method. Therefore, within the scope of this part, the objectives were to:

- Create an experimental data including a wide range of main plate thickness with three different loading conditions: 4-point bending, 3-point bending and axial loading.
- Perform an optimization based on the location of the stresses used in calculating the HSS.
- Obtain new extrapolation points for the calculation of HSS that can reduce the scatter of the S-N curve.

The main objective of this thesis is to propose a new modified Hot-spot Stress Method that can reduce the scatter of the S-N curve after an optimization study performed based on the location of the stresses used in calculating the HSS. Typical HSS calculation involves measuring stresses at two/three points at specific distances from the weld toe. The main contribution of this thesis is to find the optimum distances of stress measurements for estimating the fatigue life more accurately. Considering that the current HSS method is still frequently used, this improvement in the scatter with this new modified HSS method will be crucial in fatigue life estimations.

1.3 Outlines

This thesis consists of two main sections: understanding the existing fatigue life estimation methods and proposing a new fatigue life estimation method. The chapters are organized according to this main sections. First section of the thesis is given in Chapter 2 and Chapter 3. Chapter 4 contains the details of the proposed new method.

Chapter 2 – Fatigue in welded structures: This chapter provides an introduction to fatigue of welded structures. A brief classification of fatigue life assessment methods is given and existing fatigue life estimation approaches are introduced in this chapter.

Chapter 3 – Fatigue evaluation using FEM: In this chapter the application of the 2 most preferred fatigue life estimation methods, SS and HSS methods, is presented by applying different FE modeling techniques on sample benchmark studies.

Chapter 4 – A New Approach – Modified HSS Method: This chapter introduces a new modified HSS method. This section includes details about the method consisting of the definition of the problem, the creation of the database from literature, the optimization study and the results obtained according to the new method.

Chapter 5 – Conclusion: This chapter summarizes the conclusion of the thesis work. Suggestions for future research are also given.

CHAPTER 2

FATIGUE IN WELDED STRUCTURES

American Society for Testing and Materials (ASTM) defines fatigue as the progressive, localized, permanent structural change that occurs in materials subjected to fluctuating stresses and strains that may result in cracks or fracture after a sufficient number of fluctuations. [5]

August Wöhler (1819-1914) was the first researcher to demonstrate that a single non-deforming load can cause damage under cyclic loading. On the other hand, since fatigue damage was micro-level and invisible, fatigue was thought to be a mysterious phenomenon in the 19th century [6]. In the 20th century, there has been significant progress in the knowledge of the fatigue mechanism of structures with understanding that cyclic loads can cause cracks to initiate and propagate in structures [1]. Fatigue-induced damage is now generally recognized and it is known as a type of deformation frequently encountered, especially in steel structures with welded joints.

In this chapter, classification of fatigue assessment methods are introduced with factors influencing fatigue performance and fatigue strength S-N curves. The most common fatigue life estimation methods are also described briefly. Since HSS and SS methods are taken into consideration in more detail within the scope of the thesis, these two methods are introduced in more detail in this section.

2.1 Classification of Fatigue Assessment Methods

Various fatigue assessment methods have been introduced to estimate the fatigue life of welded steel structures over time. Life of the structures is typically expressed by number of cycles N. On the other hand, there has been some different assessment concepts (S-N, e-N, W-N etc.) over time and proposed to be classified according to different criterions and parameters such as global/local approaches, stress/strain/energy criterion, geometry parameter (intact/crack damaged) and process zone (point, line or area/volume) as depicted in Figure 2.1. ([7]–[13])

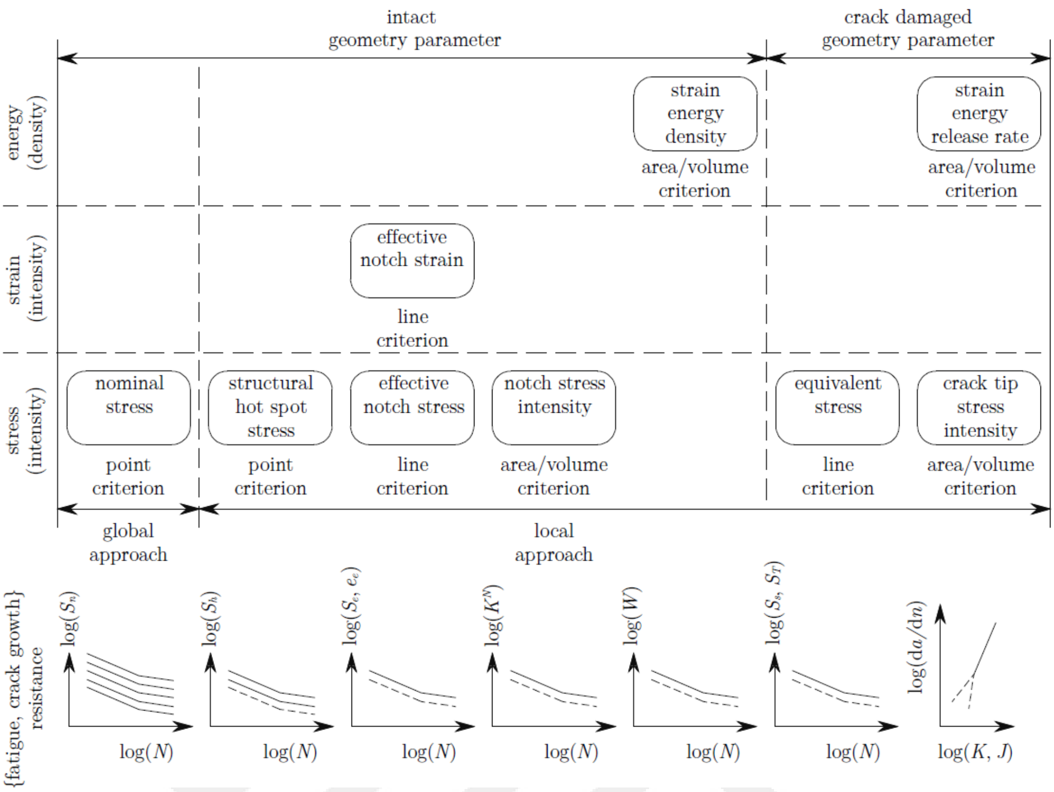


Figure 2.1 Fatigue assessment concept overview [14]

It is possible to increase the accuracy of fatigue life estimation, however this causes more complexity and it requires more effort as depicted in Figure 2.2. Considering this situation when choosing a method for fatigue life estimation, the most preferred methods are methods based on stress.

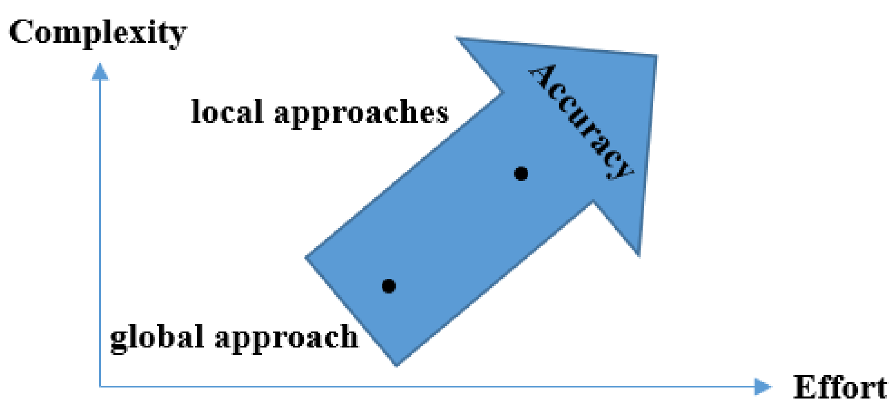


Figure 2.2 Typical complexity, effort and accuracy comparison for fatigue life estimation

2.1.1 Factors Influencing Fatigue Performance

There are many factors affect the fatigue life of welded structures. These factors are mainly expressed in four main categories [7]: structural factors, loading factors, material factors and environmental factors. Structural and loading factors are the most significant factors for the fatigue life of welded steel structures. [15]

Structures are subjected to many number of repetitive cyclic loadings during their lifetime causing fatigue damage. There are two types of cyclic loading: constant amplitude and variable amplitude cyclic loading. Although fatigue tests are generally performed under constant amplitude loading conditions, it is a well-known fact that real-life loadings mostly have variable amplitudes. However, variable amplitude loading conditions can be significantly complex loadings. In order to handle with these complex loadings, constant amplitude stress ranges of the variable amplitude loadings are taken into consideration in design calculations. There are two main parameters of loading conditions in fatigue life problems: the stress range, $\Delta\sigma$ and the stress ratio, R. Stress range and stress ratio are simply calculated as

$$\Delta\sigma = \left|\sigma_{\max} - \sigma_{\min}\right| \tag{2.1}$$

$$R = \frac{\sigma_{\min}}{\sigma_{\max}} \tag{2.2}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the maximum and the minimum stresses, respectively. Loading factor parameters are depicted in Figure 2.3.

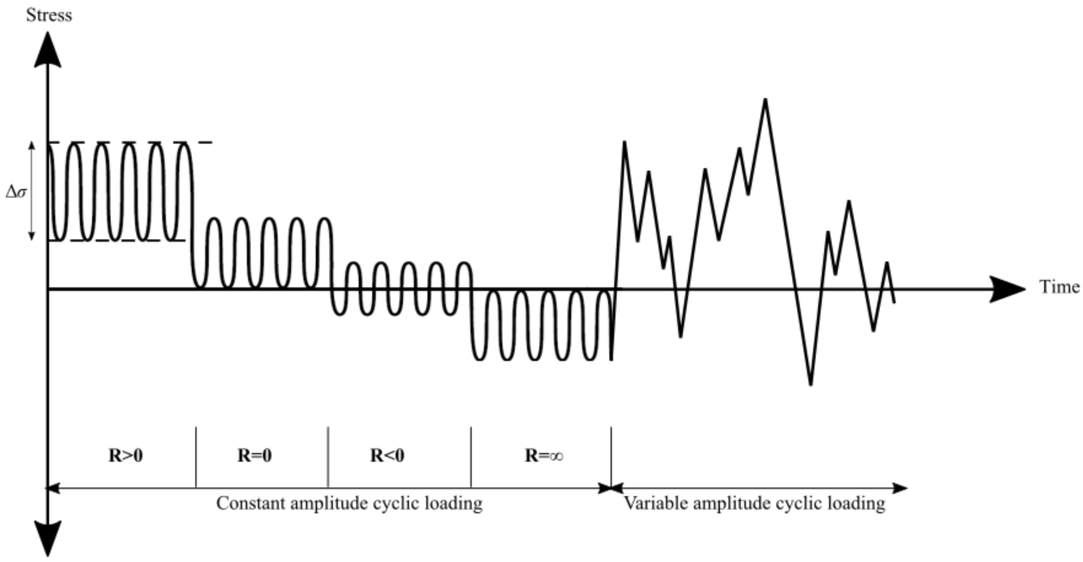


Figure 2.3 Loading factor parameters in fatigue life

Loading type (axial, bending, multi-axial etc.) is the other factor can be categorized in loading factors. Experiments have shown that proportional loading history results higher fatigue life than non-proportional loading under identical principal stress amplitudes [7]. This loading type phenomenon will be taken into consideration in more detail in this thesis in the next chapter.

Structural factors is the second main factor for the fatigue life of welded structures. The geometric configurations of the structures determine whether the welds are load carrying or not. This parameter is a decisive factor in the selection of the fatigue life estimation method. Moreover, the shapes of the structures (T-shape, cruciform plate etc.) classified in the standards are important for the selection of S-N curves, the details of which will be given in the next section. The geometric configurations of the structures, which are mostly taken into consideration and discussed in this thesis, are depicted in the Figure 2.4.

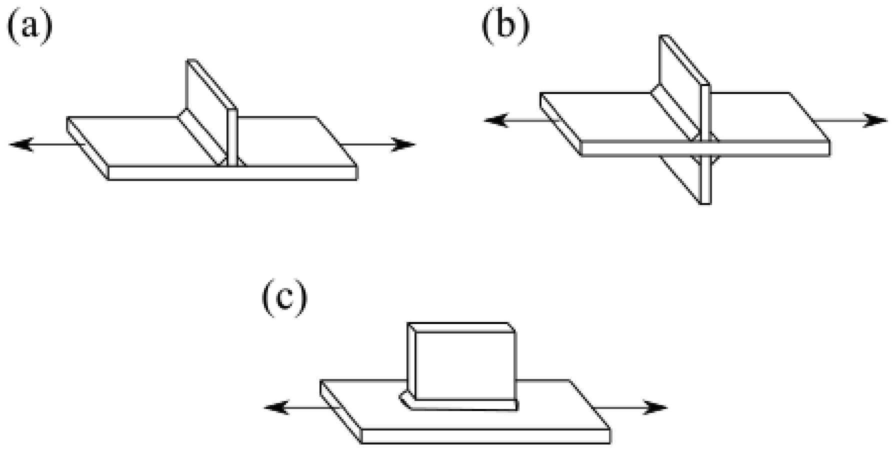


Figure 2.4 The most encountered geometric configurations for the fatigue life of welded structures: (a) Load carrying T-shape joints, (b) Load carrying cruciform joints and (c) non-load carrying longitudinal gussets

The effect of plate thickness is another structural factor that affects fatigue life. It has been demonstrated by the experiments carried out by Gurney [16] that the increase in plate thickness reduces the fatigue life. Experimental investigations also show that fatigue strength of plates with higher thicknesses are lower as depicted in Figure 2.5.

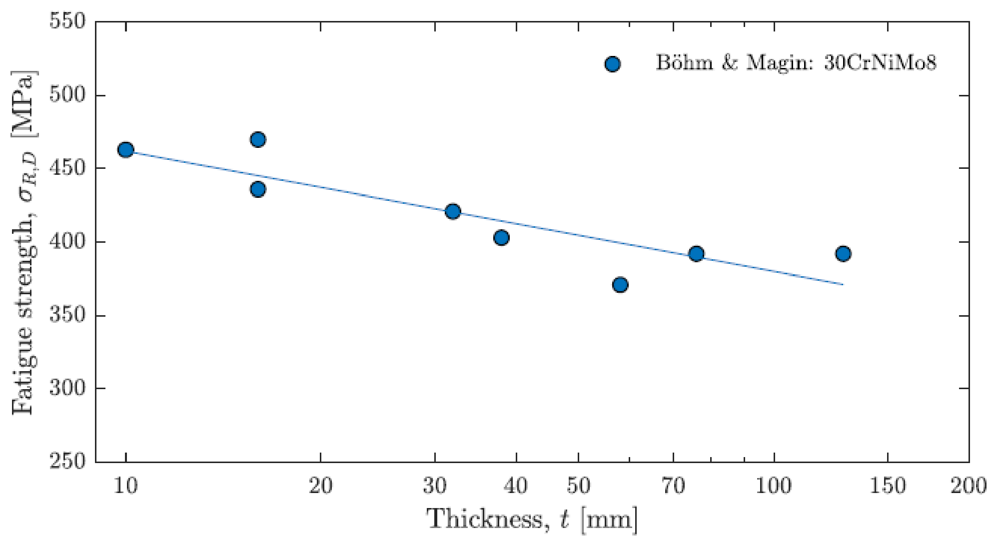


Figure 2.5 Experimental investigations of the size effect [17]

Another factor that needs to be taken into consideration under structural factors are fabrication defects. Fabrication of welds have a great influence on failure type of welded connections. In welded connections, failure typically begins at the toe, the root or the throat of the weld as illustrated in Figure 2.6. Root and throat failures generally occur due to problems in the welding operation and are mostly related to the quality of the weld. On the other hand, the highly stressed zones where cracks initiate and lead to failure are usually located at weld toe for continuously welded structures subjected to cyclic loading [18]. Therefore, only fatigue failure at the weld toe is taken into account in this study.

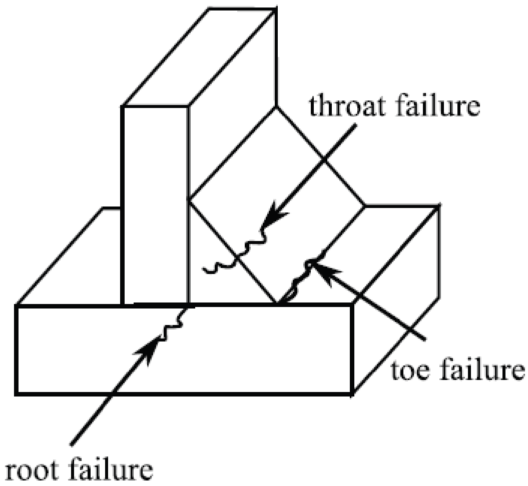


Figure 2.6 Illustration of the root, throat and toe failures in welded connections

The other two factors affect the fatigue life of welded structures are material and environmental factors. Basic material properties, effects of surface finishing, fatigue limit or threshold can be mentioned as material factors, while corrosion, temperature and maintenance can be counted as environmental factors. Since these two factors are less important than the previous factors [15], they are not included in the scope of the thesis. Details of the effects of these factors on fatigue life are given in [7].

2.1.2 Fatigue Strength S-N Curves

The fatigue life assessment of welded structures is commonly presented in the form of stress-cycle (S-N) curves which is also known as Wöhler's curve. S-N curves are derived by testing the specimens experimentally under repetitive loading cycles until failure. The amount of cycles is counted until failure and the stress ranges are measured. This procedure is repeated for different stress ranges. Plotting the results of these experiments gives the S-N curves, which are commonly given in a logarithmic scale. Relation between the stress range ($\Delta\sigma$) and the number of cycles (N) is linear in logarithmic scale with a constant slope of m.

It was stated in the previous section that fatigue strength depends on many effects such as structural, loading and environmental factors. All these effects are taken into consideration directly within the derived S-N curves by evaluating the test data with an assumption of a Gaussian log-normal distribution. On the other hand, including the effects of all factors into the S-N curves is quite tough as it requires a lot of experiments. In order to obtain a more general applicability, FAT categories are evaluated for different structural details and loading conditions. 75% two-sided confidence level of 95% survival probability is generally represented in these FAT categories. The endurable stress range at $N = 2\times10^6$ cycles gives the value of the FAT category. S-N curves are recommended by the standards, which are mainly provided for a various structural details under different FAT categories. The details of the FAT classes for each method will be presented in the next section.

2.2 Nominal Stress Approach

The nominal stress method is simply based on the stress calculated in the studied cross-sectional area. Linear theory of elasticity is assumed. Since its simplicity, this method is most commonly applied method for fatigue life estimation of welded structures. Stress raising effects of the macro-geometrical details of the component are included in this method which are depicted in Figure 2.7. However, effects of the welded joint are disregarded. Moreover, load effects such as normal stresses due to torsion are not taken into consideration in this method.

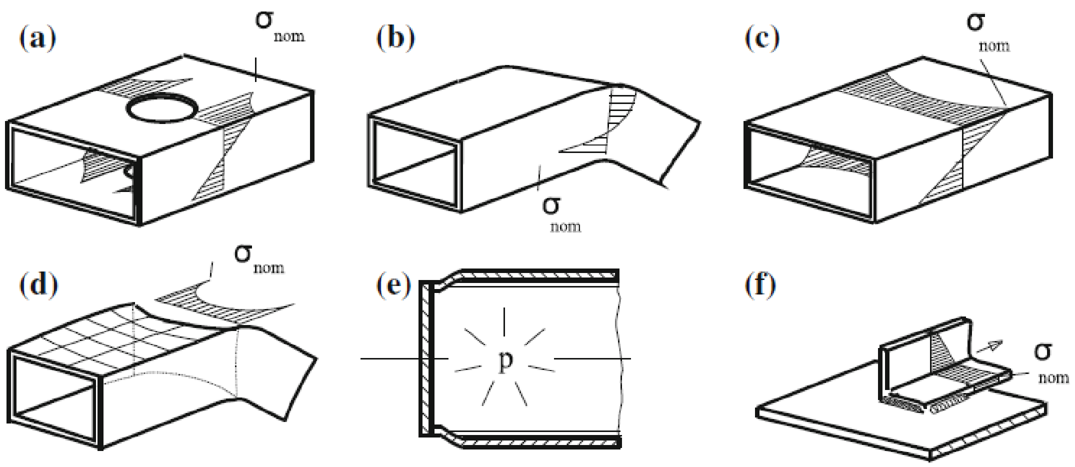


Figure 2.7 Some examples of macro-geometric details given in IIW [9]: (a) cut-out holes, (b) curved beams, (c) wide plates, (d) curved flanges, (e) concentrated loads and (f) eccentricities

For nominal stress approach, there are some design guidelines providing FAT categories and S-N curves for the specific loading and structural details. International Institute of Welding (IIW) recommendations [19], British standard BS7608 [20]. Japanese Society of Steel Construction (JSSC) recommendations [21], Eurocode 3 [22], Det Norske Veritas (DNV) [23] and Forschungskuratorium Maschinenbau (FKM) guidelines [24] are some of design guidelines representing the most common codes. In this thesis, the most commonly preferred IIW recommendations is taken into consideration. Fatigue resistance values for structural details in steel and aluminum structures for the nominal stress approach are given in tables in the IIW recommendations [11].

2.3 Hot-spot Stress Approach

Hot-spot Stress (HSS) approach is one of the most common fatigue life estimation method since it is very practical approach and can include the effect of geometric details. It is simply done by taking into consideration of the Hot-spot stress that is obtained by extrapolation from stresses located at certain distances that are some proportions of plate thickness away from the weld toe. Since it is aimed to improve this method, this section is taken into consideration in more detail.

In the 1960s, the hot-spot stress (HSS) method was first used to estimate the fatigue life of pressure vessels and welded tubular connections, then applied to various different welded structures successfully [25]. Owing to the widespread use of the finite element method and its application to many industrial problems, the use of HSS method has increased rapidly.

Over the last decades, HSS method has been extensively researched. Herein a brief literature survey will be given about the studies related to HSS. Maddox [26] evaluated a fatigue data from 1990-2000 studies for floating production storage and offloading (FPSO) structures as the basis of hot-spot stress design S-N curves. Three different extrapolation methods were considered for determining the hot-spot stress and the feasibility of using the category of FAT90 S-N curve was demonstrated in the mentioned study. Doerk et al. [27] reviewed different computation of the structural hot-spot stress methods for four different geometries and compared the methods with each other. They found that two alternative surface stress extrapolation methods proposed by International Institute of Welding (IIW) yield almost the same results. Fricke [28] studied modelling aspects, the stress evaluation and extrapolation techniques for HSS method in fatigue life estimation with a few examples. They concluded that large deviations from the actual lifetime can occur due to neglected influence factors and simplified modelling. Poutiainen et al. [29] applied HSS method for two dimensional (2D) and three dimensional (3D) structural details. They reported that HSS method was sensitive to the mesh variations when more than one element used in thickness directions. Tveiten et al. [30], [31] carried out fatigue tests of ship components to draw an S-N curve. It is suggested that in 3D modeling, the weld should be modeled with a 45 degree flank angle and without a weld toe radius. Lee et al. [32] illustrated differences in the HSS evaluation methods on four different welded structural components. It was recommended that solid elements should be used when the loading condition is out-of-plane bending. Aygul et al. [33] applied structural HSS approach to estimate fatigue life of orthotropic steel bridge decks. They presented that the use of solid element models including the weld details yielded more representative stress ranges at the hot spot points. Büyükbayram et al. [34] conducted constant amplitude fatigue tests under four-point bending loadings and compared their results with the calculated fatigue life under different discretization methods. They showed that meshing with 2D shell elements can predict fatigue life accurately for thick structures. Zamzami and Susmel [35] studied the accuracy of the various methods in designing aluminium welded joints against fatigue. They claimed that HSS method could be used for fatigue life estimation of aluminum welded joints. Alencar et al. [36] evaluated the fatigue life of steel-concreted bridge using HSS method. They pointed out that the probability of fatigue damage in the longitudinal weld is higher than in the transverse weld. Chen et al. [37] conducted fatigue tests on circular hollow profiles (CHP) with K-joints, which are often used in offshore platforms. Their results showed that HSS of a CHS K-joint can be obtained with a linear extrapolation. Masheyekhi and Santini-Bell [38] studied the fatigue assessment of a complex welded steel bridge connection using HSS method. A multi-scale model was proposed to clarify the amplitude and location of the maximum hotspot stress along the weld toe.

HSS method has been developed to evaluate the fatigue strength of welded structures including the geometric and/or loading effects and excluding the local stress peak caused by local weld profile as illustrated in Figure 2.8. This is done by extrapolating the stresses located at certain distances that are some proportions of plate thickness away from the weld toe.

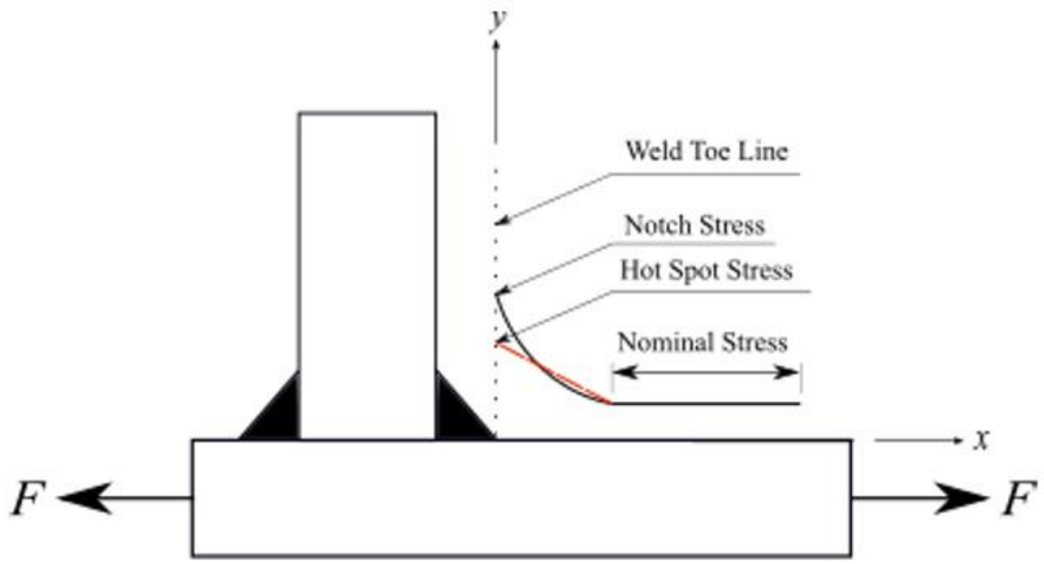


Figure 2.8 Definition of the stresses along the weld toe

The procedure for extrapolating the stresses is well documented in the guidelines reported by IIW [19]. Establishing the reference points is the first step of the procedure. IIW stated that identification of the reference points can be made by experience of existing failed components or analyzing the finite element method results or measuring several different points. The determination of reference points varies depending on the type of hot-spots, the quality of discretization and whether the extrapolation is linear or quadratic.

IIW [19] defines two types of hot spots, type (a) and type (b), according to the their orientation in respect to the weld toe and their location as depicted in Figure 2.9. In type (a), the weld toe is on the plate surface, while in type (b), the weld toe is on the plate edge. The type of hot-spot is a factor influencing the selection of the S-N curve and the extrapolation technique.

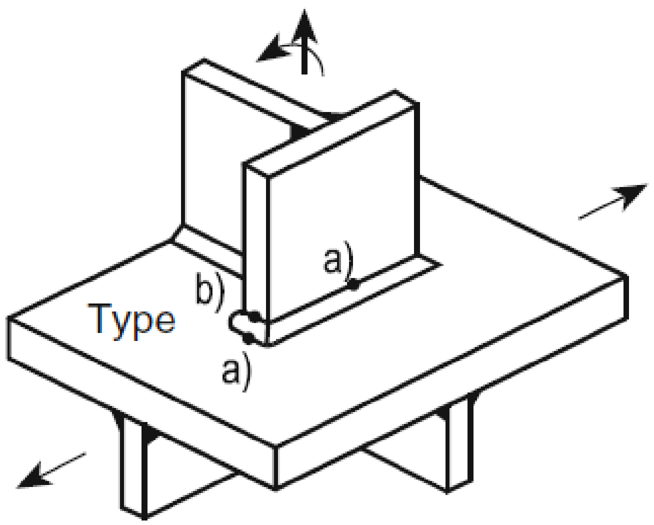


Figure 2.9 Hot-spot types defined in IIW [9]

It is not possible to obtain the hot-spot stress (HSS) by analytical methods. Finite element analysis (FEA) is generally needed to obtain HSS. Choosing a proper mesh size (element lengths) is the first step of the discretization. The extrapolation points positions play a crucial role in determining the element size.

Establishing the reference points is the first step of the procedure. The first reference point which is closest to the weld toe must be chosen to avoid any stress raising effect due to the weld itself. This is stated in the IIW recommendations as it is practically at a distance of $0.4 \times t$ from the weld toe, where t is plate thickness. There can be two or three reference points depending on the linear or quadratic extrapolation technique. The chosen of other reference point (or points) is determined by whether the generated mesh quality is fine or coarse and whether the hot spot type is type-a or type-b.

There is no specific limitations to use shell or solid elements and to model weld or not. Although solid elements can be used for fatigue analysis, they are usually not preferred in the finite element models of large structures in industrial applications due to the increased modeling effort and computational cost. Shell elements offer a computationally less expensive alternative whenever the thickness is much smaller than other dimensions. Furthermore, it is usually less time consuming to mesh complicated CAD geometries with shell elements. As such, shell elements are commonly used in structural analyses including fatigue life estimation [39]. However, the use of shell and solid elements still needs comparisons especially in structures with stiffeners. Thus, the effect of element type on the normal stress distribution near the weld for structures subjected to tension, four point bending and bi-axial tension are investigated in this thesis and will be given in next sections.

Niemi et al. [40] recommended the selection of element size and extrapolation points as given in Table 2.1. Locations of the stress readout points for extrapolation, in terms of the thickness for finely and coarsely meshed models are illustrated in Figure 2.10.

Table 2.1 Recommended element size and extrapolation points [40]

Element types and extrapolation points according to fine/coarse models		**Relatively Coarse Models**		**Relatively Fine Models**	
		Type a	**Type b**	**Type a**	**Type b**
Element size	Shells	$t \times t$ $\max t \times w / 2$	$10 \times 10mm$	$\leq 0.4t \times t$ or $\leq 0.4t \times w / 2$	$\leq 4 \times 4mm$
	Solids	$t \times t$ $\max t \times w$	$10 \times 10mm$	$\leq 0.4t \times t$ or $\leq 0.4t \times w / 2$	$\leq 4 \times 4mm$
Extrapolation points	Shells	$0.5 \times t$ and $1.5 \times t$ mid-side points	5 and 15 mm mid-side points	$0.4 \times t$ and $1.0 \times t$ nodal points	4, 8 and 12 mm nodal points
	Solids	$0.5 \times t$ and $1.5 \times t$ surface center	5 and 15 mm surface center	$0.4 \times t$ and $1.0 \times t$ nodal points	4, 8 and 12 mm nodal points

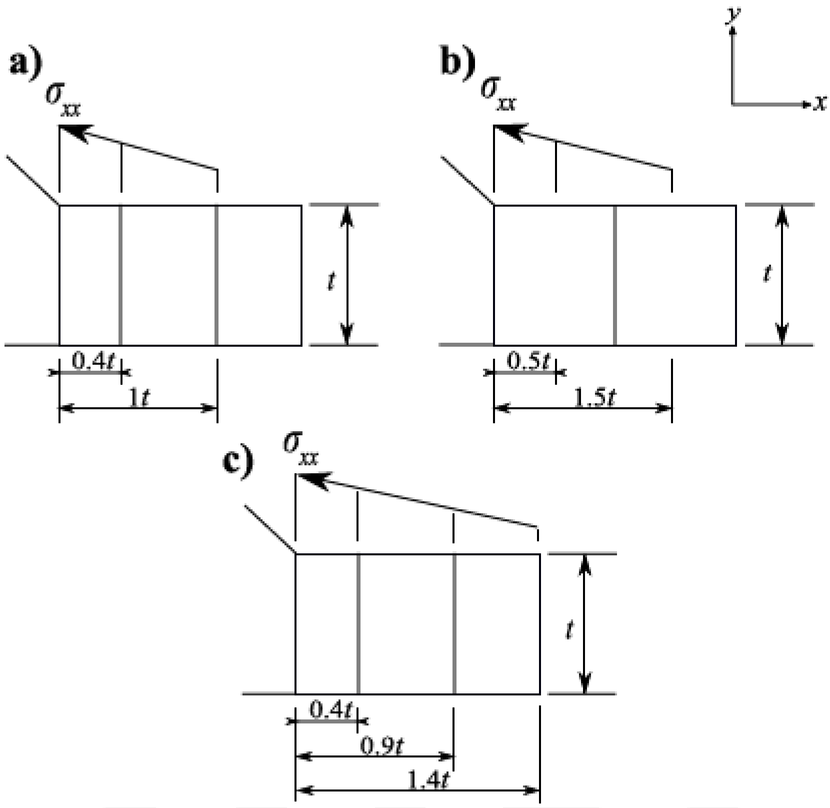


Figure 2.10 Reference points used in the calculation of the HSS (a) for linear extrapolation with fine meshing, (b) for linear extrapolation with coarse meshing, (c) for quadratic extrapolation

The main advantage of the HSS method compared to the Nominal Stress method is in the topic of S-N curves. The FAT category concept is similar with the previous one. However, there are only two FAT classes (FAT90 and FAT100) recommended for steels when using HSS method. These FAT categories depend on the joint type, HSS type (type-a or type-b) and some other structural details as given in Table 2.2. The schematic representation of the Hot-spot S-N curve is depicted in Figure 2.11.

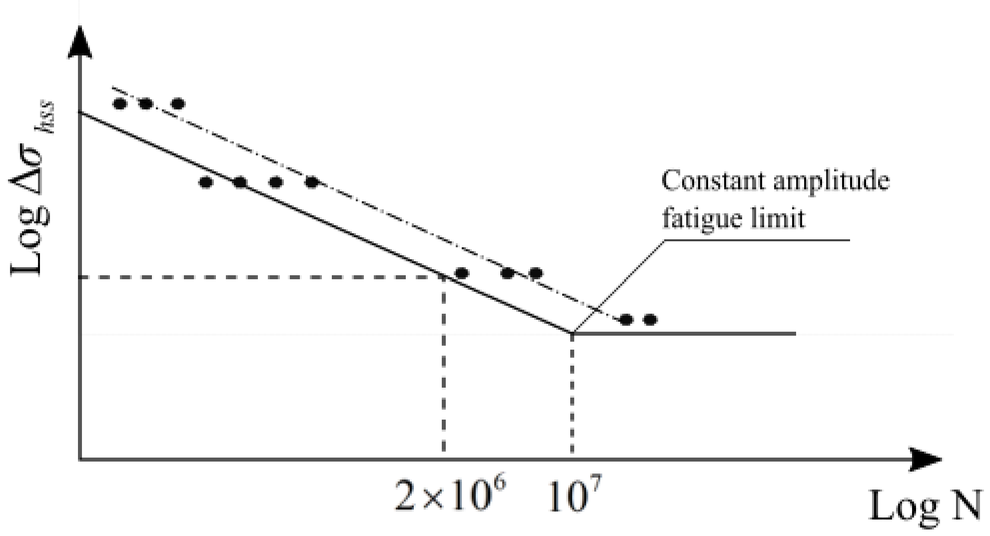


Figure 2.11 Schematic representation of the HSS S-N curve

The equation of this curve that can be used to calculate the number of cycles at any hot-spot stress range is

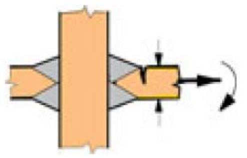
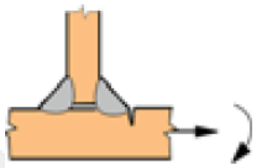
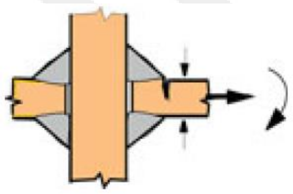
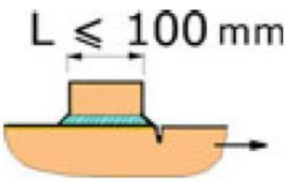
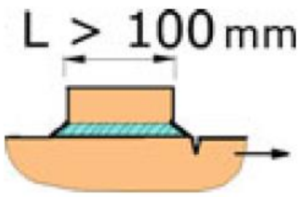
$$\Delta\sigma_{hss}^{m} \cdot N = C \tag{2.3}$$

or in logarithmic form

$$m \cdot \log(\Delta\sigma_{hss}) + \log(N) = \log(C) \tag{2.4}$$

where $\Delta\sigma_{hss}$ is the hotspot stress range, *m* is the slope of the S-N curve (generally it is taken as $m = 3$), N is the number of cycles to failure and C is the design value of the FAT category/class which can be 90 or 100 for steels. According to the weld shape, structural details and loading types, FAT classes are recommended as given in Table 2.2.

Table 2.2 Recommended FAT classes for steels according to the some structural detail for HSS method [11]

Structural Detail	Description	Requirements	FAT class
	Cruciform or T-joint with full penetration K-butt welds	K-butt welds, no lamellar tearing	100
	Non-load carrying/ load carrying fillet welds	Transverse non-load carrying attachment, not thicker than main plate, as welded	100
	Cruciform joints with load-carrying fillet welds	Fillet welds, as welded	90
	Type-b joint with short attachment	Fillet or full penetration weld, as welded	100
	Type-b joint with long attachment	Fillet or full penetration weld, as welded	90

2.4 Structural Stress Approach

Local stress concentration at welded joints dominate the fatigue behavior of welded structures. This stress concentration can be estimated through finite element (FE) method. However, the result from FE analysis is highly sensitive to modelling techniques as the stresses are often calculated in an area of high strain gradients. The resulting stresses may differ depending on the size, type and integration order of elements. To overcome this dependency, Battelle [41] has developed a mesh insensitive Structural Stress (SS) Method.

The SS method is presented in detail by Dong [42]. This approach is consistent with elementary structural mechanics theory. The method is based on the calculation of the stresses using the balanced nodal forces and moments at the weld toe location from the FE solutions. The resulting stresses will be regardless of size, type and integration order of elements.

The structural stresses are calculated in terms of membrane and bending components. First of all, a local coordinate system has to be defined, which is appropriate for calculating the structural stresses with respect to weld. According to this coordinate system nodal forces and moments are substituted in to

$$\sigma_{ss} = \sigma_m + \sigma_b = \frac{f_{x'}}{t} + \frac{6m_{y'}}{t^2} \qquad \textbf{(2.5)}$$

where σ_m is the membrane stress and σ_b is the bending stress which are obtained from line forces $f_{x'}$ and line moment $m_{y'}$ computed in a work equivalent manner from the elements' nodal forces. Local line force and moment for an element along weld toe is represented in Figure 2.12.

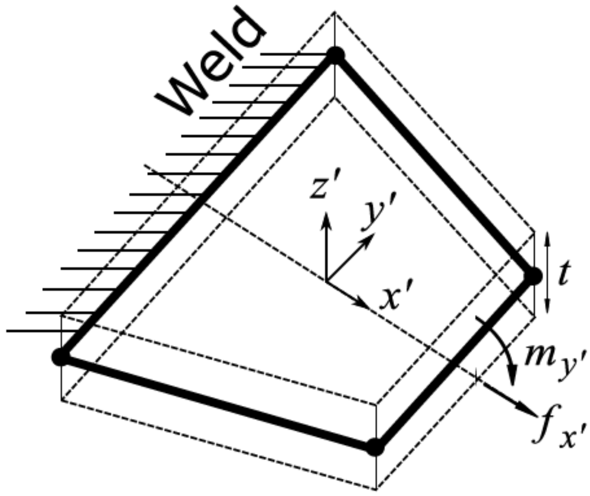


Figure 2.12 Representation of the line force and moment used in Equation (2.5) for the SS method

The second main advantage of this method is that it facilitates the use of the S-N curves mentioned in the previous sections, which can increase in number depending on various variables such as weld type and structural details. A single master S-N curve is suggested by using an equivalent structural stress parameter and this S-N curve can be used for all types of weld with plates with different thicknesses under all types of loading. An equivalent structural stress parameter is defined as

$$\Delta S_s = \frac{\Delta\sigma_{ss}}{t^{(2-m)/2m} \cdot I(r)^{1/m}} \tag{2.6}$$

where $\Delta\sigma_{ss}$ is the SS range calculated from Equation (2.5), t is the plate thickness, m is the slope of a Paris-like crack propagation curve and $I(r)^{1/m}$ is a dimensionless polynomial function of the bending ratio r and defined as

$$I(r)^{1/m} = 0.0011r^6 + 0.0767r^5 - 0.0988r^4 + 0.0946r^3 + 0.0221r^2 + 0.014r + 1.2223 \tag{2.7}$$

where the bending ratio r defined as

$$r = \frac{|\sigma_b|}{|\sigma_m| + |\sigma_b|} \tag{2.8}$$

Based on a large number of experiments [42], [43] it is derived to take the slope of the curve, *m* as 3.6.

After the calculation of the equivalent structural stress parameter, it is similar to estimate the fatigue life like the previous approaches. The master S-N curve is expressed in the form of

$$\Delta S_s = A \cdot N^b \tag{2.9}$$

where *N* is the number of the cycles to the failure and *A* and *b* material parameters which are tabulated for different prediction intervals in [43].

Table 2.3 The Master S-N curve parameters

Statistical basis	*A*	*b*
Mean Curve	19930.2	-0.32
Upper 95% Prediction Interval, $+2\sigma$	28626.5	
Lower 95% Prediction Interval, -2σ	13875.8	
Upper 99% Prediction Interval, $+3\sigma$	31796.1	
Lower 99% Prediction Interval, -3σ	12492.6	

As described above, it can be said that there are two main advantages of the SS method: mesh insensitivity and a single master S-N curve. These two features of the SS method have attracted the attention of researchers and various studies have been carried out. Here, it will be given some of them. One of the major contributors in this section is P. Dong. He defined the SS method and proved its mesh insensitive with many examples such as pressure vessels [43]–[45], butt welded pipes [42] and rectangular hollow section joints [46]. Mesh insensitivity feature is taken into consideration in the studies for different structural types such as bulk carriers [47], box-welded joints [48], T-shaped attachments [49], plate lap fillet weld [27], longitudinal gusset on a plate [29], breathing web panels [50] and some other various structures[51]–[53]. In this thesis, an existing problem of mesh sensitivity is considered to illustrate the application of the SS approach in Section-3.

2.5 Effective Notch Stress Approach

Notch geometries often cause stress singularities in FE calculations. However, these singular stresses cannot be taken into consideration directly for fatigue life estimations. The average stress over a certain distance for 2-dimensional models or a certain volume for 3-dimensional models is defined as the effective notch stress to overcome this problem by Radaj et al. [10]. The Effective Notch Stress (ENS) method evaluates the stress-increasing effects caused by the weld geometry and estimates the fatigue life. Efficient modelling of the weld is the first step of this method. As the failure in welded connections can be at the toe or at the root as illustrated in Figure 5, an effective notch root radius of $r = 1$mm has been defined [11] as can be seen in Figure 2.13.

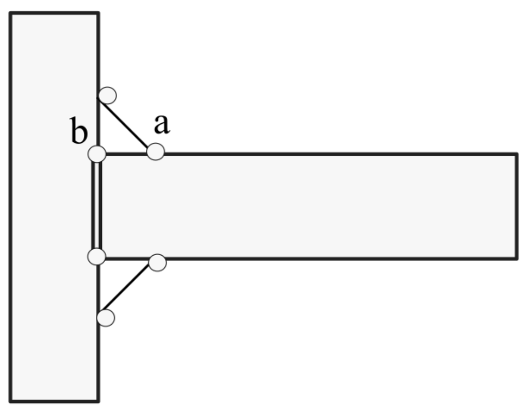


Figure 2.13 Fictitious rounding of a (weld toes) and b (roots)

In order to determinate effective notch stresses from FEA, there are some recommendations for discretization in the IIW [11] for element sizes in case of linear and high order elements as depicted in Figure 2.14 and given in Table 2.4. It is noted that ENS method needs relatively fine mesh sizes comparing with the previous methods.

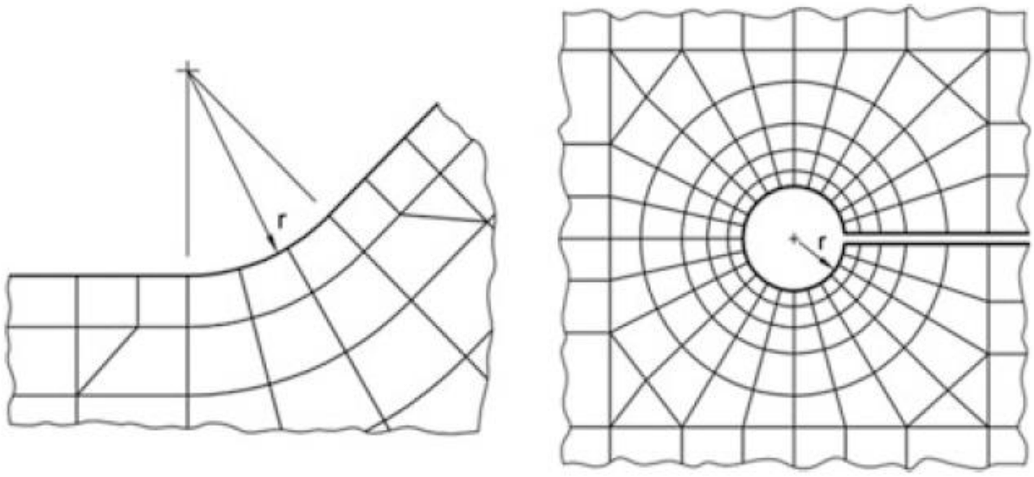


Figure 2.14 Recommended discretization from IIW [11] at weld toes on the left and at weld roots on the right side

Table 2.4 Recommended element sizes on the surface [11]

Element type	Mesh size	Number of elements in 45° are	Number of elements in 360° are
Linear	$\leq r/4$ or $\leq 0.25mm$	≥ 3	≥ 24
Quadratic	$\leq r/6$ or $\leq 0.15mm$	≥ 5	≥ 40

2.6 Other Approaches

There are some other approaches to estimate the fatigue life of welded structures such as total stress concept by den Besten [14], peak stress method by Meneghetti et al. [54], 3R Method by Nykanen et al. [55], a hot-spot energy indicator by Feng and Qian [56] and strain energy density methods (firstly it was used in fatigue life of welded structures by Lazzarin and Zambardi [57], [58]). Den Besten defined a total stress range as an equivalent structural response parameter which includes SS methods parameters with extension of notch crack growth integral. This integral includes the size effects. Peak stress method is relied on the singular linear elastic peak stresses calculated from FE analyses. 3R method is proposed as a combination of three methods: ENS method, Smith-Watson-Topper (SWT) method and local strain methods. Local stress ratio is defined with R_{local} and calculated with SWT damage parameter. Then local strain methods used for complete definition in this method. A volume-based hot-spot energy indicator is proposed to reduce the scatter of the S-N curves. With this parameter high stress gradient near the weld toe is taken into consideration. It is showed that, the scatter of the S-N curves can be reduced with this parameter. Strain energy density is taken as a damage parameter for two phases of fatigue failures: the initiation of cracks and the propagation of cracks. It is calculated using an averaged strain energy over a specified area.

CHAPTER 3

FATIGUE EVALUATION USING FEM

3.1 2D & 3D Modelling and Comparison

The Finite Element Method (FEM) is a great tool for solving engineering problems. A finite element analysis (FEA) helps engineers with problems that cannot be solved by analytical methods or are very time-consuming, although possible. This is done by dividing the solution region into a large number of simple, small, interconnected, subregions called finite elements. A large model is divided into parts that can be solved more easily and connected to each other with many nodal points. This is called discretization in FEM. Discretization can be made by different element types such as 1D rigid elements, 2D shell elements and 3D solid elements.

Although solid elements can be used for fatigue analysis, they are usually not preferred in the finite element models of large structures in transport industry due to the increased modeling effort and computational cost. Shell elements offer a computationally less expensive alternative whenever the thickness is much smaller than other dimensions. Furthermore, it is usually less time consuming to mesh complicated CAD geometries with shell elements. As such, shell elements are commonly used in structural analyses including fatigue life estimation. However, the use of shell and solid elements still needs comparisons especially in structures with stiffeners.

In this section, FE analyses are performed to study the effect of element type on the normal stress distribution near the weld for structures subjected to tension, four point bending and bi-axial tension. For each loading type, the structure is modeled with solid elements with and without the weld geometry, and with shell elements without the weld geometry.

A linear elastic material behavior with modulus of elasticity E = 210 GPa and Poisson’s ratio ν = 0.3 corresponding to steel was assigned to the entire geometry including the welds. 8 Noded quadrilateral elements (ANSYS element SHELL281) are used for shell models and 20 Noded hexahedral elements (ANSYS element SOLID186) are used for solid models. Finally, the hot spot stresses computed with linear and quadratic extrapolation using the stress field obtained from different models are compared, for each loading condition.

3.1.1 Tensile Loading: Non-load Carrying Longitudinal Attachments

Fatigue failures at fillet weld termination are investigated using non-load-carrying longitudinal welded attachments. Due to the symmetry of this attachment, bending effects can be neglected. Moreover, this type of attachment allows a simple method for reproducible notch severity [59]–[61]

Here, a longitudinal attachment was modeled as shown in Figure 3.1 (a) following the work of Heshmati [62]. where the length and the thickness of the attachment are L = 150 mm and t' = 12.7 mm, respectively. The width and the thickness of the plate are w = 100 mm and t = 12.7 mm, respectively. The geometry is discretized with solid elements including the weld, solid elements without modeling the weld geometry and shell elements as shown in Figures 3.1 (b), 3.1 (c) and 3.1 (d), respectively. An element size of 1.27 mm is chosen such that the HSS extrapolation stresses can be read at nodes placed at $0.4\times t$ and $1\times t$ distance away from the weld toe. A line pressure of 100 N/mm is applied on both sides of the attachment and the stress distribution along a path perpendicular to the attachment thickness direction is recorded. Figure 3.1 (e) shows the normal stress σ_{yy} as function of the distance to the weld toe as well as the extrapolation points used to compute the HSS for each model. It is seen that the extrapolated HSS obtained from solid and shell models without the weld geometry are almost indistinguishable and very close to the HSS computed from the solid model with weld geometry (see Figure 3.1 (e)).

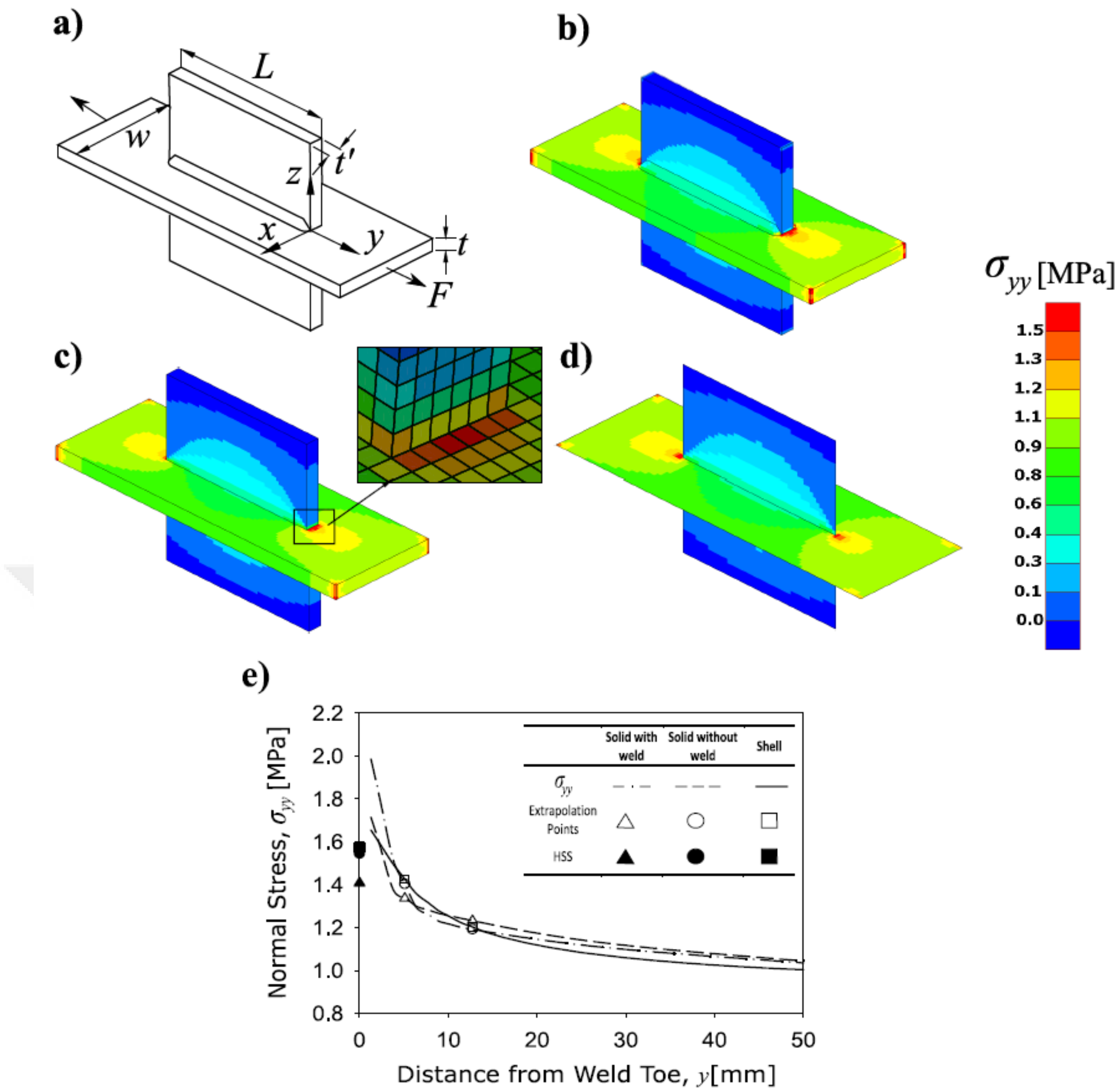


Figure 3.1 Non-load carrying longitudinal attachments under tensile loading (a) Geometry of the attachment. Normal stress σ_{yy} distribution in the model discretized with (b) solid elements including the weld, (c) solid elements without modeling the weld, and (d) shell elements. (e) Normal stress σ_{yy} as function of the distance from weld toe. The HSS extrapolation points are marked with open triangles, circles and squares for solid (including the weld), solid (without the weld) and shell element models, respectively

3.1.2 Four Point Bending: Cruciform Welded Plate

Three and four point bending tests are commonly used to determine the bending strength of a specimen. The highest stress achieved in the three-point bending test is at the midpoint of the specimen. On the other hand, the four-point bending test produces peak stress over a range of specimen surface and accordingly it becomes possible to observe potential deformations on the pure bending region. Hence, it is one of the most widely used test to study fatigue life performance.

In this section, four point bending simulation is conducted using the cruciform welded plate having the geometry described in [56] (see Figure 3.2 (a)). Note that, the beam thickness is t = 15 mm, thickness of the attachment is t' = 16 mm and the length of the beam is L = 210 mm. Two forces having a magnitude of F = 5 kN were applied both sides of the beam leading to 186.7 MPa nominal stress in the pure bending region. The plate is modeled using solid elements (with weld), solid elements (without weld) and shell elements as can be seen on Figure 3.2 (b), 3.2 (c) and 3.2 (d), respectively. The models are meshed with an element size of 0.15 mm such that the HSS extrapolation stresses can be computed at $0.4\times t$ and $1\times t$ distances from the weld toe. Figure 3.2 (e) shows the normal stress distribution obtained from solid element and shell element models as function of the distance from the weld toe, as well as the extrapolation points of HSS. Note that the element type used in the model does not have a significant effect on the extrapolated HSS results.

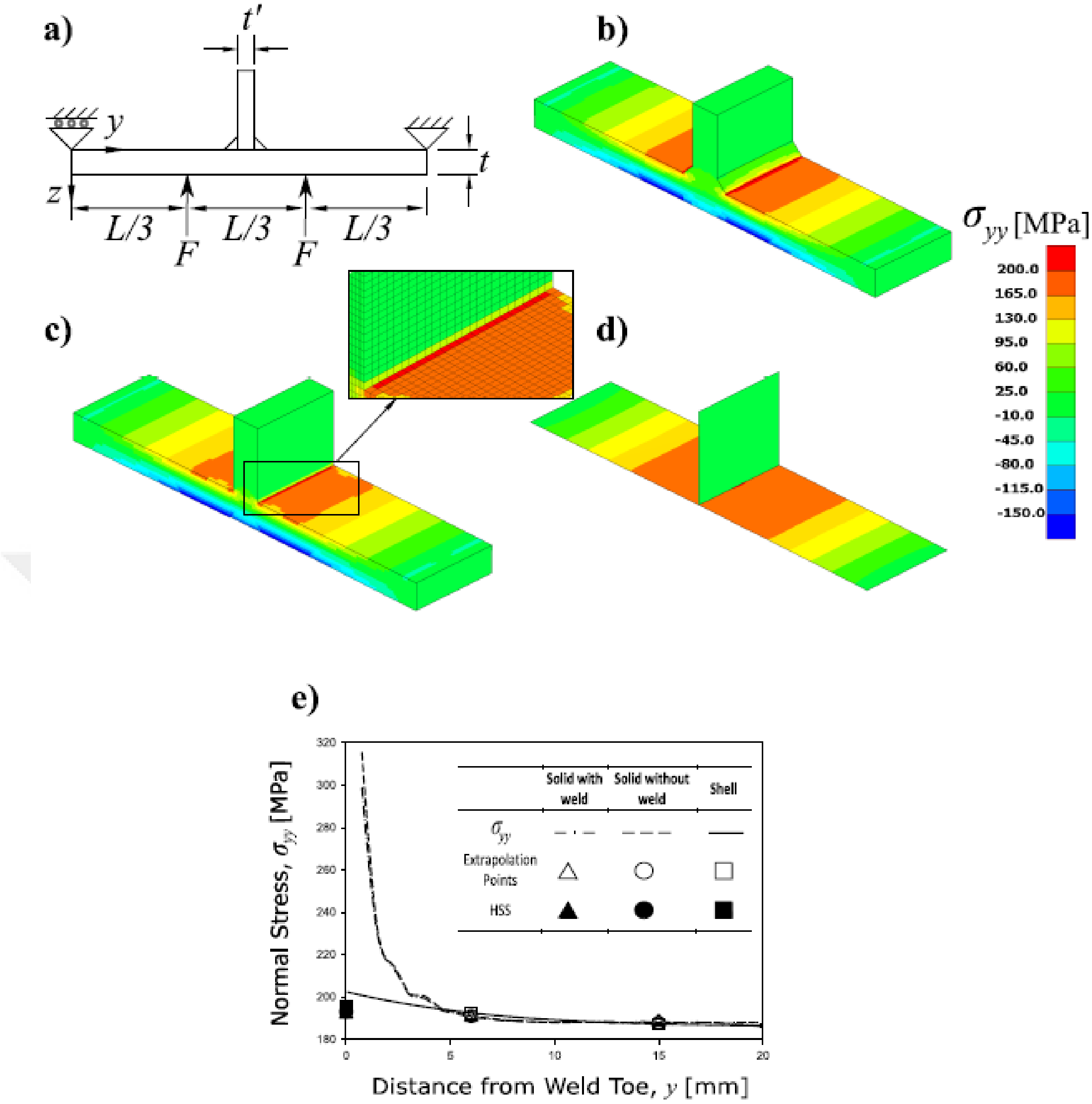


Figure 3.2 Cruciform welded plate under four point bending (a) Sketch showing the geometry and dimensions of the plate. Normal stress σ_{yy} distribution in the model discretized with (b) solid elements including the weld, (c) solid elements without modeling the weld, and (d) shell elements. (e) Normal stress σ_{yy} as function of the distance from weld toe. The HSS extrapolation points are marked with open triangles, circles and squares for solid (including the weld), solid (without the weld) and shell element models, respectively

3.1.3 Bi-axial Loading: Box-welded (Wrap-around) Joint

In the previous two sections tension and bending loadings producing a uniaxial stress state, were investigated. On the other hand, tram structures is also subjected to multiaxial loads which can be critical in determining the fatigue life of tram structures. Hence, in this section, the fatigue behaviour of a specimen under biaxial tensile loading was investigated. Figure 3.3 (a) shows the half geometry of a box-welded joint specimen following [63]. The dimensions of the specimen attachments are L_1 = 400 mm, L_2 = 200 mm and L_3 = 50 mm with the thickness t = 12mm. Bi-axial loading condition is imposed by applying forces along x and y directions as F_x = 354.6 kN and F_y = 104.9 kN, respectively.

The geometry is discretized with solid elements with and without the weld and shell elements as seen on Figure 3.3 (b), 3.3 (c) and 3.3 (d), respectively. An element size of 2.4 mm such that the HSS extrapolation stresses can be computed at $0.4\times t$ and $1\times t$ distances from the weld toe. Figure 3.3 (e) shows the extrapolation points for the HSS method and HSSs for each modeling type. The HSS obtained from the solid element model with weld and shell element model are very close and similar to the HSS computed using the solid model without the weld geometry.

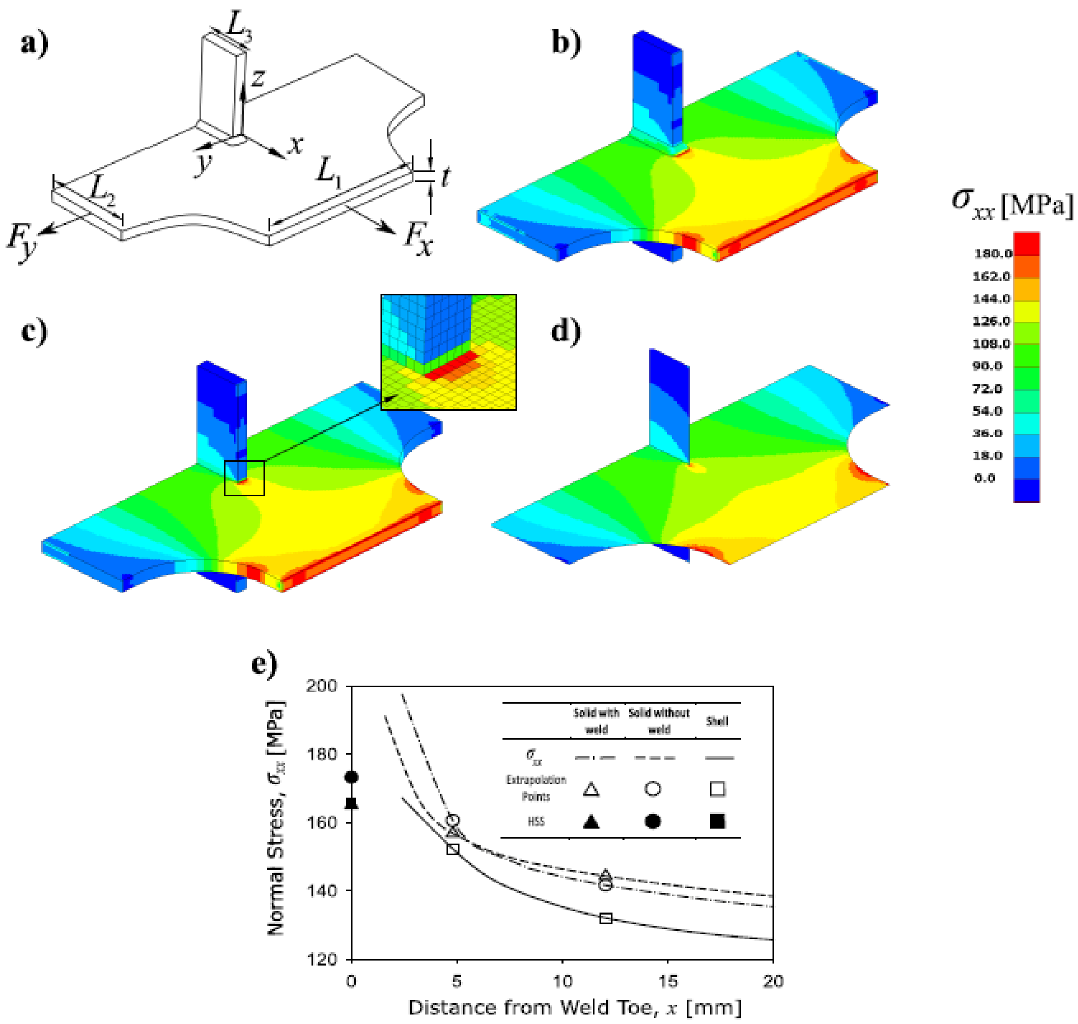


Figure 3.3 Box-welded (wrap-around) joint under bi-axial loading. (a) Sketch showing half sectioned view and dimensions of the attachment. Distribution of the normal stress σ_{xx} distribution in the model discretized with (b) solid elements including the weld, (c) solid elements without modeling the weld, and (d) shell elements. (e) Normal stress σ_{xx} as function of the distance from weld toe. The HSS extrapolation points are marked with open triangles, circles and squares for solid (including the weld), solid (without the weld) and shell element models, respectively

3.1.4 Comparison of the HSS Results Obtained Using Solid and Shell Elements

The hot spot stresses for different loading conditions explained in the previous sub-sections are listed in Table 3.1. For the tensile loading case, the % error in HSS computed using the shell model with reference to the solid model with weld geometry increased from 12.1% to 23.2%, with the thickness increasing from 12.7 mm to 25 mm. This is expected since shell elements produce more accurate results for smaller thicknesses. For the four point bending and biaxial loading cases, the difference between HSS results obtained from the shell models and solid models with weld is 0.6% and 0.4%, respectively. It is clear that for the given welded specimens and loading conditions the HSS obtained from shell element models are very close to the HSS obtained from solid element models. From this observation we can conclude that FE models meshed with shell elements can be used in HSS fatigue analysis for large structures in order to reduce computational work. In addition, notice that the differences between the HSS values computed using linear and quadratic extrapolations are very small (see Table 3.1).

Table 3.1 Comparison of extrapolated hot spot stresses at the weld toe obtained from the solid and shell models for various loading conditions

		HSS Results [MPa]					
		Solid with weld		**Solid without weld**		**Shell**	
	***t* [mm]**	**Linear**	**Quad.**	**Linear**	**Quad.**	**Linear**	**Quad.**†
	12.7	1.57	1.62	1.54	1.67	1.58	1.68
Tension	15	1.48	1.57	1.44	1.53	1.51	1.59
	25	1.31	1.36	1.28	1.33	1.38	1.46
Four Point Bending	15	191.74	191.03	192.94	195.4	195.87	198.81
Biaxial Loading	12	165.16	170.1	173.29	183.5	165.87	175.2

3.2 Mesh Sensitivity of Structural Stress Method

It is usual to use stiffeners in body of machines to reinforce structures in industry. For this reason benchmark geometry is chosen as shown in Figure 3.4. In geometry, 200×200×600 mm square base profile are connected to 600×200 mm plate and for better strength 160×200 stiffener is welded as shown in Figure 3.4. The thickness for all parts are 5 mm.

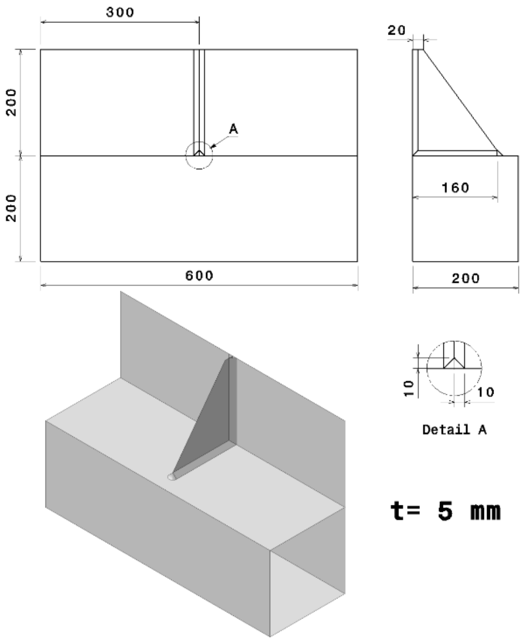


Figure 3.4 Benchmark geometry detail

Then this geometry is discretized with four different mesh sizes. Models with different mesh sizes (10, 5, 2.5 and 1.25 mm) are shown in Figure 3.5. Considering the accuracy of the results, it is vital to use quad elements at the weld toes for all models. SHELL181 element type from ANSYS element library is used for shell elements. Number of the nodes and elements for overall finite element models of four different models are depicted in Table 3.2. Models are made from Structural Steel and the parameters for the material model are listed in Table 3.3.

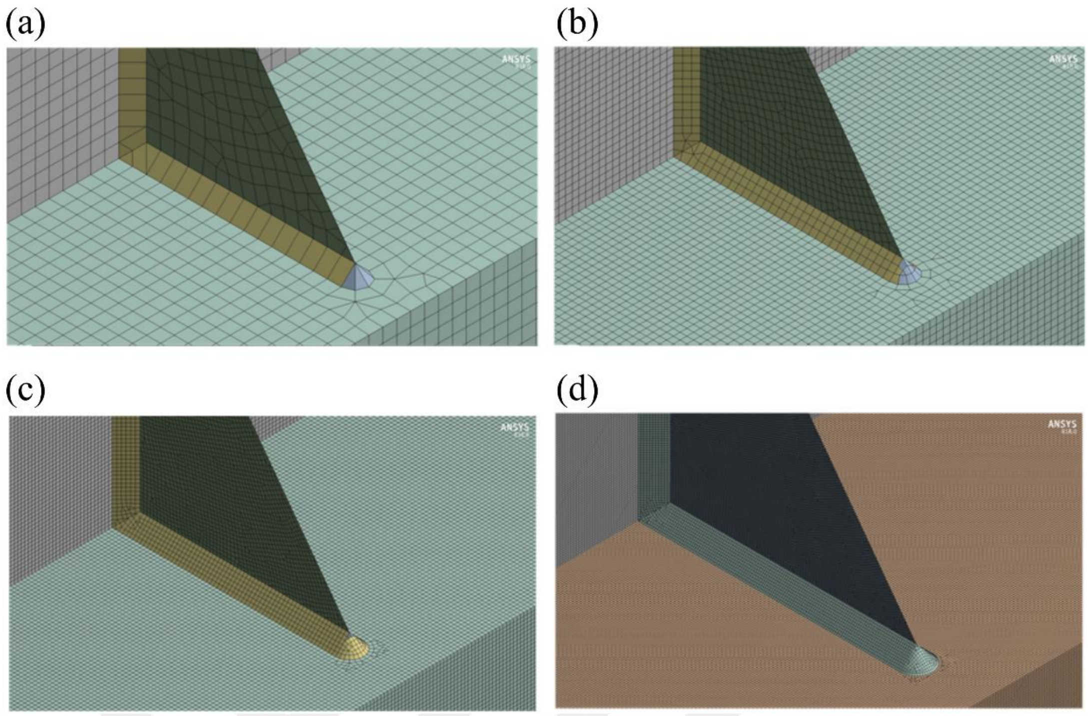


Figure 3.5 Models with mesh sizes of (a) 10 mm, (b) 5 mm, (c) 2.5 mm and (d) 1.25 mm.

Table 3.2 Details of finite element models

Mesh size (mm)	**Element type**	**Number of Nodes**	**Number of Elements**
10	Linear Quad & Tria	6283	6280
5	Linear Quad & Tria	25243	25232
2.5	Linear Quad & Tria	100815	100797
1.25	Linear Quad & Tria	402485	402458

Table 3.3 Material properties

Properties	Value	Units
Density	7850	kg/m^3
Young’s Modulus	200000	MPa
Poisson’s Ratio	0.3	-
Yield Strength	250	MPa
Ultimate Strength	460	MPa

The finite element analysis of this study is conducted using ANSYS Mechanical V.18. The loading and boundary conditions are defined as shown in Figure 3.6. The model is fixed from the end of 200×200 square base profile and 5000 N force are applied to 200×600 plate in –*y* direction.

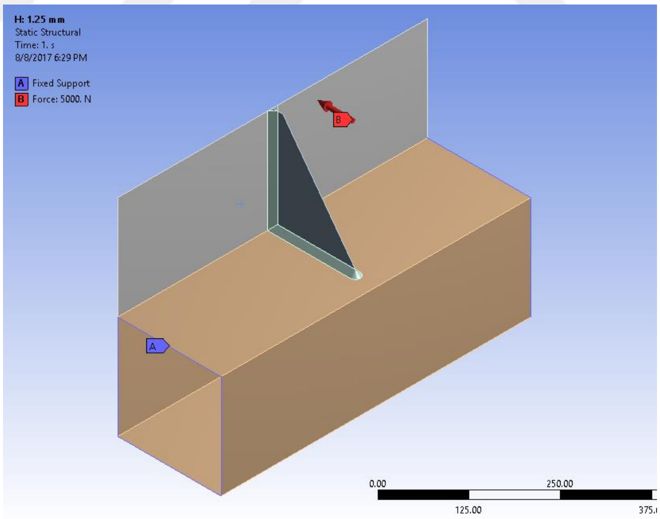


Figure 3.6 Loading and boundary condition of model

Under these conditions, problems are solved with finite element solver. Figure 3.7 shows the von Mises stress results of the models. Elements with the highest von Mises stresses are pointed with an arrow.

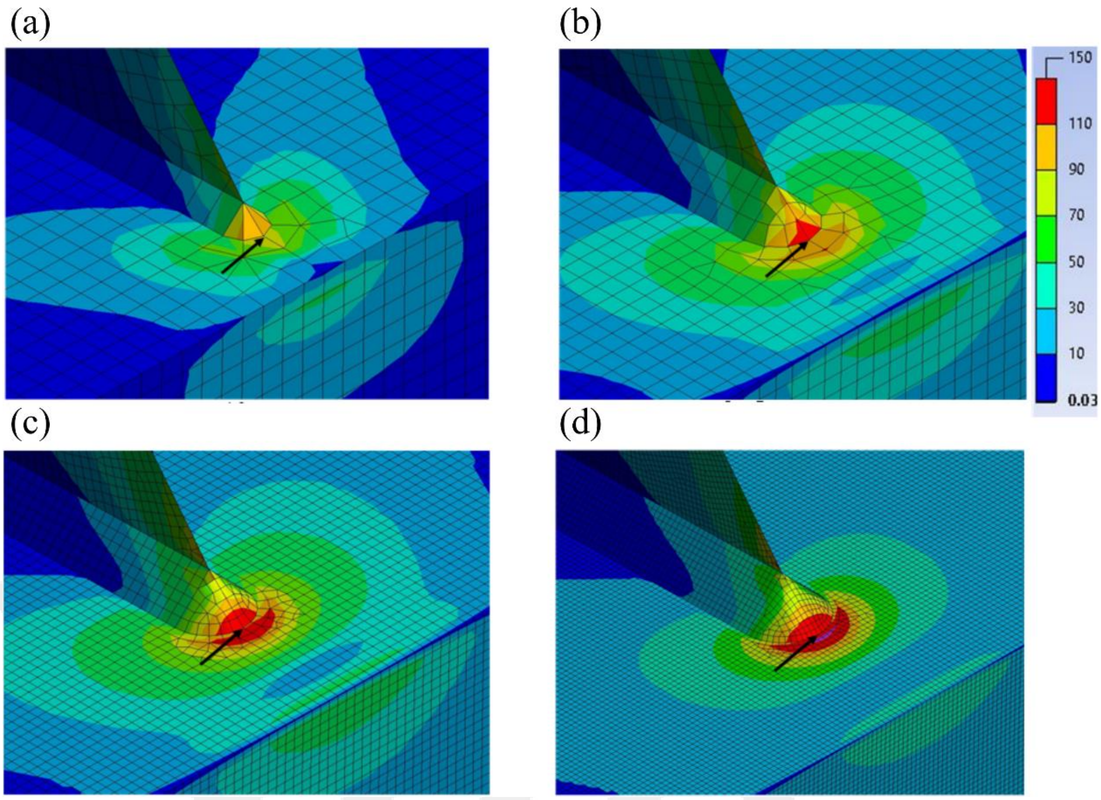


Figure 3.7 Von Mises stress results from FEA: (a) Model with an element size of 10 mm, (b) model with an element size of 5 mm, (c) model with an element size of 2.5 mm and (d) model with an element size of 1.25 mm

Mechanical APDL Module of ANSYS V.18 is used for reading the nodal forces and the nodal moments. Using Equation (2.5), Structural Stresses are calculated for each model separately. The results are depicted in Table 3.4. Variation in the Structural Stresses with different element sizes is 0.88% whereas von Mises stresses at the critical points with same mesh sizes have quite large range of stress. Besides, it can be seen from the table that with decreasing the element size –improving the mesh quality- von Mises stresses are increasing considerably. On the other hand, SS results are nearly same at higher stress values from the von Mises stresses.

Table 3.4 Variation of the von Mises and Structural Stress results with different elements sizes

Element size	**von Misses Stress (MPa)**	**Variation (%)***	**Structural Stress (MPa)**	**Variation (%)***
10	107.14	-16.22	151.39	-0.0726
5	129.30	1.11	151.36	-0.0924
2.5	135.72	6.13	150.50	-0.6600
1.25	139.35	8.96	152.73	0.8118
Average Stress (MPa)	127.88	**25.19** **	151.5	**0.8844** **

* Variation (%) =[(Stress - Average Stress) / Average Stress] ×100

** Total Variation = Max Va. – Min Va.

This section considers an existing problem of mesh sensitivity to illustrate the application of the SS approach. A workbench geometry is discretized with four different element sizes and finite element analysis of these models were successfully conducted. Based on the results obtained in this section, it is proved that SS method is mesh insensitive and it can be used for an alternative fatigue assessment method.

3.3 Structural Stress & Hot-spot Stress comparison on a Tram Fatigue Life

The fatigue life of four critical regions in a tram are estimated using the hot-spot stress and structural stress methods. The tram model considered in this section consists of five wagons ("modules" hereafter). The modules are labeled as MA, RA, P, RB, and MB as depicted in Figure 3.8. The bogies are located in the front, middle, and back modules, which are MA, P, and MB, respectively. Loading and tram weight definitions are given in Table 3.5 and Table 3.6. Loading conditions are defined according to the VDV 152 regulations [64] as listed in Table 3.6. For the loading conditions of interest in this study, m_1 and m_3 mass condition is applied from Table 3.5 and given in detail in Table 3.7.

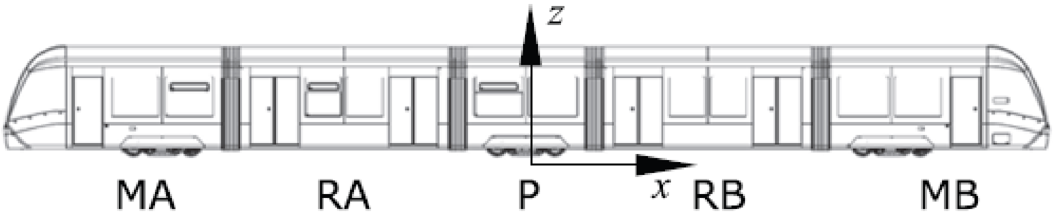


Figure 3.8 Tram model showing the module labels and global coordinate axes.

Table 3.5 Tram weight definitions from VDV-152 [64]

Definition	Symbol	Description
Mass of the entire tram	m_0	Empty mass of the entire vehicle without consumables (sand, water, etc.) and passengers
Ready for service body mass	m_1	Empty mass m_0 with consumables and train personnel (80 kg) without bogie masses
Mass of a frame	m_2	The mass of the equipment under the body suspension. The mass of the connecting elements between the body and frame is distributed between m_1 and m_2.
Mass of passengers in a normal passenger load	m_3	The total mass of the number of passengers in a body as the partial load of the maximum mass possible: $m_3 = 2m_4/3$
Mass of passengers in the case of unusual passengers load	m_4	Maximum possible total mass of standing and seated passengers in a body

Table 3.6 Fatigue-related operating loads from VDV-152 with body loads

Loading Condition ($m_1 + m_3$)	a_x (m/s²)	a_y (m/s²)	a_z (m/s²)	Side wind (Pa)	Twisting
Departure/switch	-	± 1.6	$-g \pm 2.4$	± 200	-
Curve Passing	-	± 3.3	$-g \pm 1.2$		$\pm 1/250$
Docking/braking	± 2.0	± 0.9	$-g \pm 1.2$		-

Table 3.7 Loading conditions investigated

Loading Condition ($m_1 + m_3$)	a_x (m/s²)	a_y (m/s²)	a_z (m/s²)	Side Wind (Pa)	Twisting
S01	-	-1.6	-12.21	-200	-
S08	-	$+1.6$	-7.41	$+200$	-
S09	-	-3.3	$+8.61$	-	$-1/250$
S16	-	$+3.3$	-8.61	-	$+1/250$
S17	-2.0	-0.9	-11.01	-	-
S24	$+2.0$	$+0.9$	-8.61	-	-

Since the motion of the tram in the lateral and downward directions is restricted by the rails, all the wheels of the tram are fixed in the *y* and *z* directions, as shown in Figure

3.7. For the curve passing scenario, 7.2 mm displacement in the *z* direction is defined on the right front wheels of the M and P modules on the tram body.

In order to achieve sufficient accuracy without excessive computational cost, a minimum mesh size of 5 mm is used at high stress regions to reduce the computational time. The FE mesh is generated using ANSA [65]. The overall tram FE model consists of 4,424,099 shell elements, 221,105 solid elements, and 436 1D rigid and mass elements, with a total of 4,645,640 elements. 4-noded shell element type SHELL181 is used for shell structures in this study. The SOLID185 8-noded brick element type is selected for solid structures of the tram because of its ability to simulate deformations of nearly incompressible elasto-plastic materials as well as fully incompressible hyperelastic materials [66]. An elasto-plastic material model from the ANSYS Material Library is used for the main body frames of the modules. S355J2 Steel material model with modulus of elasticity of 200 GPa, Poisson's ratio of 0.3 and yield strength of 355 MPa is used.

The structures of railway vehicle bodies are exposed to a large number of dynamic loads of varying magnitudes throughout their operational life. The effects of these loads are most apparent at critical regions in the vehicle body structure. Geometry changes in door and window corners cause an increase in stress concentration (see e.g., Figures 3.11 (d) and 3.12 (d)). In this section four critical regions (CR) are identified likely to fail due to fatigue. The first three regions which are named as CR-1, CR-2 and CR-3 hereafter are located on the MA Module. The fourth region CR-4 is located on the RA Module.

The weight of the equipment such as air conditioner, pantograph, battery box, etc. located on the roof of the MA module is transferred to the chassis through columns. Critical region 1 (CR-1) is located at the base of the column near the lower articulation joint of the MA Module with RA Module (see Figure 3.9 (a) and (b)). The maximum stress range in this region occurs during S08 and S17 load cases which correspond to departure/switch and docking/braking scenarios, respectively as shown in Figure 3.9 (c). The longitudinal accelerations during braking cause stress concentrations in this region. Similarly, during load case S17 higher stresses develop in the corner of passenger footrest area of the MA Module which is designated as critical region 2 (CR-2) (see Figure 3.10 (b) and (e). The deformed shapes of the module shown in Figures 3.9 (e) and 3.10 (e) indicate that the dominant mode of deformation in CR-1 and CR-2 under load case S17 is bending.

The third critical region (CR-3) shown in Figure 3.11, is located on the corner of the window of the MA module. The maximum stress range in this region occurs when the loading alternates between load cases S01 and S08 corresponding to the departure/switch scenario, as shown in Figure 3.11 (c). The side wind loads defined in loading case S01 and S17 cause lateral deformations and stress concentrations at the window corners (see Figure 3. 11 (d) and (e)).

Figure 3.12 (a) shows the normal stress σ_x distribution in the RA module under S24 load case (docking/braking scenario). Largest stresses develop near the corner of the door which is selected as critical region 4 (CR-4). The stress range in this region (see Figure 3.12 (c)) is given by the difference between the stress values of load case S24 and load case S16 which corresponds to the curve passing scenario. In contrast to other modules there is no bogie parts in this module. Consequently, it is less stiff and undergoes relatively large deflections as shown in Figure 3.12 (d) and (e). Similar to the critical regions CR-1 and CR-3, the S24 (docking/braking) load case leads to bending dominated deformation in region CR-4.

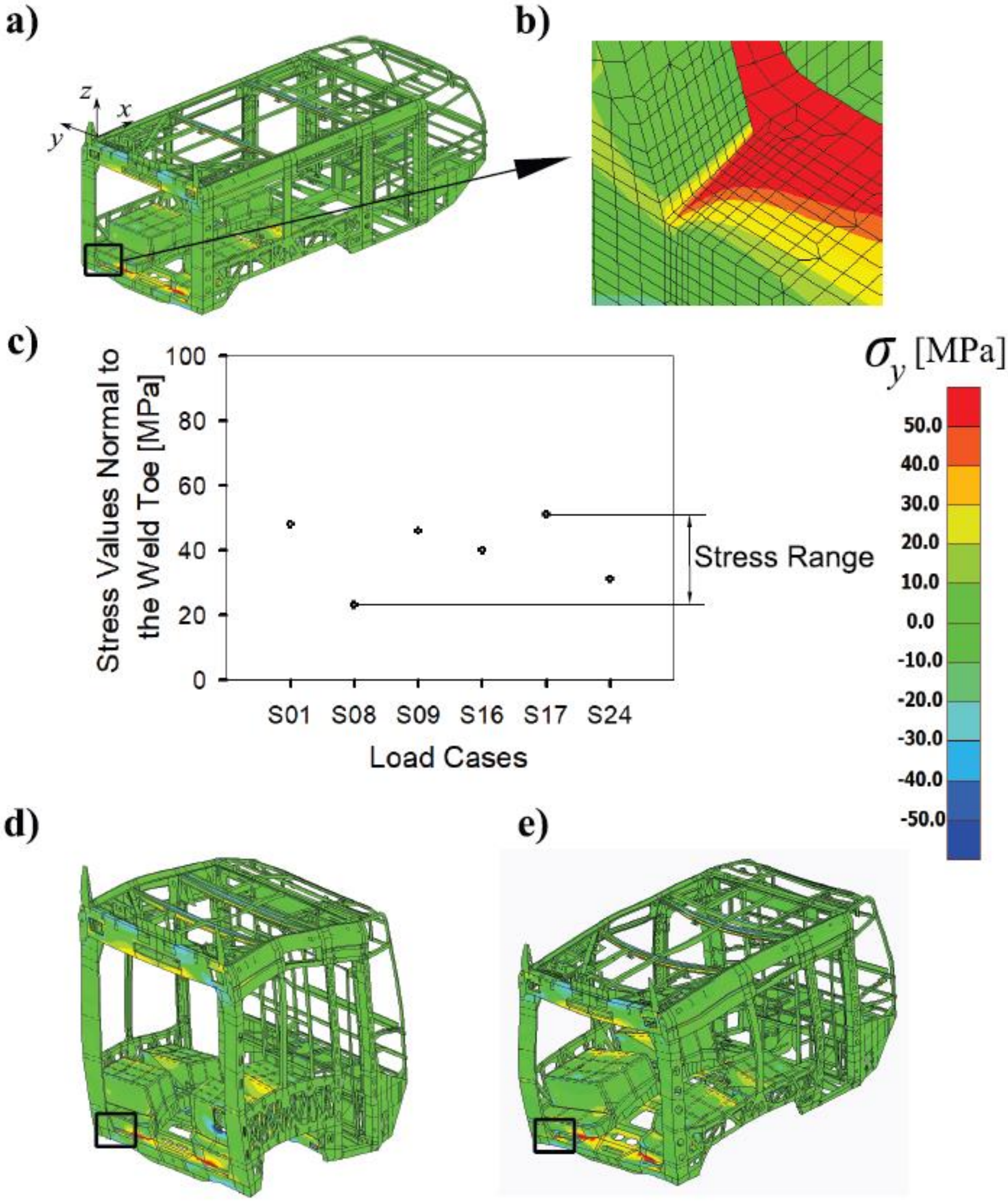


Figure 3.9 (a) Critical region 1 (CR-1) is located near the lower articulation joint of the MA Module with RA Module. (b) Close-up view of CR-1. (c) Normal stress values and the stress range for different load cases. (d) Deformed geometry of MA Module under (d) S08 load case and (e) S17 load case (displacements are scaled by a factor of 150×)

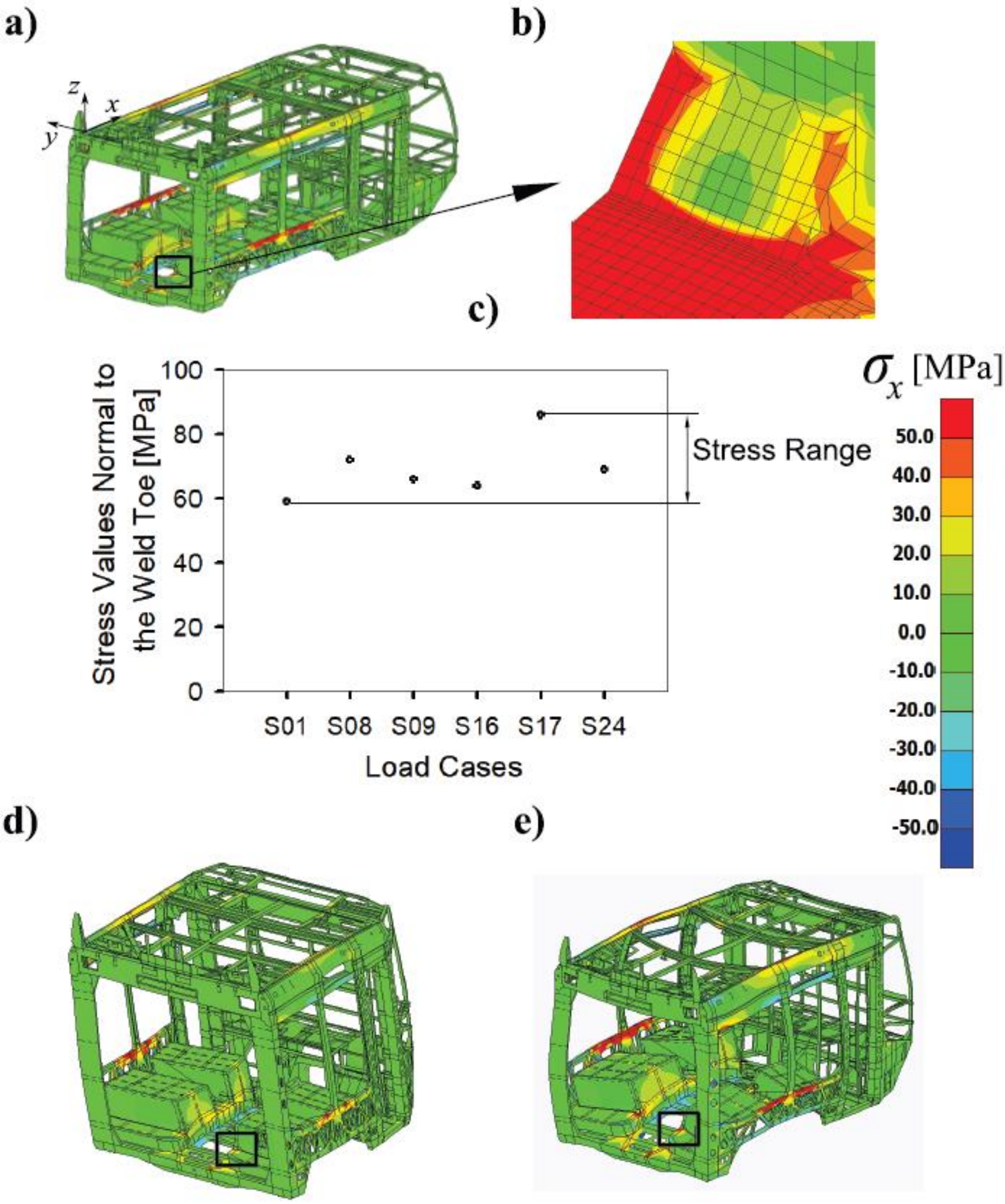


Figure 3.10 (a) Critical region 2 (CR-2) is near the passenger footrest of the MA Module. (b) Close-up view of CR-2. (c) Normal stress values for different load cases and the stress range. Deformed geometry of MA Module under (d) S01 load case and (e) S17 load case (displacements are scaled by a factor of 150×)

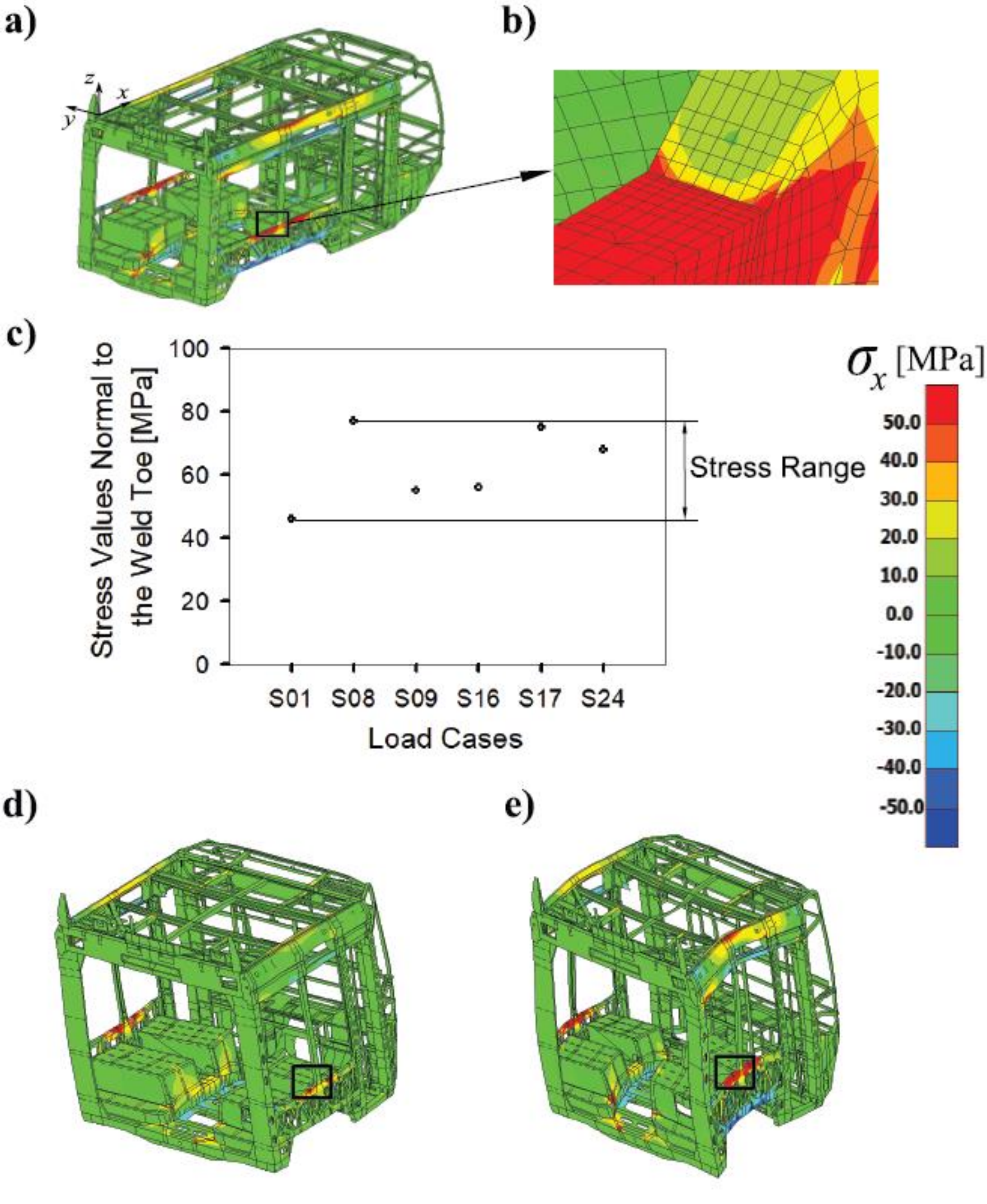


Figure 3.11 (a) Critical region 3 (CR-3) is located at the corner of the window of MA Module. (b) Close-up view of CR-3. (c) Maximum stress range and normal stress values for different load cases. Deformed geometry of MA Module (d) under S01 load case and (e) S08 load case (displacements are scaled by a factor of 150×)

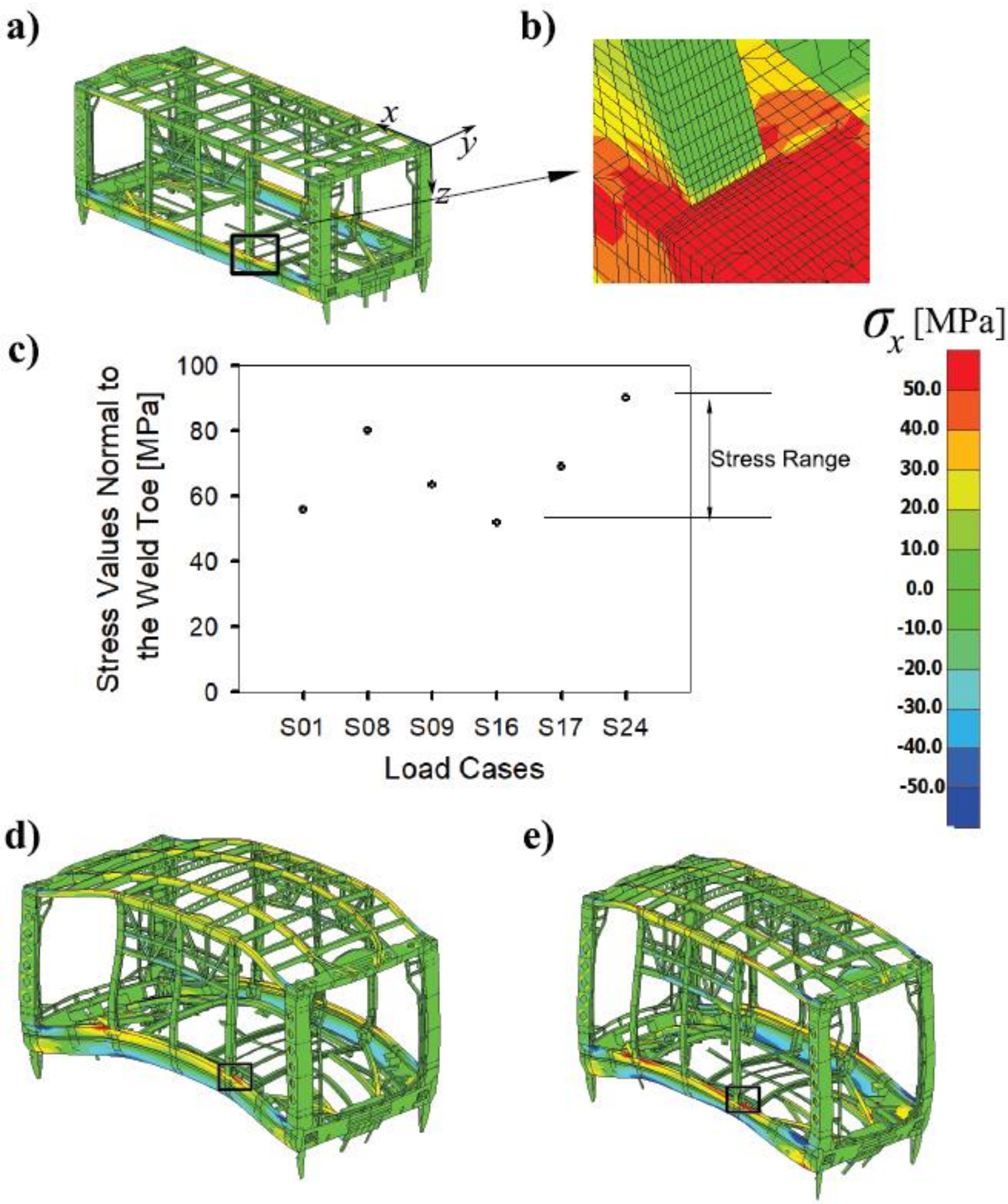


Figure 3.12 (a) Critical region (CR-4) is at the corner of the door of RA Module. (b) Close-up view of CR-4. (c) Maximum stress range and normal stress values for different load cases. Deformed geometry of RA Module under (d) S16 load case and (e) S24 load case (displacements are scaled by a factor of 150×)

The fatigue lives of the welded connections in aforementioned critical regions are computed using the HSS and structural stress methods and are listed in Table 3.8. The fatigue lives are also computed with the commercial package NCode which can employ the HSS method by selecting a suitable FAT class following the IIW recommendations [19]. NCode generated results are included in Table 3.8 for comparison. As expected manually computed fatigue lives using the HSS method are close to the lives reported by NCode for all critical regions. On the other hand, the structural stress method produces categorically lower fatigue lives for all regions. Therefore, it can be inferred that the SS method is more conservative than the HSS method.

Table 3.8 Fatigue lives of welded connections in critical regions computed using different methods

Calculation Method	CR-1	CR-2	CR-3	CR-4
SS	1.563	1.396	1.211	1.140
HSS (from FEA)	1.811	1.655	1.725	1.382
NCode (based on HSS)	1.920	1.798	1.758	1.432

Critical regions are identified susceptible of fatigue failure in an entire tram model subjected to load cases specified in the VDV-152 regulation [64]. Then, the fatigue lives of welded connections are compared in these regions computed using the HSS and SS methods. It is found that the regions of a tram likely to fail due to fatigue are located in places such as window and door corners where the rapid change in geometry causes stress concentrations. Among the load cases investigated, the maximum stress range in three of the four critical regions is reached when the tram is subjected to the load cases corresponding to the docking/braking scenario. Overall, the structural stress method predicts more conservative fatigue lives compared to the HSS method in critical regions investigated in this section.

3.4 Mesh Sensitivity of Hot-spot Stress Method

A mesh convergence study is conducted to investigate the mesh size sensitivity for a specimen given in Figure 3.13 under three loading types: 3-point bending, 4-point bending and axial loading. Figure 3.14 (a) shows the categorization of the mesh types as relatively coarse- 1 and 2 (rc-1 and rc-2); and relatively fine- 1 and 2 (rf-1 and rf-2) and size of each mesh type is given in Table 3.9. In the case of axial loading, it is observed that the element type did not affect the normalized stress significantly, while in the case of bending, normal stresses are obtained as below the normalized values with coarse element types as depicted in Figure 3.14 (b) - (d). On the other hand, it is determined that the difference between the rf-1 and rf-2 element types is negligible, except for the high stress concentration point at the weld toe which has not taken into consideration for HSS calculations. As a result, the use of rf-1 element type with a mesh size of $0.1\times t$ is found suitable for FE analysis.

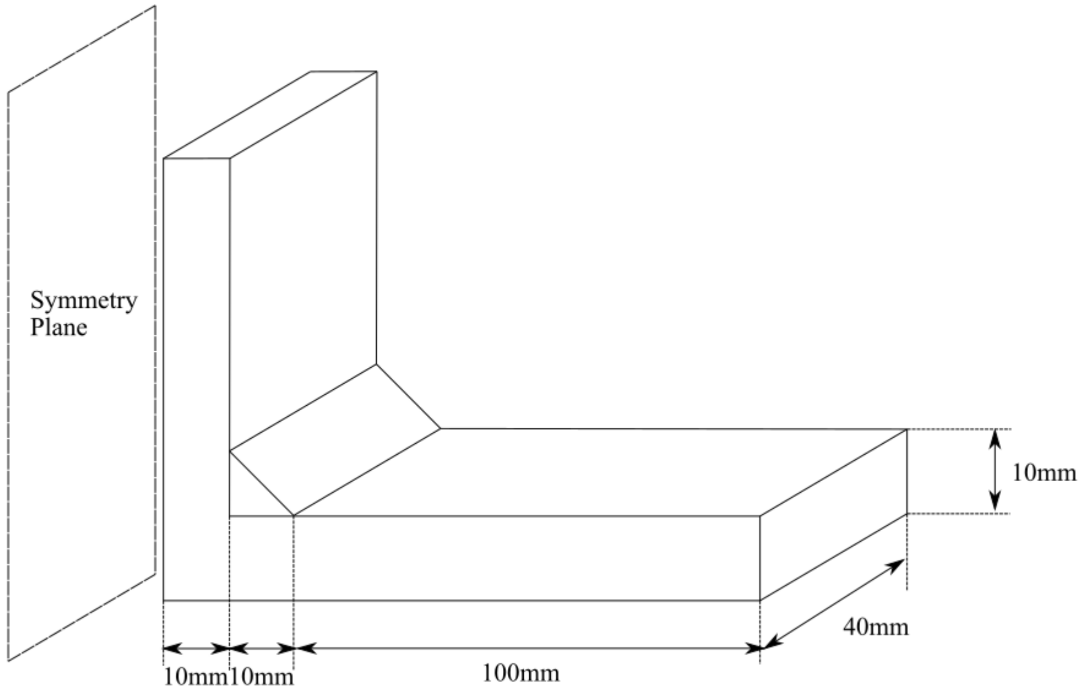


Figure 3.13 Geometrical configuration of the specimen

Table 3.9 Discretization details for mesh convergence study

Mesh type	Mesh size
rc-1	$0.5\times t$
rc-2	$0.25\times t$
rf-1	$0.1\times t$
rf-2	$0.05\times t$

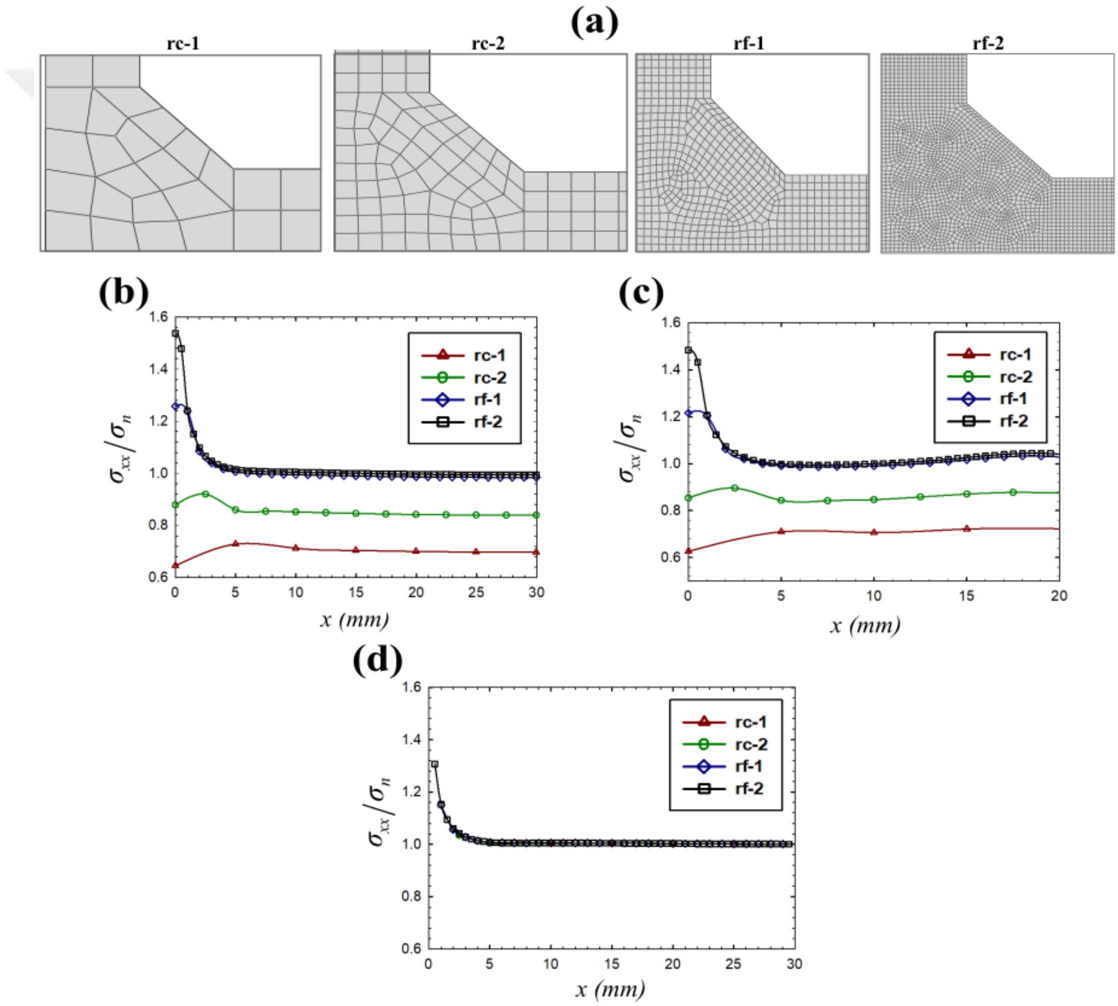


Figure 3.14 (a) Four mesh types for the critical fatigue driving zone; Normalized stress-distance from weld toe graphs for: (b) 4-point bending, (c) 3-point bending and (d) axial loading

CHAPTER 4

A NEW APPROACH – MODIFIED HOT-SPOT STRESS METHOD

4.1 Problem Definition

HSS method evaluates the fatigue strength of welded structures including the geometric and/or loading effects and excluding the local stress peak caused by local weld profile as illustrated in Figure 4.1. This is done by extrapolating the stresses located at two or three extrapolation points away from the weld toe.

The procedure for extrapolating the stresses is well documented in the guidelines reported by IIW [19]. Establishing the reference points is the first step of the procedure. IIW stated that identification of the reference points can be made by experience of existing failed components or analyzing the finite element method results or measuring several different points. The determination of reference points varies depending on the quality of discretization and whether the extrapolation is linear or quadratic. In this context, this study covers the linear extrapolation technique with fine mesh quality.

The first reference point must be chosen to avoid any stress raising effect due to the weld itself. This is stated in the IIW recommendations as it is practically at a distance of $0.4 \times t$ from the weld toe (t: plate thickness). The second reference point is at a distance of $1 \times t$ and linear extrapolation is done by the following equation:

$$\sigma_{hs} = 1.67\sigma_{0.4t} - 0.67\sigma_{1t} \quad \textbf{(4.1)}$$

where σ_{hs} is hot-spot stress, t is the thickness of the plate, $\sigma_{0.4t}$ and σ_{1t} are the normal stresses at distance of $0.4 \times t$ and $1 \times t$ from the weld toe, respectively. Stresses along the weld and reference points for HSS method can be seen in Figure 4.1.

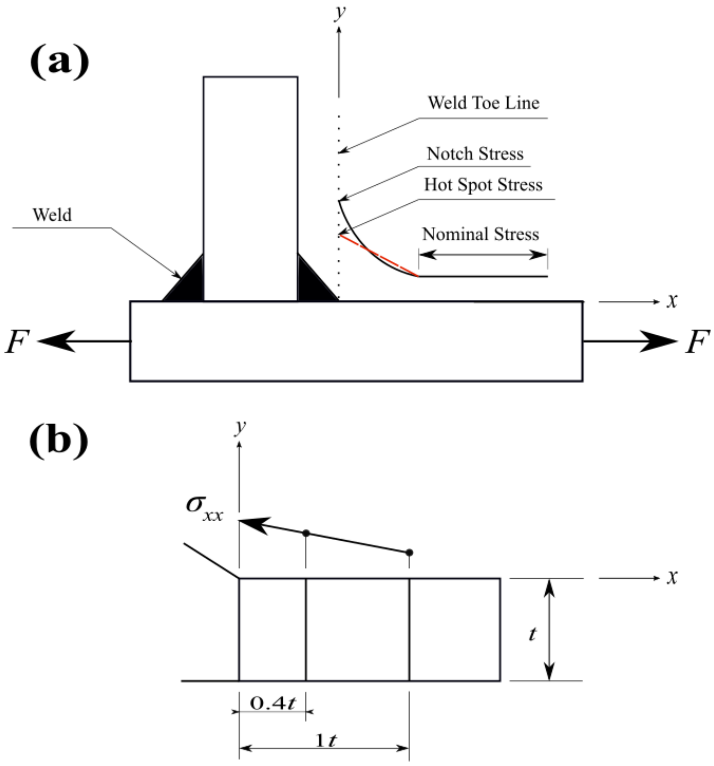


Figure 4.1 (a) Definition of the stresses along the weld toe and (b) linear extrapolation points for HSS method

It has been mentioned before that there is no parametric study in the determination of extrapolation points in the HSS method. Since it is thought that a systematic study in the determination of these points will be of great importance in the use of the HSS method, the focus is on the determination of these points. Here, the success criterion will be reduced scatter of the S-N curves according to the HSS values calculated with the new extrapolation points to be used. The narrower distribution in the S-N curves will minimize the risk of wide distributions caused by the factors affecting the fatigue life mentioned in Chapter 2, thus enabling more accurate fatigue life estimations in welded structures.

In the next sections, a new modified HSS method is proposed which can be used to determine the distances of the extrapolation points from the weld toe for estimating the fatigue life more accurately by reducing scatter. To achieve this a combination of finite element analyses (FEA) and design of experiments and optimization techniques are employed. Near 100 experimental results from previous studies have been collected to construct the polynomial response surface (PRS), which is then used as the basis for the optimization problem.

4.2 Creating a Database from the Literature

The experimental data was collected from the literature including a wide range of main plate thickness with 3 different loading conditions as shown in Figure 4.2. The geometry and the loading type as well as the reference to which the data belongs are listed in Table 4.1. It is reported that fatigue cracks initiate at the weld toe for all of the specimens.

The data belonging to the 4-point bending fatigue experiments used in this study were carried out by Miki et al.[67]. SM58 steel specimens with five different main plate thicknesses ranging from 9 mm to 50 mm were tested in the mentioned study. Other significant dimensions are listed for the specimens in Table 4.1. Geometric configuration of the transverse fillet welded joint under 4-point bending loading depicted in Figure 4.2 (a).

For the case of 3-point bending, the experimental data is obtained from Cheng [68] who studied cruciform welded plate joints under 3-point bending. S550 steel specimens with three different main plate thicknesses 20, 40 and 60 mm used in the mentioned study (see Table 4.1 for complete information). Geometric configuration of the cruciform welded plate joint under 3-point bending loading is shown in Figure 4.2 (b).

Lastly, the experimental database pertaining to axial loading is taken from Refs [69] and [16]. The thickness of HSLA80 specimens is 11 mm, and the thicknesses of BS 4360, grade 50 specimens have a large range from 13 mm to 100 mm. The geometric configurations for the axial loading studies are schematically shown in Figure 4.2 (c) (see Table 4.1 for geometric information).

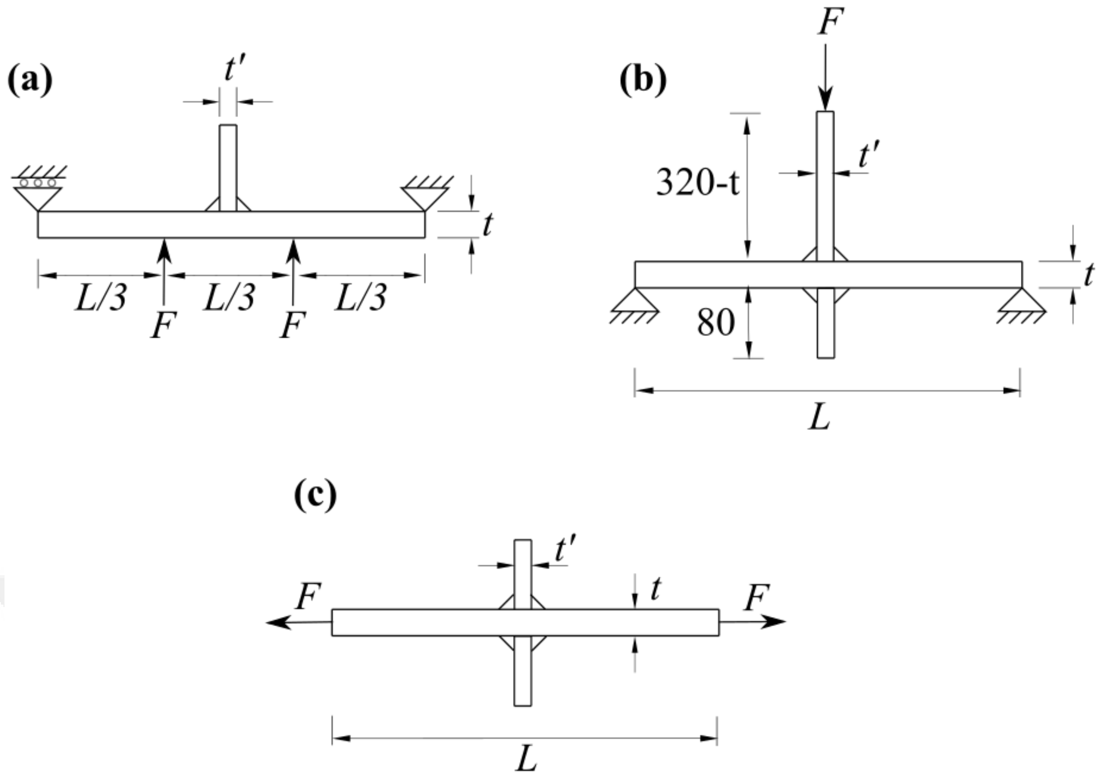


Figure 4.2 Geometric configurations, loading and support conditions under (a) 4-point bending [67]; (b) 3-point bending [68]; and (c) axial loading [16], [69]

Table 4.1 Experimental dataset details

Specimen	Load types	Number of specimens	*L* (mm)	*t*(mm)	*t'*(mm)	weld size (mm)	Refs
PC9	4p. bending	8	180	9	16	6	[67]
PC15		7	240	15	16	6	
PC24		6	300	24	16	6	
PC34		6	400	34	16	6	
PC50		7	500	50	16	6	
BT20	3p. bending	5	340	20	20	10	[68]
BT40		6	340	40	20	10	
BT60		6	340	60	20	10	
KS11	Axial loading	7	112.6	11	11	4	[69]
G13	Axial loading	12	730-900	13	10	8	[16]
G25-1		7	730-900	25	3	5	
G25-2		5	730-900	25	13	10	
G50		6	730-900	50	3	5	
G100-1		5	730-900	100	13	8	
G100-2		5	730-900	100	200	15	

*Stress Ratio, $R = 0.1$ (with constant amplitude loading)

4.3 Statistical Design and Analysis of Experiments

Statistics have a crucial role in the evaluation of experimental results. In particular, the fatigue life evaluation of welded structures is based on many experiments and the interpretation of the results of these tests. Fatigue life is indicated as scatter in S-N curves as stated in the previous sections. In this section, statistical terminology, which plays an important role in the interpretation of scatters, will be covered. Note that here it will be sufficient to give a summary and the sections related with the S-N scatters.

A scatter diagram is a plot of independent *X* variable and dependent *Y* variable for pairs of observations (X_i, Y_i) obtained from the sample [70]. The S-N curves of the experimental studies have usually a scatter which can be evaluated by performing a linear regression. In statistics, fitting a response variable to a linear function of one or more predictor variables is called linear regression [71]. The general form of the linear regression model is expressed as

$$\hat{Y} = a + bX \tag{4.1}$$

where $\hat{Y}$ is the response value of the predictor variable *X*, *a* is the intercept and *b* is the slope of the regression line. *a* and *b* are calculated by using least square methods as

$$b = \frac{N\sum_{i=1}^{N} X_i Y_i - \sum_{i=1}^{N} X_i \sum_{i=1}^{N} Y_i}{N\sum_{i=1}^{N} X_i^2 - \left(\sum_{i=1}^{N} X_i\right)^2} \tag{4.2)(a}$$

$$b = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{N}(X_i - \bar{X})^2} \tag{4.2)(b}$$

$$a = \bar{Y} - b\bar{X} \tag{4.3}$$

where Y_i is i^{th} value of the response variable, X_i is i^{th} value of the predictor variable, $\bar{Y}$ and $\bar{X}$ are the mean values of the response and the predictor variables, Y and X, respectively. $\bar{Y}$ and $\bar{X}$ are calculated for N number of datapoints as

$$\bar{Y} = \frac{\sum_{i=1}^{N} Y_i}{N} \tag{4.4}$$

$$\bar{X} = \frac{\sum_{i=1}^{N} X_i}{N} \tag{4.5}$$

A sample scatter with a linear regression line is depicted in Figure 4.3. Random error for any point in the linear regression line, residual term, e_i, is calculated as

$$e_i = Y_i - \hat{Y}_i \tag{4.6}$$

where $\hat{Y}_i$ is i^{th} value of the predicted response value.

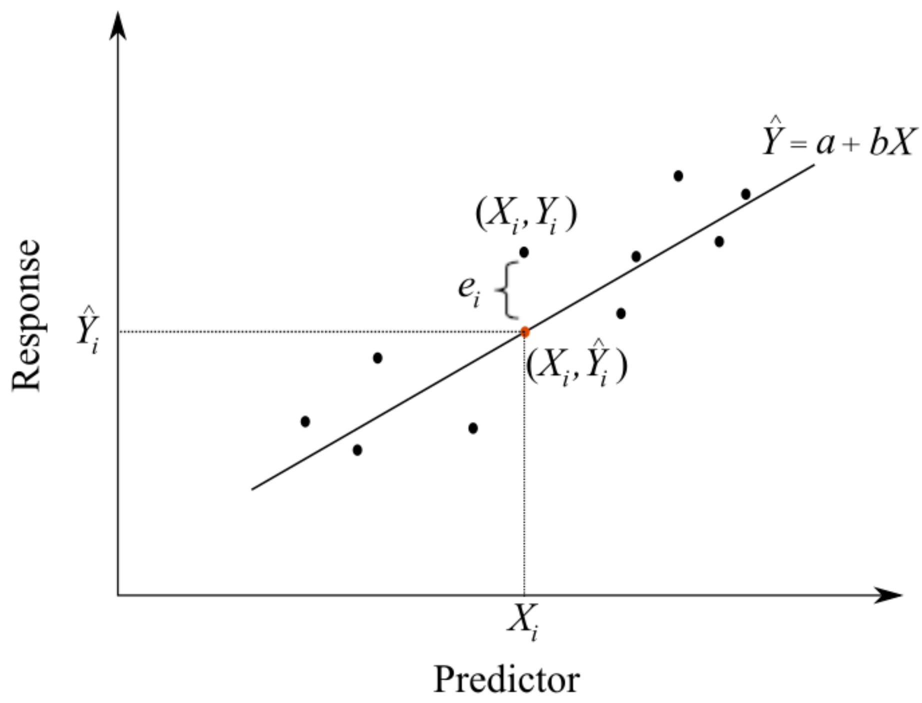


Figure 4.3 A sample scatter with a linear regression line

Using the error term indicated in the figure, the sum of squared errors (SSE) is expressed by the following equation:

$$SSE = \sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2 \qquad \textbf{(4.7)}$$

Standard deviation of calculated errors (s^2) which equals mean squared errors (MSE) is calculated as

$$s^2 = MSE = \frac{SSE}{N-2} = \frac{\sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2}{N-2} \qquad \textbf{(4.8)}$$

where $N-2$ is represents the degrees of freedom for the regression model. Since one degree of freedom will decrease for $\bar{Y}$ and $\bar{X}$ separately, it is obtained as $N-2$. Standard error of the regression, *s*, which also named as root mean squared error (RMSE) is calculated as

$$s = RMSE = \sqrt{MSE} = \sqrt{\frac{SSE}{N-2}} = \sqrt{\frac{\sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2}{N-2}} \qquad \textbf{(4.9)}$$

The percentage of data closest to the regression line and the degree of variation predicted for one variable from the other is represented by coefficient of determination, R^2. This value can be calculated as

$$R^2 = \frac{\sum_{i=1}^{N} (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^{N} (Y_i - \bar{Y})^2} \qquad \textbf{(4.10)}$$

The value of R^2 ranges from 0 to 1 and is expressed as a percentage. The higher value of R^2 means that the higher representation of the scatter with the regression line.

Another important part of interpreting scatter diagrams is the representation of confidence and prediction intervals. The confidence intervals are represented with a percentage (90%, 95%, 97.7% i.e.). For a predictor variable (X^*), the confidence interval (CI) values of the response variable (on the upper CI, Y_u^* and on the lower CI, Y_l^*) may be formulated as

$$Y_{u,l}^* = \hat{Y}^* \pm t_{\alpha/2} s_{\hat{Y}^*} \tag{4.11}$$

where $\hat{Y}^*$ is the response value of the regression line at $\hat{X}^*$, $t_{\alpha/2}$ is the constant of the t-normal distribution which is determined with respect to the desired confidence level value of α and $s_{\hat{Y}^*}$ is the estimated standard deviation of the predicted values which can be calculated as

$$s_{\hat{Y}^*} = s\sqrt{\frac{1}{N} + \frac{(X^* - \bar{X})^2}{\sum_{i=1}^{N}(X_i - \bar{X})^2}} \tag{4.12}$$

where s is the standard error of the regression which can be calculated from Equation (4.9).

For a predictor variable (X^*), the prediction interval (PI) values of the response variable (on the upper PI, Y_u^{**} and on the lower PI, Y_l^{**}) may be formulated as

$$Y_{u,l}^{**} = \hat{Y}^* \pm t_{\alpha/2} s_{pred} \tag{4.13}$$

s_{pred} is the standard deviation as to prediction of the value of $\hat{Y}^*$ which can be calculated as

$$s_{pred}^2 = s^2 + s_{\hat{Y}^*}^2 \tag{4.14}$$

CIs and PIs are schematically shown in Figure 4.4.

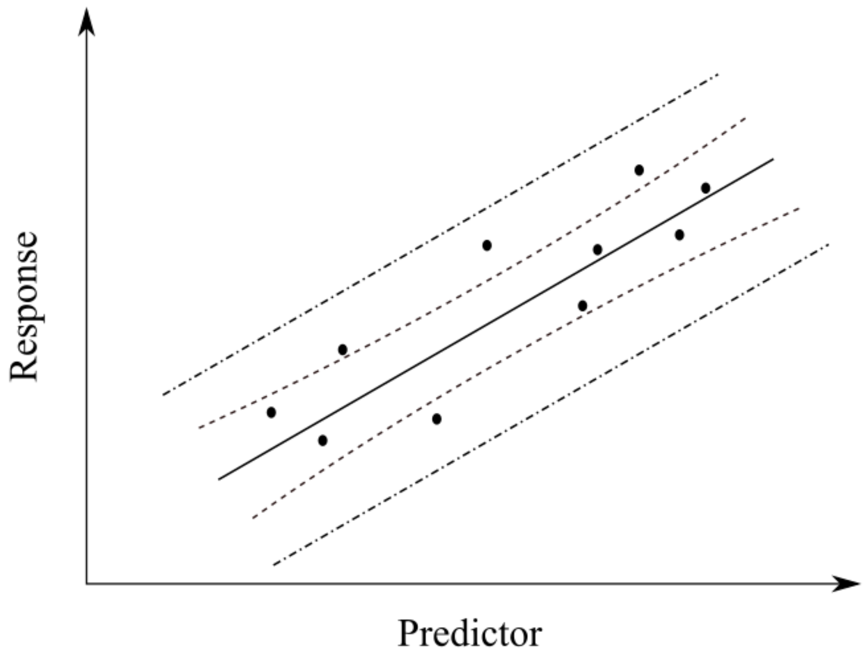


Figure 4.4 A sample scatter with a regression line and CIs (dash-dash lines) and PIs (dash-dot lines)

The sections above are the statistical sections on regression analyzes performed on the scatter diagrams. For fatigue life estimations, it is necessary to evaluate the uncertainty in the statistical estimators [72]. Design S-N curves relies on probability of survival (P_s) with a confidence level (γ) which can be expressed as

$$Y_{(P_s,\gamma)} = Y_{50\%} - q \cdot s \tag{4.15}$$

where $Y_{(P_s,\gamma)} = \log N_{(P_s,\gamma)}$, $Y_{50\%} = \log N_{50\%}$ and q is a statistical parameter which can be determined using the tolerance interval approach [19], [73] as

$$q = \Phi^{-1}_{(P_s)} + t_{(\gamma;N-2)} \frac{\sqrt{2}}{\sqrt{N-2}} \tag{4.16}$$

where $\Phi^{-1}_{(P_s)}$ is the P_s-percentile from the inverse standard normal cumulative distribution, $t_{(\gamma;N-2)}$ is the one-tail γ-percentile of Student's t-distribution with (N-2) degrees of freedom and γ is the confidence level (typically taken as γ = 95%). Note that, as the sample size N increases, the last term of the above equation decreases and becomes negligible. A sample S-N curve with regression, P_s line and P_s line with confidence level is depicted in Figure 4.5. Note that contrary to the classical notation, the vertical and horizontal axes for the S-N curves are reversed for historic reasons [74].

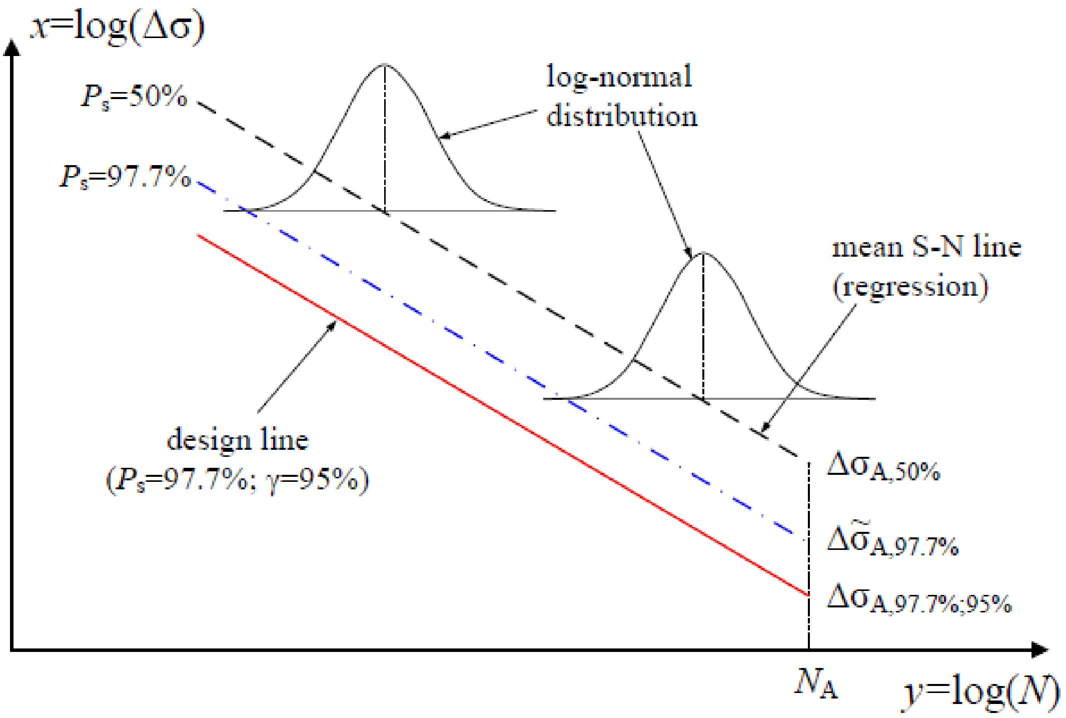


Figure 4.5 Mean S-N curve from the regression analysis and the design S-N curve at the prescribed probability P_s of survival (97.7%) and confidence γ (95%) [72]

4.4 Optimization

Although processors (CPU), memory and storage capacities of computers have reached enormous levels in recent years, analysis times performed with acceptable accuracy are still computationally costly. Moreover, it may be necessary to repeat the analyzes many times to solve an optimization problem. To overcome this, surrogate model-based optimization techniques can be used for this type of computationally costly problems [75], [76].

Surrogate models provide a mathematical relationship between the inputs and the responses of a problem. With the use of these mathematical models, it is possible to predict the results of engineering problems that require a large number of analyzes and have long computation times. In order to determine this mathematical relationship, a certain number of design points must be created for different values of the inputs and the responses at these design points must be obtained by experiments or analyzes. Firstly, design points are determined by experimental design methods (Design of experiment, DoE). Then, the necessary analyzes are conducted for each design point and the response values are calculated. Finally, using the design points and the responses obtained for these points, a mathematical model is created between the inputs and the responses. Once this mathematical model is created, response estimation can be easily made for different values of inputs very quickly and simply.

The first step in creating a surrogate model is to determine the experimental design type. Experimental design refers to experiments to be performed in a series, expressed through design variables with certain values [77]. There are two categories for Design of Experiment (DoE) types: classical methods and space filling methods. The most preferred classical experimental designs are central composite design [78], factorial design or fractional factorial design [78], Plackett-Burman design [79] and Box-Behnken [80] designs. Comprehensive information on classical designs are given in [78]. Space filling designs are preferred because they consider all regions of the design space equally when creating sampling points. When the effects of sampling method selection are examined, it is seen that space filling design methods are more suitable for large-sized problems [81]. The most well-known space-filling designs are Latin hypercube (Latin hypercube) sampling design [82], minimax and maximin designs [83] and maximum entropy designs [84]. In this study, the Latin hypercube sampling design type was used.

In Latin hypercube method, each design point is divided into n partitions with equal probability. The design space consists of n^m cells with equal probability for m variables. For example, when we consider two variables with eight partitions, the design space is divided into 64 cells as can be seen in Figure 4.6. The next step is to select 8 cells from these 64 cells as the design point. For the first selection, random sampling is taken and the cell number of this point is calculated. The cell number indicates the section number to which the sample belongs, according to each variable. For example, the cell number (2,1) specifies the sample in the second partition created based on the variable in the first part. At each step, the random variable is generated to be different from the previous sample. The shaded cells in Figure 4.6 show the design points selected with the Latin hypercube sampling for the example given above.

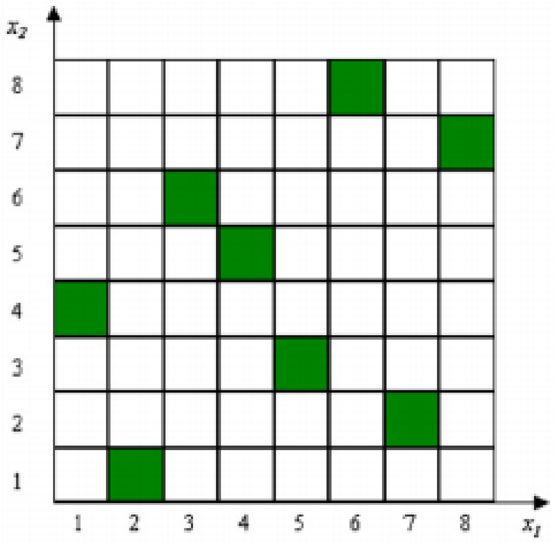


Figure 4.6 Latin hypercube sampling for two design variables and eight design points [85]

Many surrogate modeling techniques have been developed from past to present. Among these techniques, the most preferred ones are; polynomial response surface [78], radial basis functions [86], Kriging [87], neural networks [88] and support vector regression [89]. The polynomial response surface model (PRS) is used in many types of problems due to its easy mathematical model. In this study, second-order PRS approximation is used which is the most commonly used surrogate model. For an N number of variables, second-degree algebraic polynomial function, $\hat{y}(x)$, can be written as

$$y(x) = k_0 + \sum_{i=1}^{N} k_i x_i + \sum_{i=1}^{N} k_{ii} x_i^2 + \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} k_{ij} x_i x_j \tag{4.17}$$

where k_0, k_i, k_{ii}, k_{ij} are the coefficients to be found using a least-squares method, **x** is the input vector.

The first step of the optimization study with surrogate models is the determination of the objective function and design variables. As mentioned in the problem definition of this section, there is no established way of determining extrapolation points for the calculation of Hot-spot Stress (HSS). In order to have a systematic approach and at the same time to make the fatigue life estimation more reliable, it is aimed to determine the optimum locations for the extrapolation points in this newly proposed modified HSS method. Hot spot stress (σ_{hss}) can be written for any two points located at $x = x_1$ and $x = x_2$ as:

$$\sigma_{hss} = \frac{x_2}{x_2 - x_1}\sigma_1 - \frac{x_1}{x_2 - x_1}\sigma_2 \qquad \textbf{(4.18)}$$

where σ_1 and σ_2 are the normal stresses at $x = x_1$ and $x = x_2$ from the weld toe, respectively.

First it is necessary to determine the range of the design variables, x_1 and x_2. Note that x_1 should be at a distance that is not affected by the stress increase from the weld itself. This was taken into consideration in determining the lower bound for x_1. Feng et al. [56] determined that this point is $0.067 \times t$ away from the weld toe for a plate with a thickness of $t = 15$ mm. Upper bound of x_1 and boundaries of x_2 are related with the global geometry effects. It is mentioned ([26], [28], [29], [90]) that geometrical effects vanishes at a distance $2.5 \times t$ from the weld toe. Therefore, the lower and upper boundaries of the design parameters x_1 and x_2 in the optimization study were determined as $0.1 \times t$ and $2.5 \times t$, respectively.

The objective function of the optimization study is to maximize the R^2 value, which is called the coefficient of determination in linear regressions. R^2 represents the percentage of data closest to the line of best fit and shows the degree of variation predicted for one variable from the other [56]. This value can be calculated as given in Equation 30.

$$R^2 = \frac{\sum_{i=1}^{N} (\hat{Y}_i - \overline{Y})^2}{\sum_{i=1}^{N} (Y_i - \overline{Y})^2} \tag{4.19}$$

Higher values of R^2 means higher level of confidence for the fatigue life estimations. Thus, optimization problem for maximum R^2 value can be stated as:

Find $x = \{ x_1, x_2 \}$

Max R^2

Such that $x_2 \geq x_1$

$2.5 \times t \geq x_1, x_2 \geq 0.1 \times t$

Optimization problem is solved by "fmincon" built-in function of MATLAB [91]. This function uses sequential quadratic programming to obtain a constrained optimum of a scalar of several variables. The flowchart for surrogate-based optimization approach followed in this study is shown in Figure 4.7.

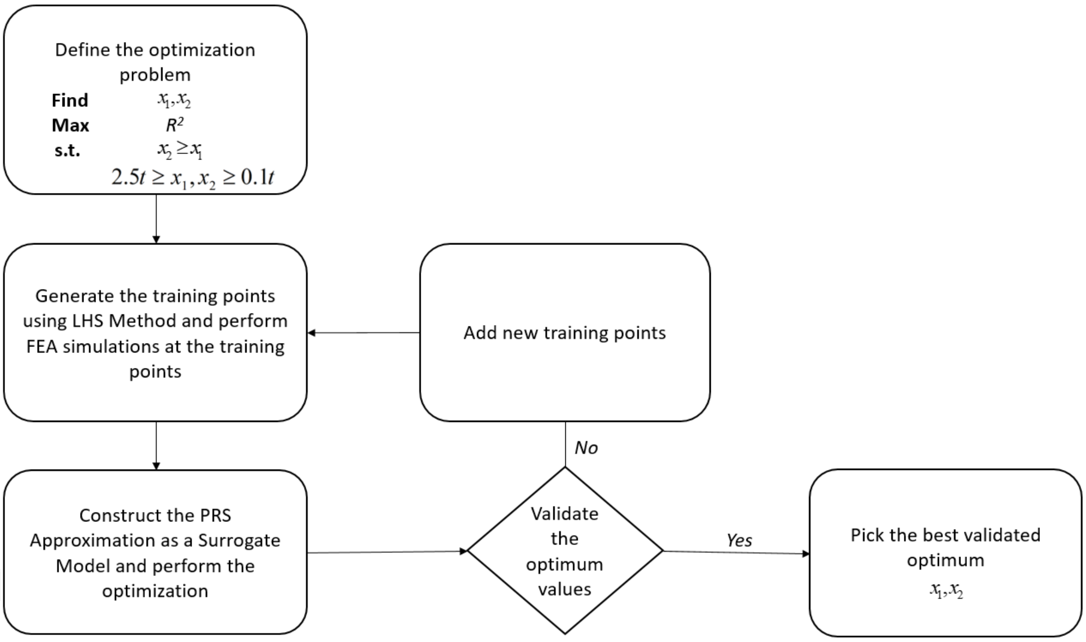


Figure 4.7 Flowchart for performing surrogate-based optimization

20 design points were created (see Table 4.2 and Figure 4.9) for the two design variables (x_1 and x_2) between $0.1\times t$ and $2.5\times t$ thicknesses as illustrated in Figure 4.8. Then, for the experimental dataset given in Table 15, the R^2 values are obtained by using these training points and thus the response values at the training points are generated. The surrogate model's response is estimated at any random point within the specified limits by these training points and the corresponding response values.

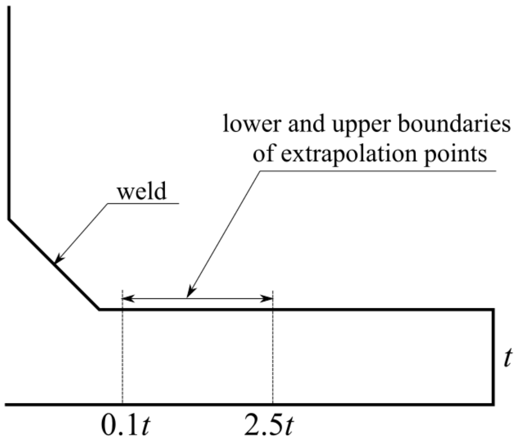


Figure 4.8 Illustration of the lower and upper boundaries of extrapolation points

Table 4.2 Design of experiments (DoE) of x_1 and x_2 and the response values (R^2) at the DoEs

	$x_1(\times t)$	$x_2(\times t)$	R^2
DoE #1	0.8	1.1	0.59
DoE #2	1.8	2.0	0.32
DoE #3	1.2	1.6	0.56
DoE #4	0.6	1.4	0.59
DoE #5	0.9	1.5	0.60
DoE #6	0.2	0.4	0.56
DoE #7	0.1	1.3	0.55
DoE #8	1.7	2.3	0.28
DoE #9	1.9	2.2	0.30
DoE #10	1.1	1.7	0.56
DoE #11	0.5	1.0	0.57
DoE #12	0.2	0.3	0.56
DoE #13	1.5	2.3	0.35
DoE #14	0.3	1.8	0.57
DoE #15	0.5	2.1	0.56
DoE #16	0.7	1.8	0.59
DoE #17	1.1	2.1	0.50
DoE #18	0.4	2.4	0.56
DoE #19	1.2	2.5	0.41
DoE #20	2.2	2.5	0.23

As mentioned in Section 4.2, the experimental fatigue test data used in this study [16], [67]–[69] were carried out for 3 different loading conditions. The numerical investigations of this data is done using ABAQUS [92]. In order to obtain consistent results from the FEA simulations, mesh sizes were chosen as $0.1\times t$, where t is the thickness of the main plates. In cases where symmetry can be used while creating models, the definition of symmetry is made with appropriate boundary conditions.

An 8-node linear brick elements (C3D8R) are used in the ABAQUS element library. Since the integration points of the C3D8R elements are located in the middle of the element, discretization with fine mesh is needed in the regions having high stress values. It is observed from the FE analysis results that $0.1\times t$ mesh size can capture the stress concentration around the weld toe as can be seen in Figure 4.10 (a), Figure 4.11 (a) and Fig 4.12 (a). FE analyzes of the experimental studies given in Table 15 were performed and the normal stress contour plots for each loading conditions under specific force ranges are depicted in Figure 4.10 (c), Figure 4.11 (c) and Figure 4.12 (c). Moreover, Figure 4.10 (d), Figure 4.11 (d) and Figure 4.12 (d) show the normalized stresses obtained along the weld toe for each loading conditions.

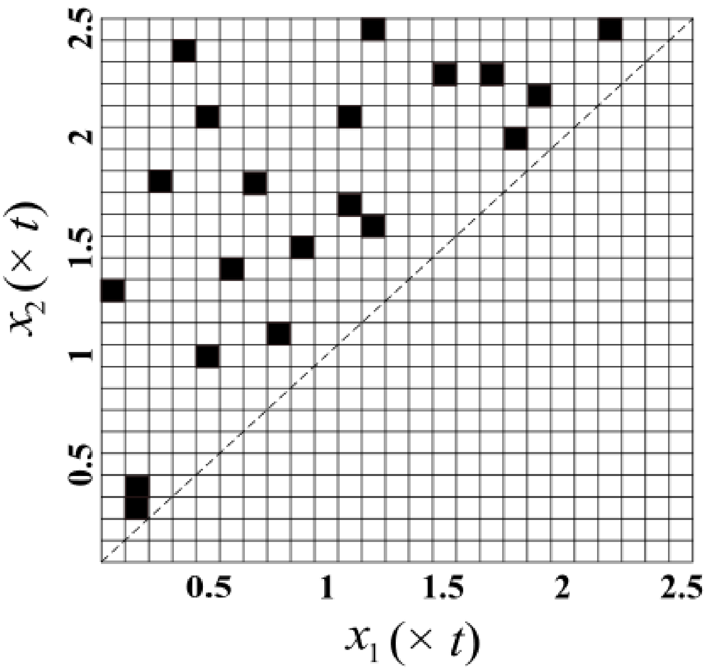


Figure 4.9 Illustration of the Design of experiments (DoE) of x_1 and x_2

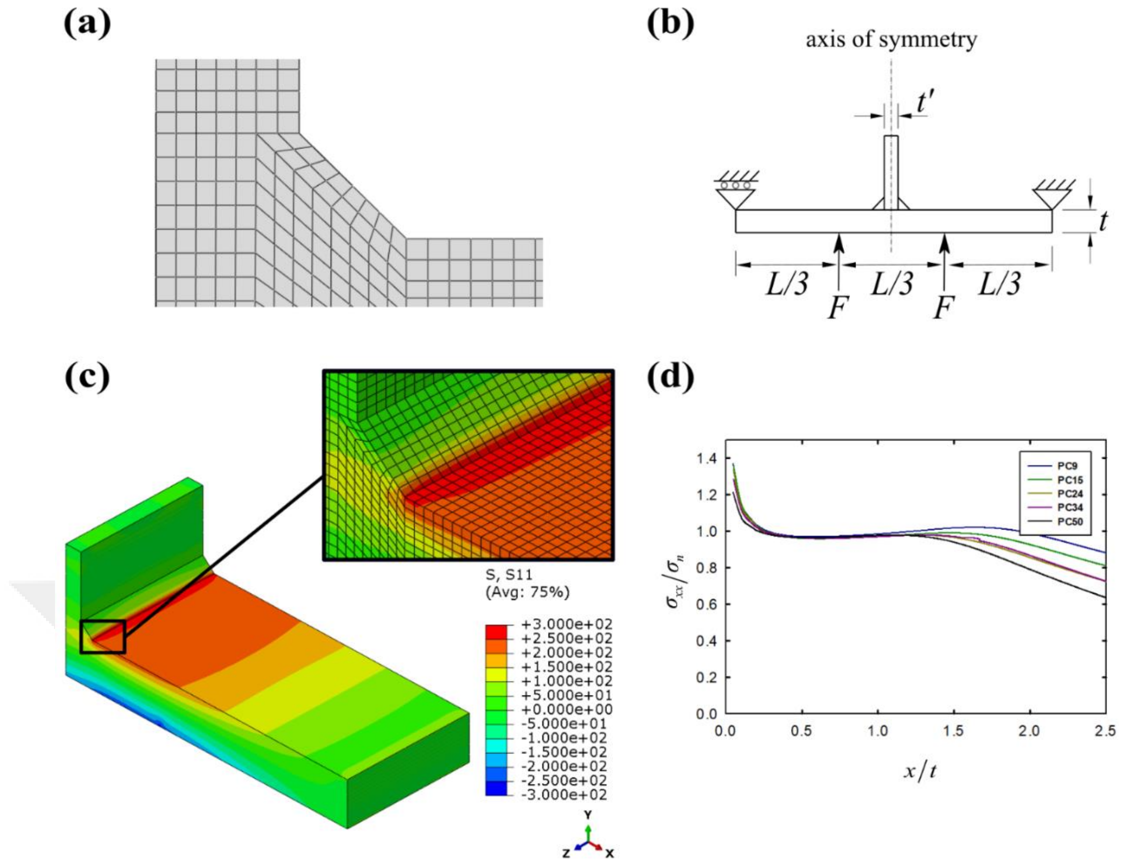


Figure 4.10 4-point bending results: (a) FE model of the weld, (b) geometric configurations, loading and support conditions, (c) normal stress results for PC15 under 6.75 kN (see Table 4.1) and (d) normalized stress results in the direction perpendicular to the weld toe

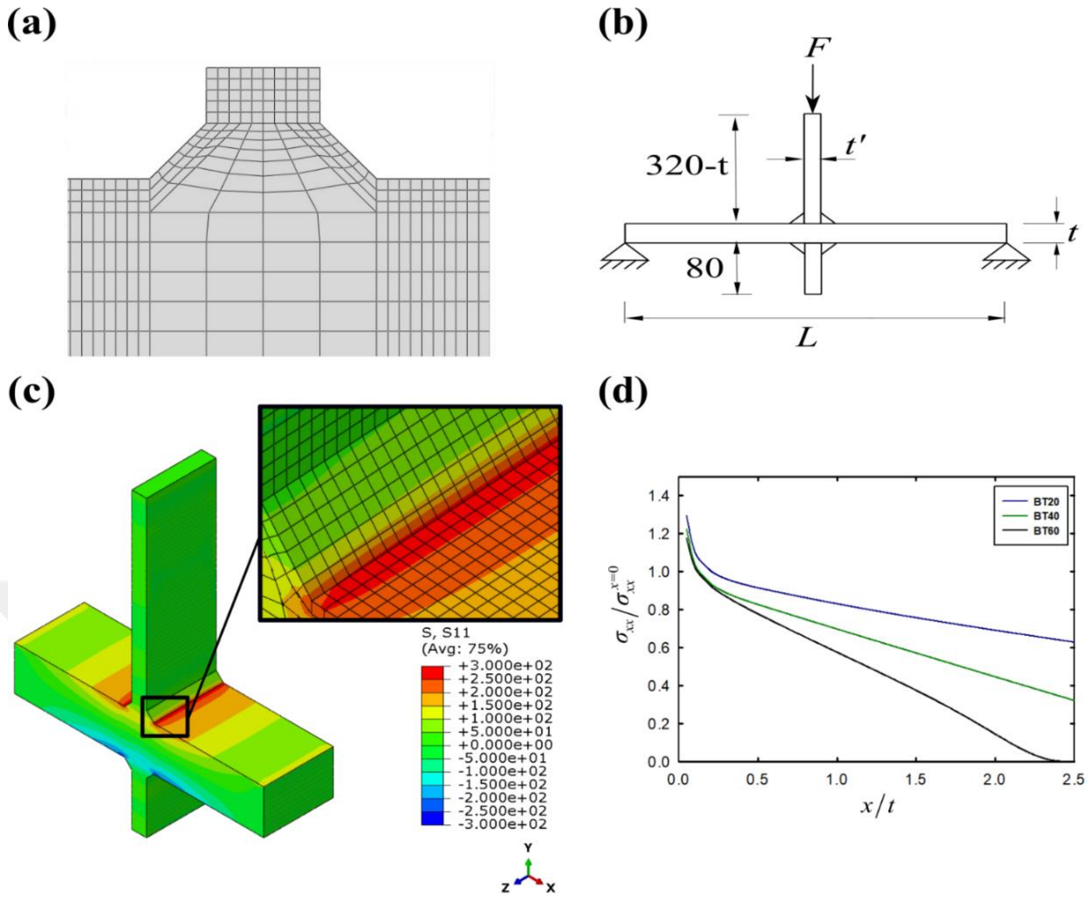


Figure 4.11 3-point bending results: (a) FE model of the weld, (b) geometric configurations, loading and support conditions, (c) normal stress results for BT40 under 76 kN (see Table 4.1) and (d) normalized stress results in the direction perpendicular to the weld toe

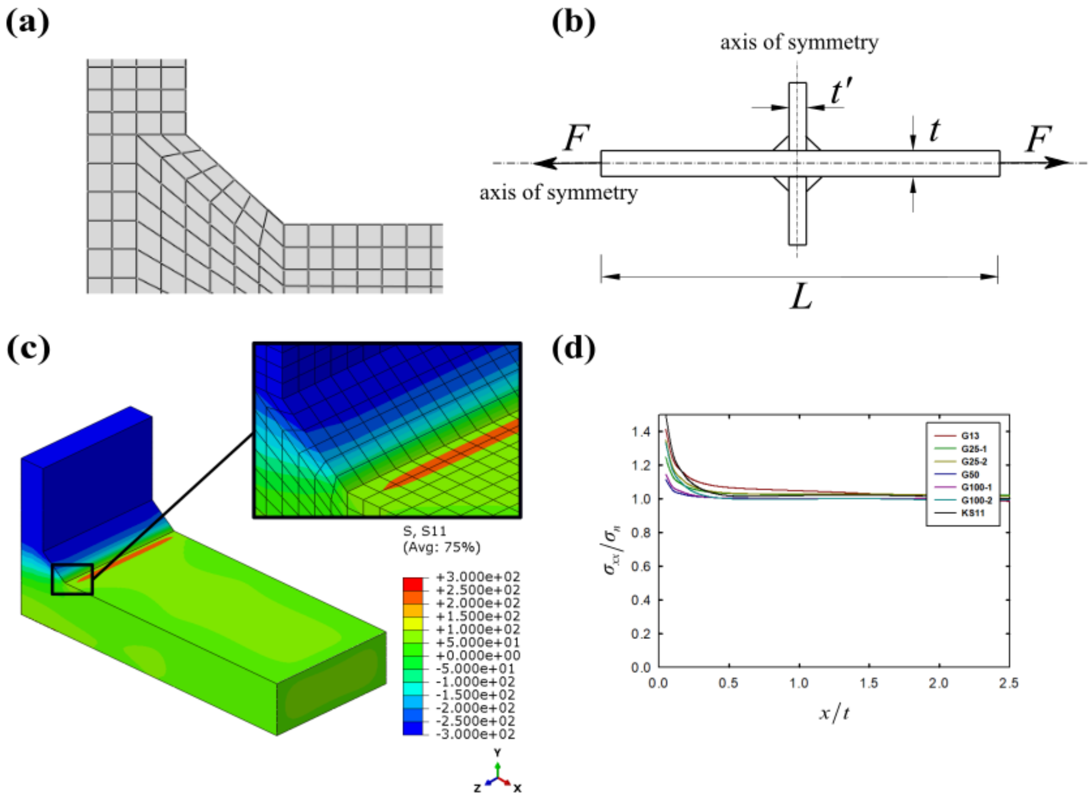


Figure 4.12 Axial loading results: (a) FE model of the weld, (b) geometric configurations, loading and support conditions, (c) normal stress results for GS25-2 under 1050 kN (see Table 4.1) and (d) normalized stress results in the direction perpendicular to the weld toe

By performing the FE analyzes of the specimens given in Table 4.1, S-N curve is created according to the existing HSS method as given in Figure 4.13. Coefficient of determination, R^2, is computed as 0.55.

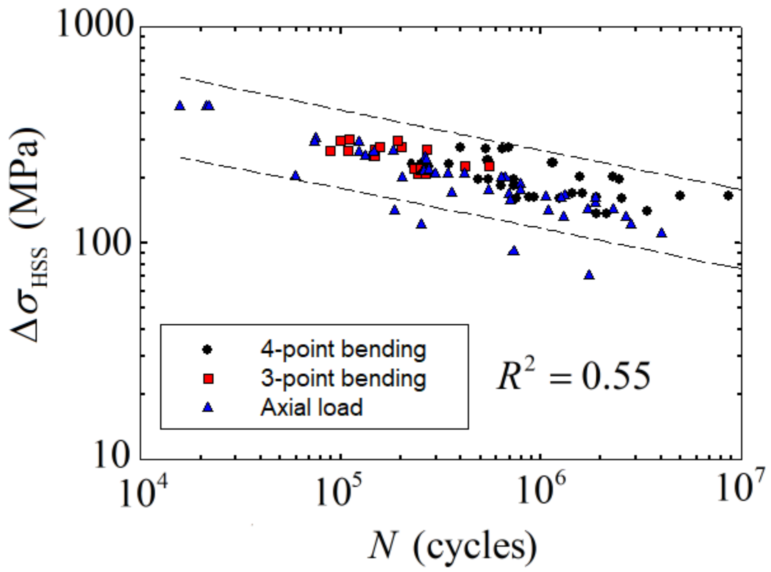


Figure 4.13 S-N curve obtained using HSS Method using the experimental dataset given in Table 4.1

Second-order PRS is constructed for prediction of the responses in constraints. The plot of the constructed surrogate model for R^2 is depicted in Figure 4.14. Two different views of surrogate model are given for easy depiction of maximum and minimum values of R^2.

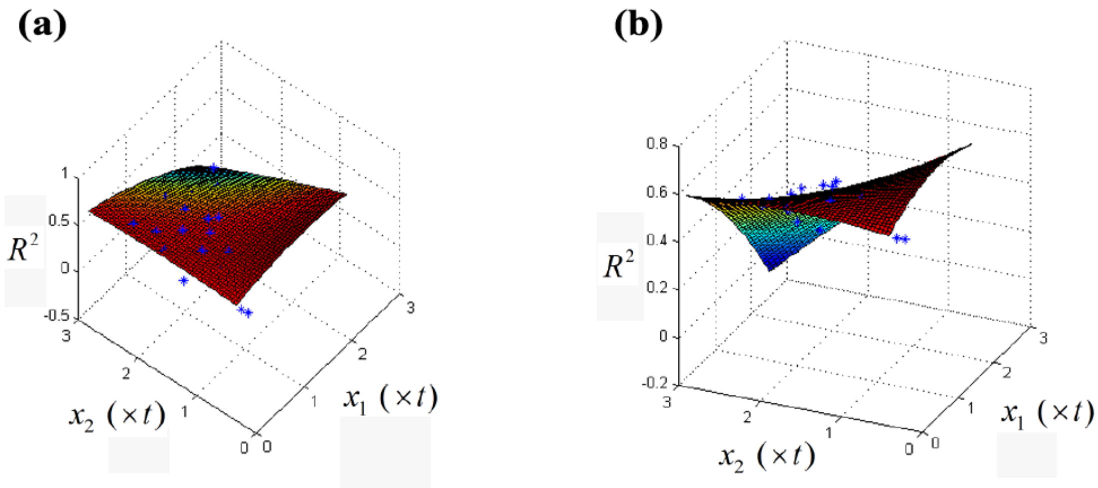


Figure 4.14 PRS model graph for R^2; a) view angle: (-55°, 60°) and b) view angle: (-65°, 30°)

Root mean square error (RMSE) is calculated to determine the accuracy of the PRS model. The difference between predicted R^2 values of PRS model and calculated R^2 values from the FEA results is taken into consideration to calculate RMSE which is defined as

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \qquad \textbf{(4.20)}$$

where N is the number of the training points ($N = 20$), $\hat{y}$ is the predicted value of R^2 from the PRS model and y is the input vector of R^2 which is obtained from FEA results. RMSE is normalized by using the maximum and the minimum values of input vector and calculated as 8% which points out that the constructed PRS model has good accuracy.

S-N curves of 20 DoE are created as depicted in Figure 4.15 - 34. It was observed that the variation of the extrapolation points under axial loading conditions did not have much effect on the HSSs and the width of the scatter of the S-N curves. In case of axial loading, since the stress concentration increase starts in the region very close to the weld toe and obtaining the nominal stress value in the regions outside this highly stressed region, makes it an expected result that the HSS values obtained under this loading condition are close to each other regardless of the extrapolation points. On the other hand, it has been determined that the variation of the extrapolation points under 3-point bending and 4-point bending loading conditions has a significant effect on the HSS values and therefore the width of the scatters.

It was obtained that in cases where the first extrapolation point (x_1) is located at a value higher than $1.5 \times t$ from the weld toe, the scattering width will be the highest, in other words, the fatigue life estimation will be performed most unsuccessfully (see Figures 4.16, 4.22, 4.23 and 4.34.) It was determined that HSS values of BT60 sample under 3-point bending loading condition caused low R^2 value to be obtained in these figures. Considering that the distance of the extrapolation point from the weld toe is obtained by multiplying the plate thickness with a certain coefficient (here

$x_1 = 1.7\times t, 1.8\times t, 1.9\times \mathrm{t}$ and $2.2\times t$), it is an expected result that this high-thickness plate (BT60 has a thickness of 60 mm) may cause a deviation in the S-N curves.

Considering the DoE points where the highest R^2 values were obtained (see Figures 4.15, 4.16, 4.19 and 4.30), it was determined that the range of the first extrapolation point (x_1) was between $0.5\times t$ and $1\times t$, and the second extrapolation point (x_2) was between $1.1\times t$ and $1.8\times t$.

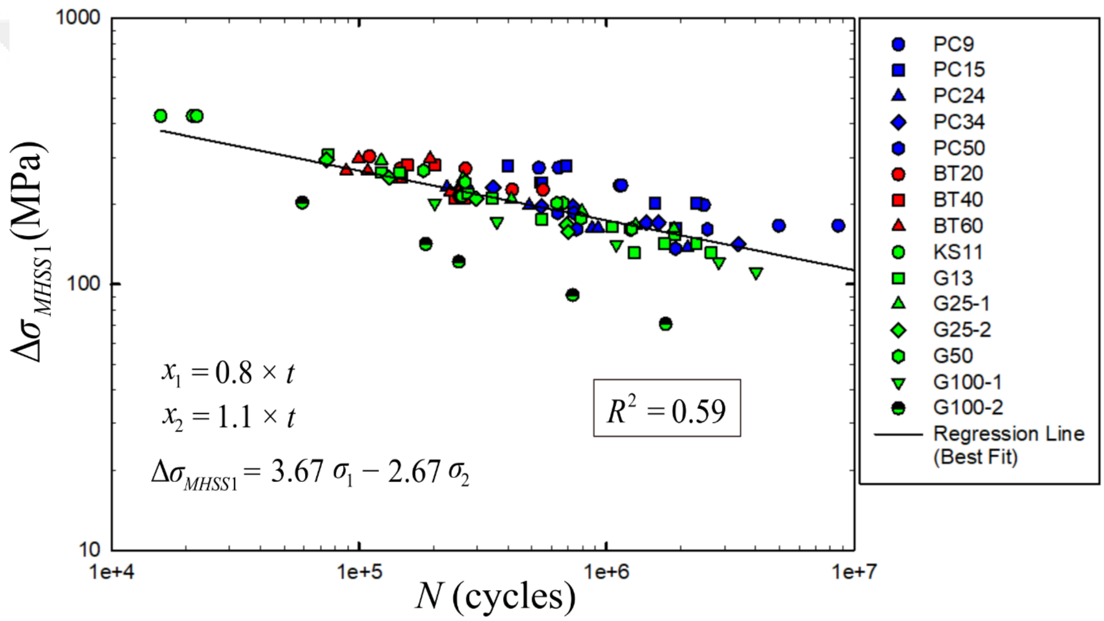


Figure 4.15 S-N curve for DoE #1

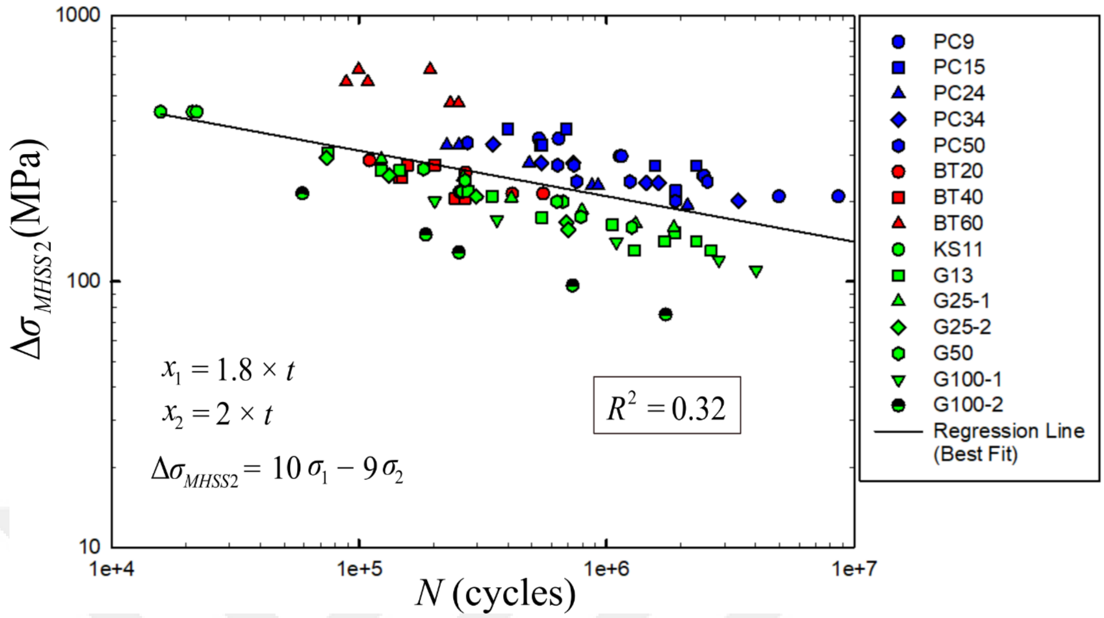


Figure 4.16 S-N curve for DoE #2

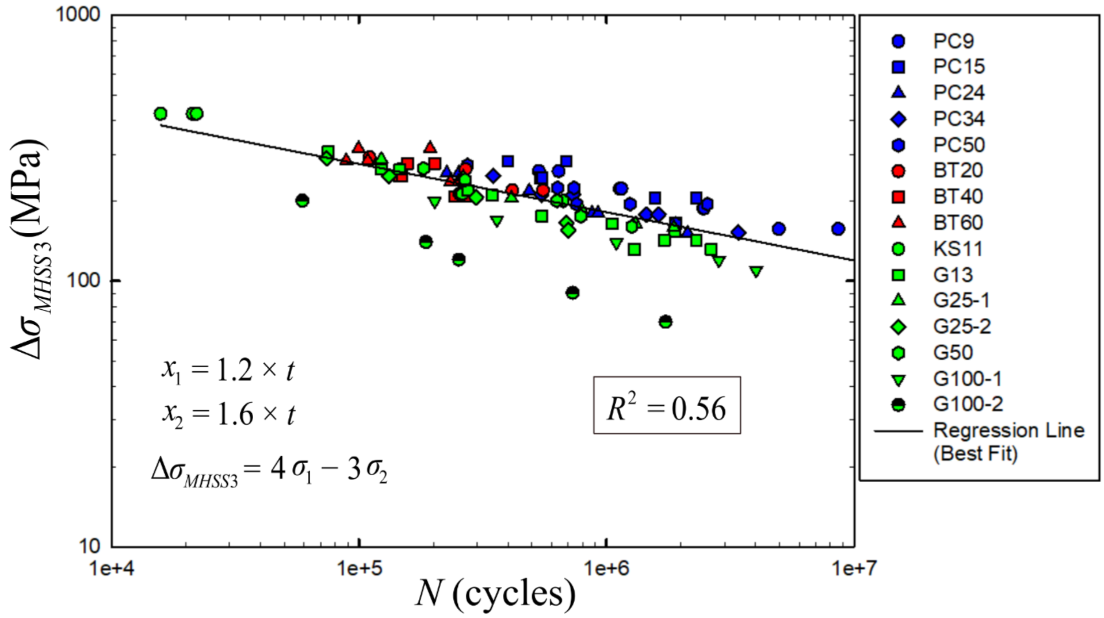


Figure 4.17 S-N curve for DoE #3

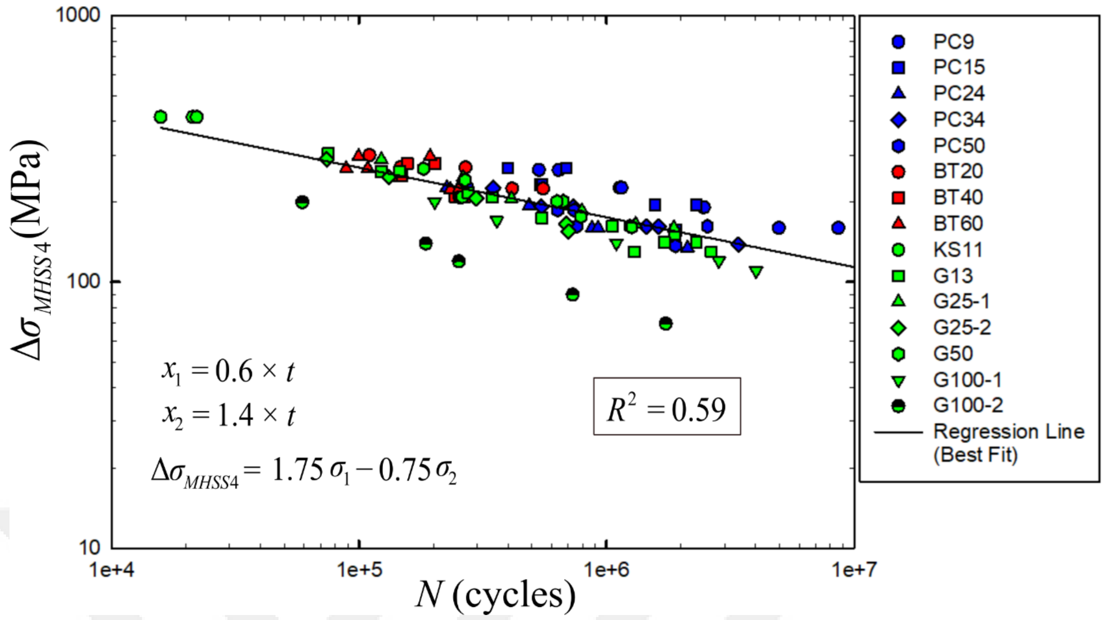


Figure 4.18 S-N curve for DoE #4

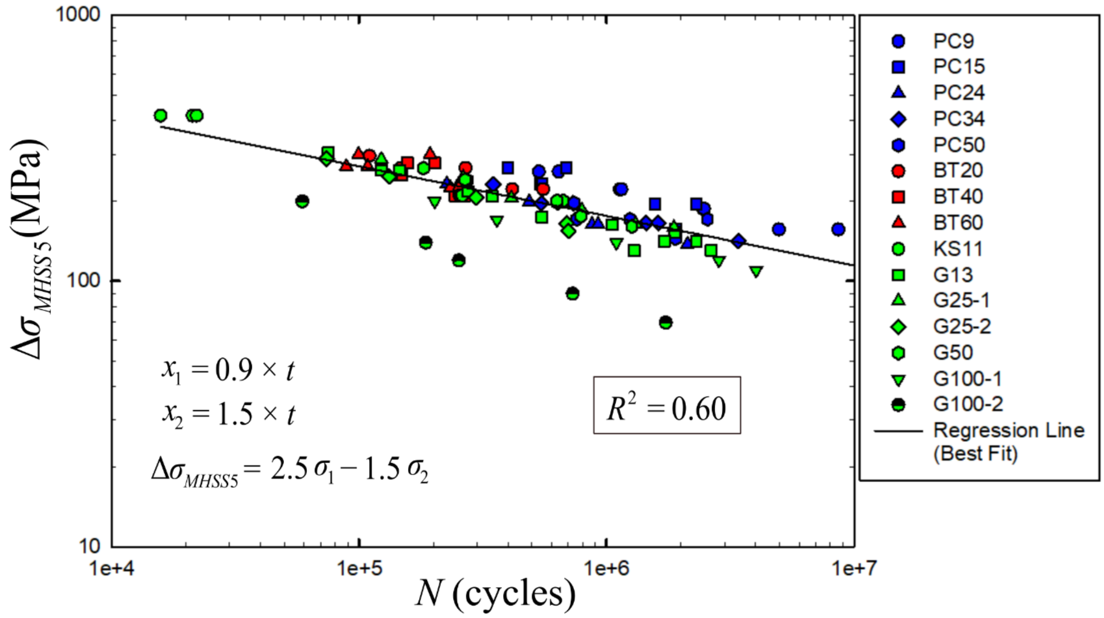


Figure 4.19 S-N curve for DoE #5

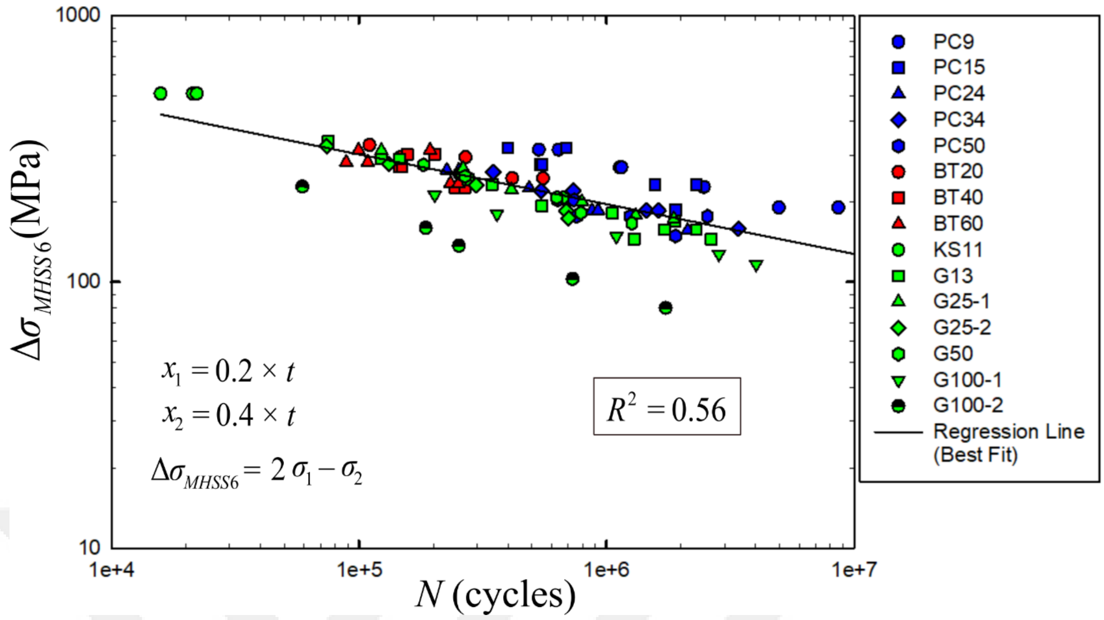


Figure 4.20 S-N curve for DoE #6

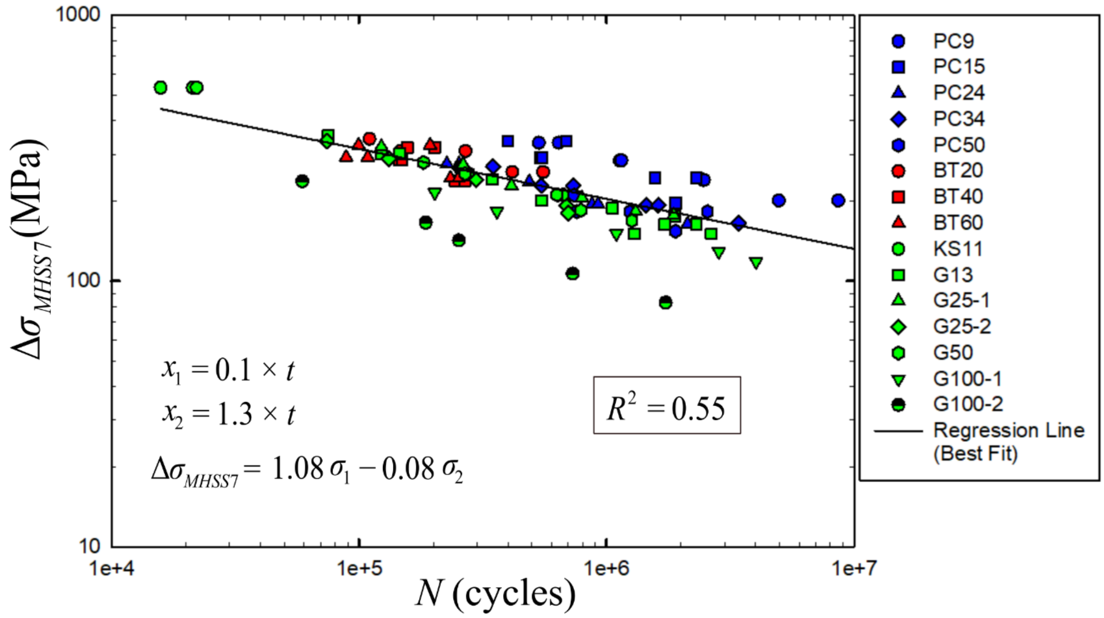


Figure 4.21 S-N curve for DoE #7

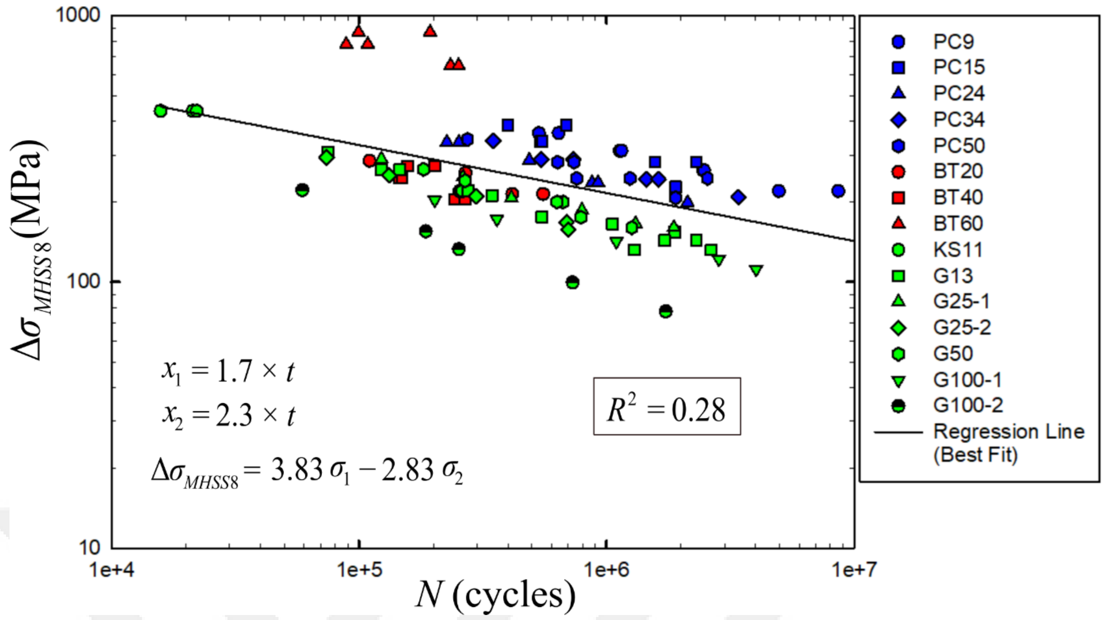


Figure 4.22 S-N curve for DoE #8

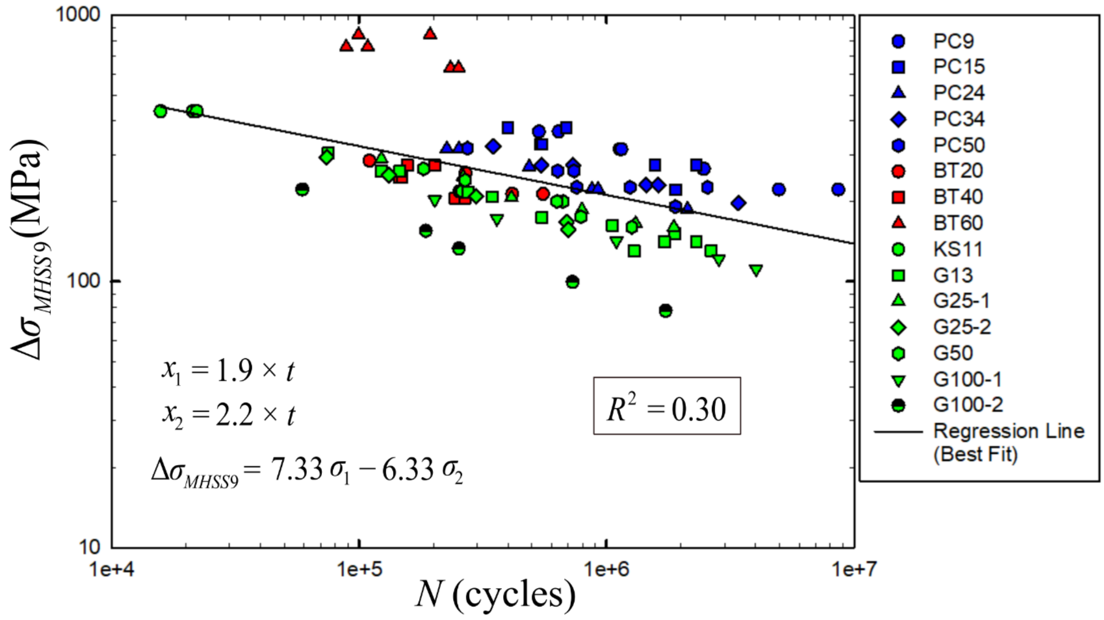


Figure 4.23 S-N curve for DoE #9

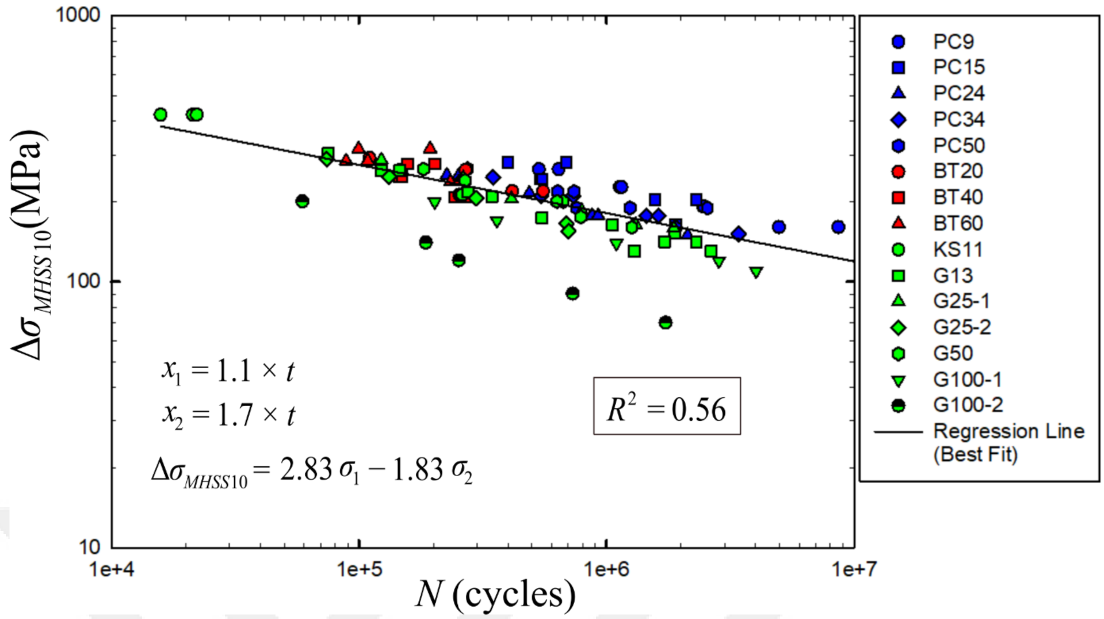


Figure 4.24 S-N curve for DoE #10

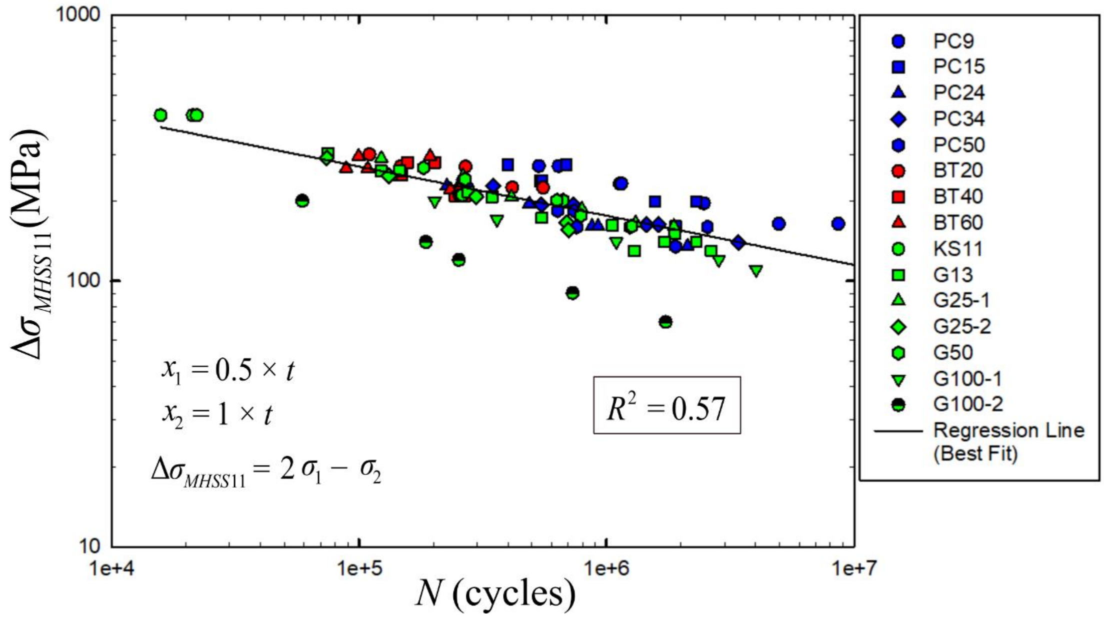


Figure 4.25 S-N curve for DoE #11

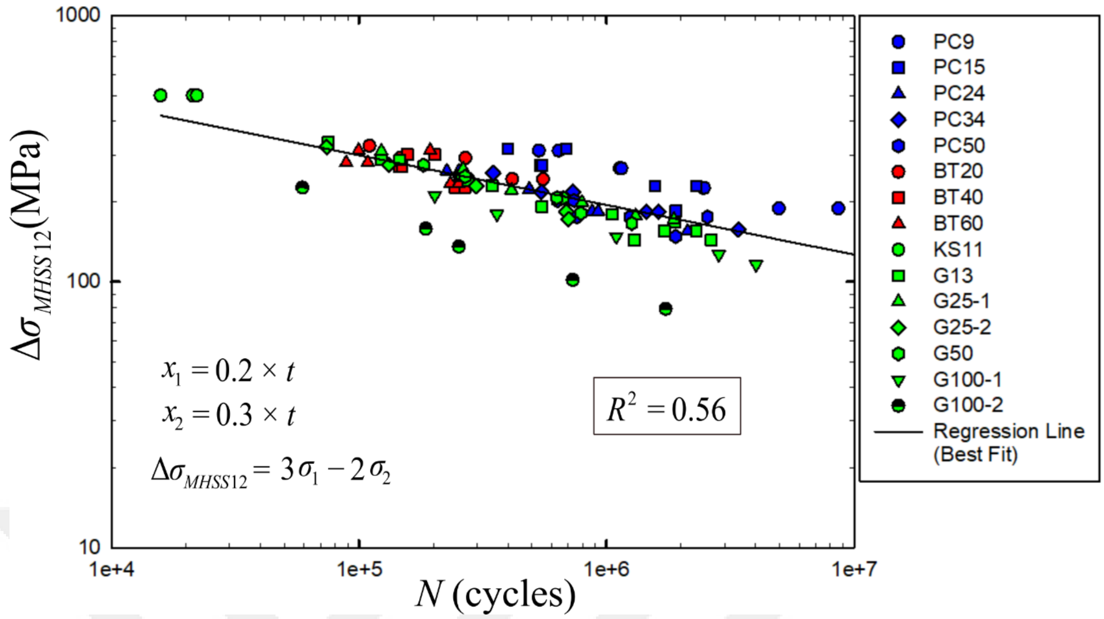


Figure 4.26 S-N curve for DoE #12

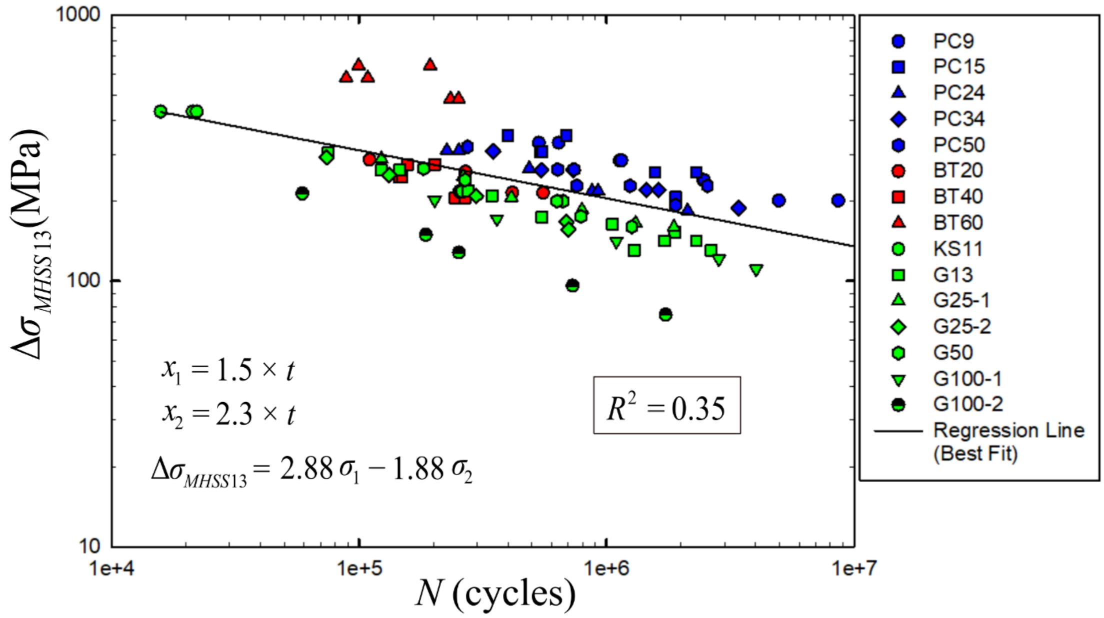


Figure 4.27 S-N curve for DoE #13

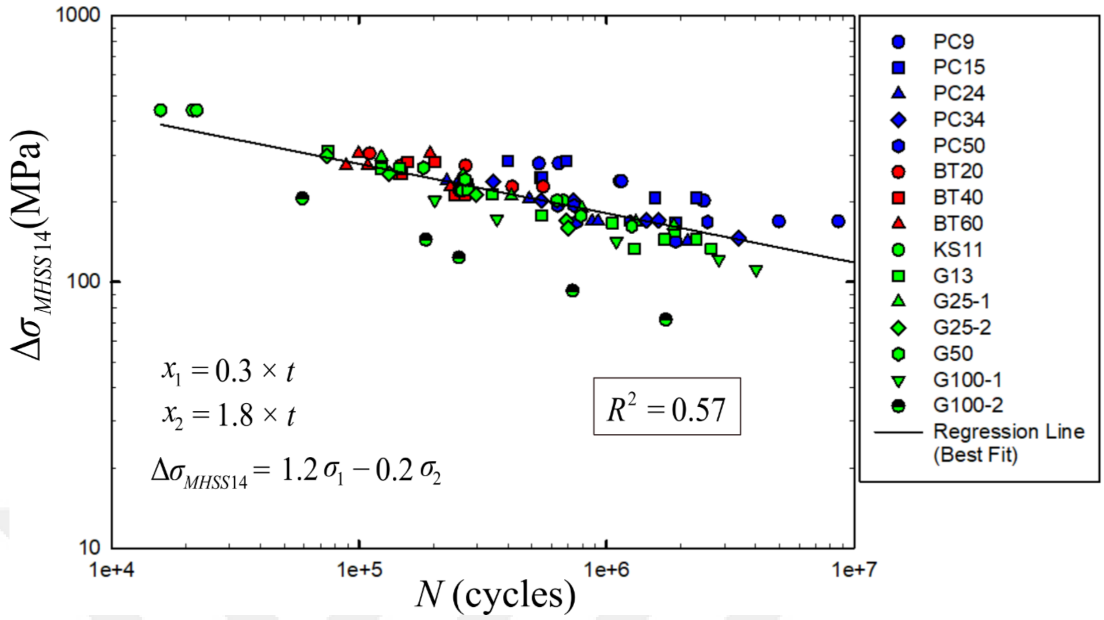


Figure 4.28 S-N curve for DoE #14

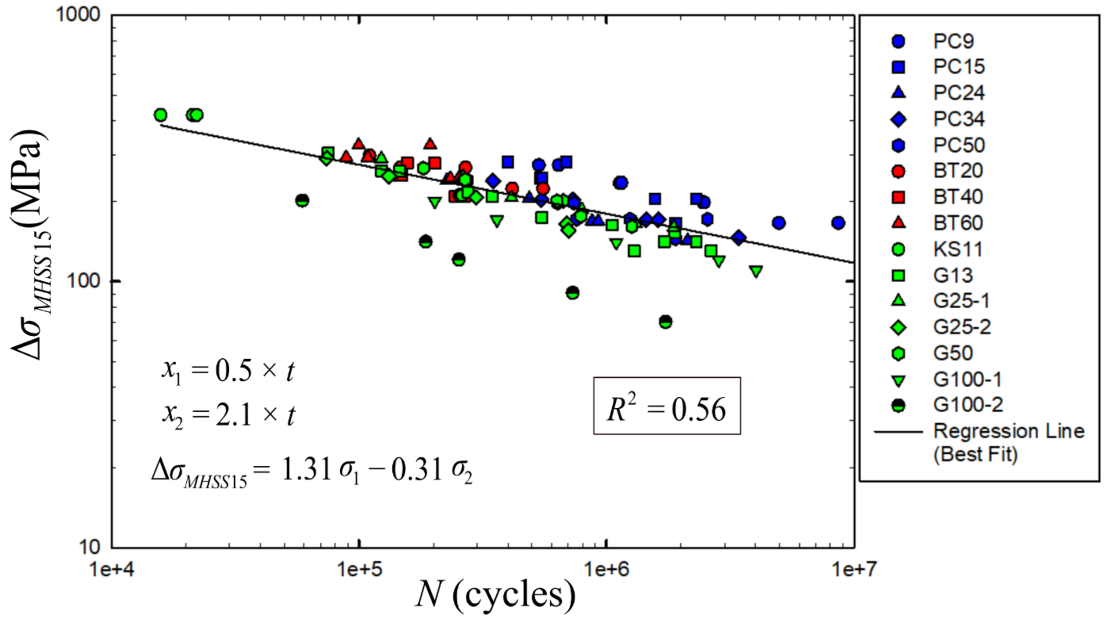


Figure 4.29 S-N curve for DoE #15

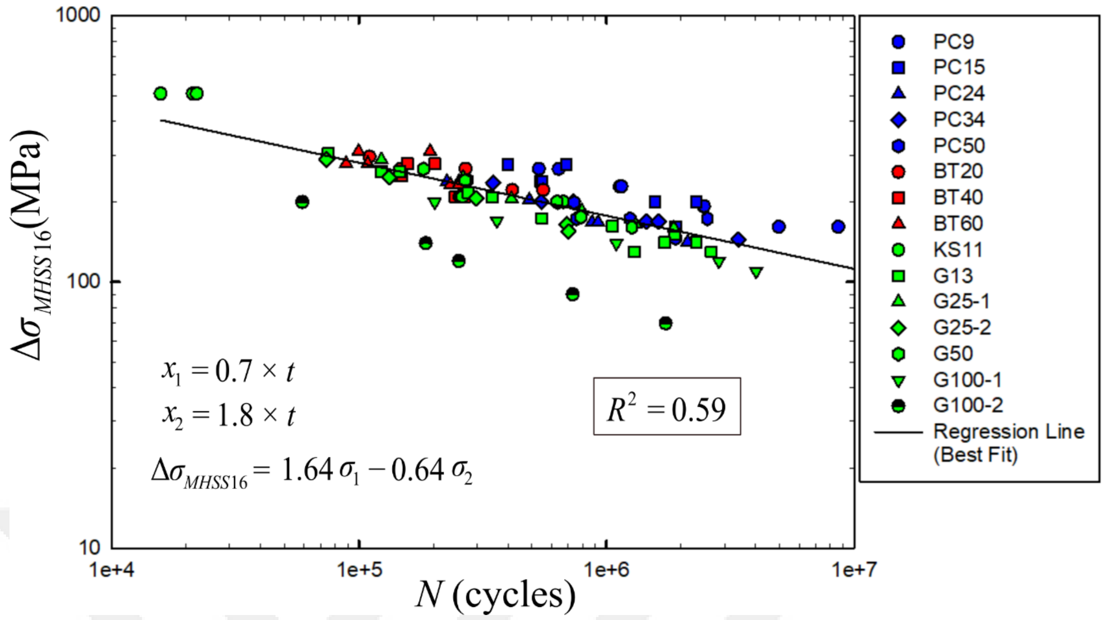


Figure 4.30 S-N curve for DoE #16

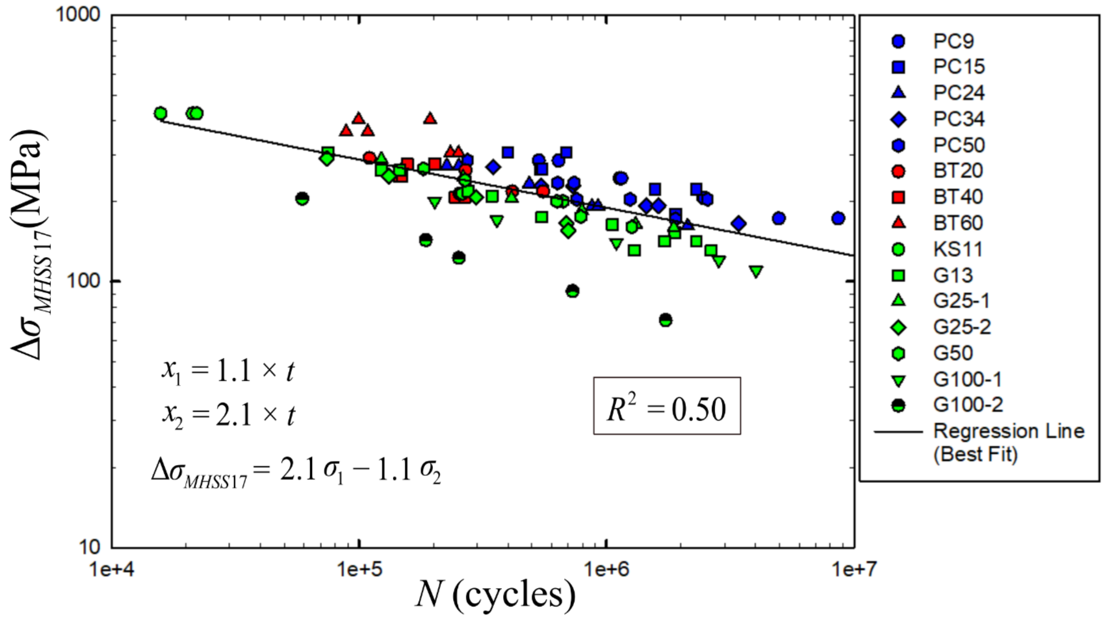


Figure 4.31 S-N curve for DoE #17

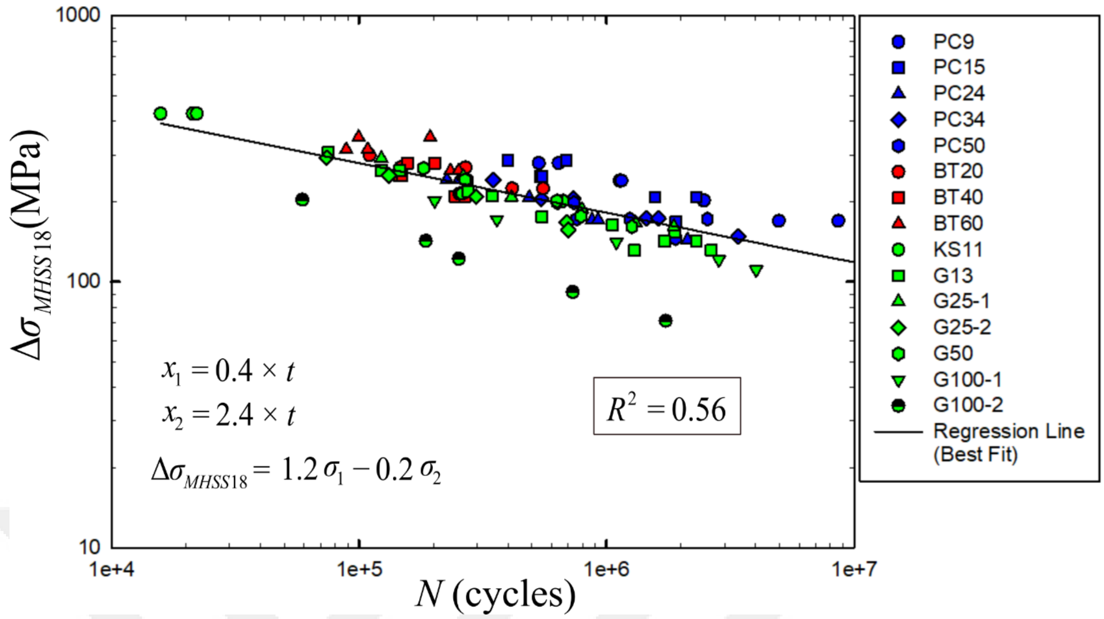


Figure 4.32 S-N curve for DoE #18

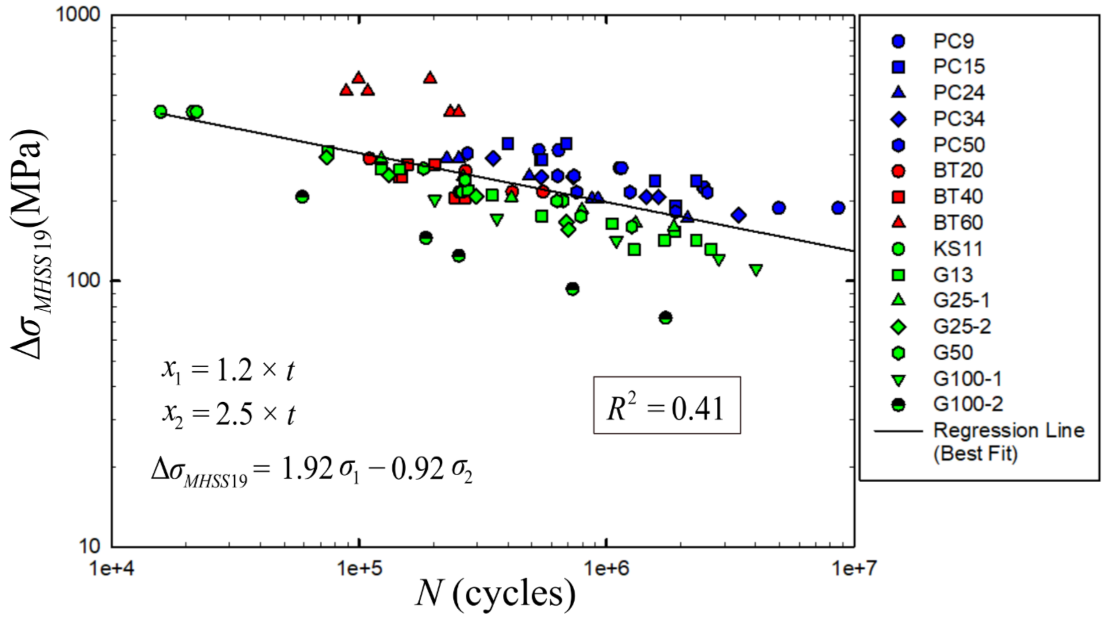


Figure 4.33 S-N curve for DoE #19

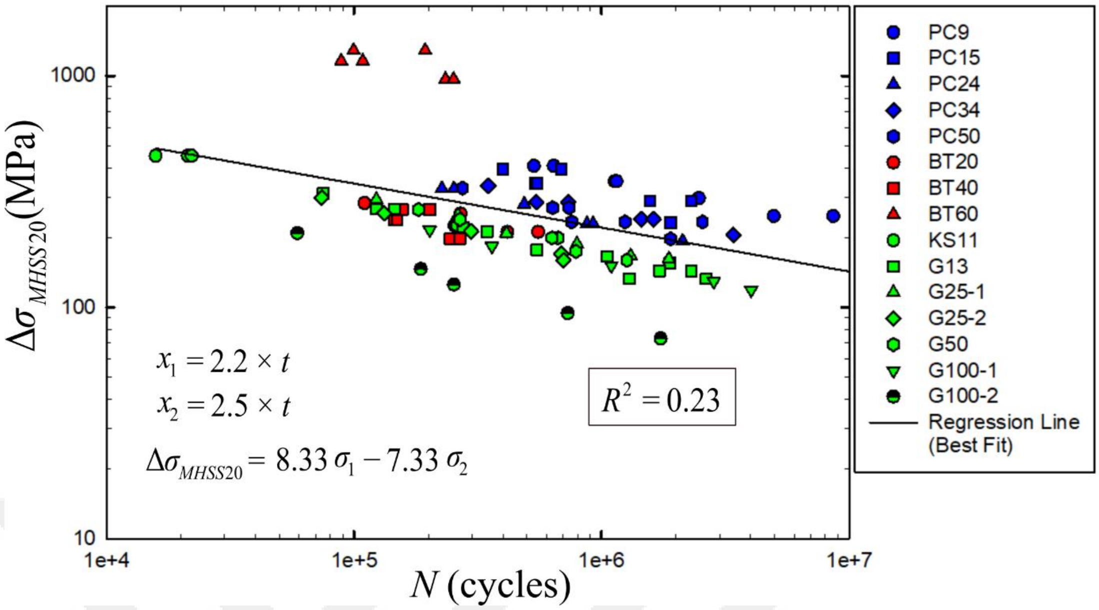


Figure 4.34 S-N curve for DoE #20

Optimization is performed by using the constructed second-order PRS model. Optimum values of design variables (x_1 and x_2) need to be validated with FEA simulations. This validation is done by recalculating the R^2 values according to the new optimum values of x_1 and x_2 from the FEA results. With the successful validation of the surrogate model, it is concluded that the optimization results are successful and the new values of x_1 and x_2 can be used as optimum values for R^2.

4.5 Results

As a result of the optimization performed using the surrogate model, the new values obtained for the design variables x_1 and x_2 are $0.9\times t$ and $1.5\times t$, respectively. The optimum value of the new R^2 value obtained from surrogate model ($R^2 = 0.6022$) is successfully validated with the R^2 value calculated from FEA results ($R^2 = 0.5958$). Therefore, the modified HSS equation can be written as

$$\sigma_{mhss} = 2.5\sigma_{0.9t} - 1.5\sigma_{1.5t} \tag{4.21}$$

where σ_{mhss} is the new modified HSS parameter and $\sigma_{0.9t}$ and $\sigma_{1.5t}$ are the normal stresses in the direction perpendicular to the weld toe at a distance of $0.9\times t$ and $1.5\times t$, respectively.

The new R^2 value obtained as a result of using this new modified HSS parameter in all experimental studies included from the literature is calculated as 0.60 with a 9.1% improvement. S-N curve is created according to the modified HSS method as given in Figure 4.34. Although the difference between S-N curves of the new modified method (see Figure 4.35) and the existing method (see Figure 4.13) is quite minimal in the graphical view, there is almost 10% increase in R^2 value.

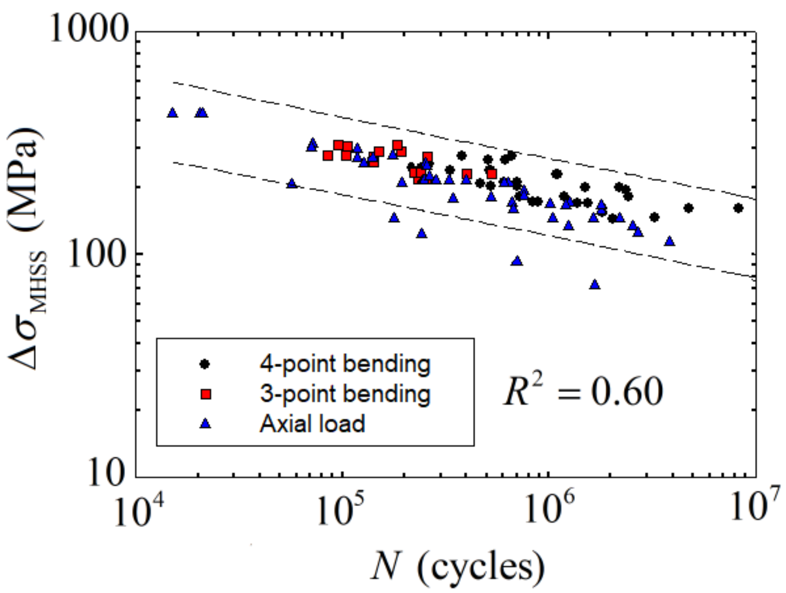


Figure 4.35 New S-N curve for modified HSS Method using the experimental dataset given in Table 4.1

CHAPTER 5

SUMMARY, CONCLUSIONS, AND FUTURE WORK

5.1 Summary

This thesis consisted of two main sections: understanding the existing fatigue life estimation methods and proposing a new fatigue life estimation method. The chapters of the thesis were organized according to this main sections. In this chapter, a comprehensive overview of the study and the findings of the chapters are provided.

In the first chapter, the importance of fatigue-induced failure in engineering applications was mentioned and general information about fatigue life estimation in welded structures was given.

The second chapter, classification of fatigue life assessment methods was given briefly with different fatigue life estimation approaches. Hot-spot Stress (HSS) method was taken into consideration in more detail since this thesis focused on the improvement of this method.

The next chapter, Chapter-3, consisted of the applications of the two most preferred fatigue life estimation methods: Structural Stress (SS) and HSS. To ascertain the accuracy of the fatigue lives obtained using shell element models, the hot-spot stresses were computed for three commonly encountered welded attachments and the results were compared with hot-spot stresses obtained from solid element models. Next, mesh sensitivity of SS method was investigated with four different mesh sizes. The fatigue lives of welded connections in a tram model were compared with using the HSS and SS methods in the next section of this chapter. Finally, a mesh convergence study for HSS method was studied in this chapter.

In the fourth chapter, a new modified HSS method was introduced. A database from the literature was created and basics of statistical design and analysis of experiments were given. The details of optimization study was provided with the results in the end of this chapter.

5.2 Conclusions

Since this study covers understanding the existing fatigue life estimation methods and proposing a new fatigue life estimation method, the conclusion is divided into two parts. The following conclusions can be drawn from the first part of this study:

- For the tensile loading case, as the plate thickness increases the difference between hot-spot stress results obtained using shell and solid elements increases, as expected.
- For the four point bending and biaxial loading cases, modelling the solid geometry without the weld did not have a significant effect on the HSS results.
- The benchmark study of the welded attachments showed that shell elements can be used in HSS fatigue analysis in the finite element models of large structures.
- Mesh insensitivity property of the SS method was proved on a benchmark geometry.
- The SS method predicts more conservative fatigue lives compared to the HSS method in critical regions of investigated tram model.
- It is determined that the difference between models discretized with $0.1 \times t$ and $0.05 \times t$ mesh sizes is negligible, except for the high stress concentration point at the weld toe which has not taken into consideration for HSS calculations. As a result, the use of rf-1 element type with a mesh size of $0.1 \times t$ is found suitable for FE analysis.

In the second part of this study, a new modified hot-spot stress method is proposed. The new modified HSS is tested using the experimental data covering 4-point, 3-point and axial loading situations. The aim of this study is to reduce the scatter of the S-N curve (increasing the R^2) by performing an optimization at the distances from the weld toe where the stresses are taken into account. An experimental data covering nearly 100 results with different geometries and loading conditions was used in the optimization study. The following conclusion can be drawn from the results of this study:

- The experimental data taken into consideration in this study includes a wide range of main plate thickness between 9mm-100mm.
- By performing the FE analyzes of the specimens an S-N curve is created according to the existing HSS method. Coefficient of determination, R^2, is computed as 0.55.
- New extrapolation locations (x_1 and x_2) are obtained as $0.9\times t$ and $1.5\times t$, respectively. This leads the modified HSS equation as $\sigma_{mhss} = 2.5\sigma_{0.9t} - 1.5\sigma_{1.5t}$
- The new R^2 value obtained as a result of using this new modified HSS method is calculated as 0.60 with a 9.1% improvement comparing the existing HSS method.

In summary, the results of this study shows that the new modified HSS method can reduce the scatter of the S-N curve. Considering that the current HSS method is still frequently used, this improvement in the scatter is crucial in fatigue life estimations.

5.3 Future Works

In this study, an optimization study performed for nearly 100 experiments and new extrapolation points were found to calculate the Hot-spot stress. The number of experiments considered can be increased with different new geometry types. The optimization study was focused on linear extrapolation technique in this study. A methodology for quadratic extrapolation technique can also be developed. Moreover, fatigue tests can be performed by placing strain gauges on new extrapolation points and the new method can be compared with experimental results.

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